

TAILWIND CSS + REACT

text-2xl font-bold bg-blue-500 rounded-lg

Tailwind CSS Master Cheat Sheet

The Practical Frontend Developer Reference

Documents **Tailwind CSS v4.3** (v4.x) · with v3 migration notes

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37 sections · CSS-to-Tailwind table · Quick finder · 19 UI patterns · React guide

How To Use This Reference

If you know...	Go to
The CSS you want	Section 32: CSS to Tailwind conversion table
The effect you want (center, hide, shadow)	Section 33: "I Want To Do X" quick finder
The class prefix , but not the values	Section 35: cheat tables
How a class name is built	Sections 1 and 34
A class that does nothing	Section 36: common mistakes
Ready-made components	Section 31: UI patterns
React specifics	Section 37

Callout legend

Tailwind v3 note: what changed from v3 to v4 and how to migrate.

Important: a trap or common surprise.

Note: useful context.

Code samples use plain HTML (`class`). In React, write `className` . All class names are valid for the documented version; version-dependent items are marked (for example "v4.1+").

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1. Tailwind Fundamentals

1.1 What Tailwind CSS Is

Tailwind CSS is a **utility-first CSS framework**. Instead of writing CSS in a separate file and inventing class names like `.card-title`, you compose your design directly in your markup using small, single-purpose classes.

```
<!-- Traditional CSS -->
<h2 class="card-title">Hello</h2>
<style>
  .card-title { font-size: 1.5rem; font-weight: 700; color: #3b82f6; }
</style>

<!-- Tailwind -->
<h2 class="text-2xl font-bold text-blue-500">Hello</h2>
```

Why people like it:

- **No naming things.** You never invent `.wrapper-inner-2` again.
- **No context switching.** Style and structure live in the same file (perfect for React components).
- **Consistent design.** Spacing, colors and font sizes come from a shared scale, so designs look cohesive.
- **Small CSS output.** Tailwind only generates the CSS for classes you actually use.
- **Responsive and state styling is built in** (`md:`, `hover:`, `dark:`).

1.2 The Utility-First Concept

One class = one (or a few) CSS declaration(s). You combine many small classes to build a design.

Class	CSS it generates
<code>p-4</code>	<code>padding: 1rem</code>
<code>text-xl</code>	<code>font-size: 1.25rem; line-height: 1.75rem</code>
<code>font-bold</code>	<code>font-weight: 700</code>
<code>bg-blue-500</code>	<code>background-color: <blue 500></code>
<code>rounded-lg</code>	<code>border-radius: 0.5rem</code>
<code>flex</code>	<code>display: flex</code>

1.3 Documented Version and Setup

Version documented: Tailwind CSS v4.3 (v4.x series). Everything in this manual targets v4 syntax. Where v3 differs, you will see a **Tailwind v3 note**.

Minimum browser support for v4: Safari 16.4+, Chrome 111+, Firefox 128+ (it relies on modern CSS such as `@property` and `color-mix()`). If you must support older browsers, stay on Tailwind v3.4.

Quick setup: React + Vite (Tailwind v4)

```
npm create vite@latest my-app -- --template react
cd my-app
npm install tailwindcss @tailwindcss/vite
```

```
// vite.config.js
import { defineConfig } from 'vite'
import react from '@vitejs/plugin-react'
import tailwindcss from '@tailwindcss/vite'

export default defineConfig({
  plugins: [react(), tailwindcss()],
})
```

```
/* src/index.css */
@import "tailwindcss";
```

```
// src/App.jsx
export default function App() {
  return <h1 className="text-3xl font-bold underline">Hello Tailwind</h1>
}
```

Make sure `index.css` is imported in `main.jsx` (`import './index.css'`). There is **no** `tailwind.config.js` and **no** `content` array needed in v4 by default: Tailwind scans your project automatically.

Tailwind v3 note: v3 used `@tailwind base; @tailwind components; @tailwind utilities;`, a `tailwind.config.js` file with a `content: [...]` array, and PostCSS + autoprefixer. In v4 you write `@import "tailwindcss";` and configure the theme in CSS with `@theme`.

Recommended tools

- **Tailwind CSS IntelliSense** (VS Code extension): autocomplete, hover previews of the CSS behind each class, and lint warnings for invalid classes.
- **prettier-plugin-tailwindcss**: automatically sorts classes into a consistent order.

1.4 How Tailwind Class Names Are Constructed

Almost every class follows this pattern:

```
[variants:] [-] utility - value [/ modifier] [!]
```

Part	Meaning	Example
variants	Conditions: screen size, hover, dark mode... (separated by <code>:</code>)	<code>md:hover:</code>
<code>-</code> (leading)	Negative value	<code>-mt-4</code>
utility	Which CSS property family	<code>bg</code> , <code>text</code> , <code>p</code> , <code>w</code> , <code>border</code>
value	From the theme scale, or arbitrary <code>[...]</code>	<code>blue-500</code> , <code>4</code> , <code>lg</code> , <code>[350px]</code>
<code>/modifier</code>	Opacity, line-height or a fraction	<code>bg-black/50</code> , <code>text-xl/8</code> , <code>w-1/2</code>
<code>!</code> (trailing)	Important	<code>p-4!</code>

Worked examples

Class	Breakdown
<code>text-xl</code>	utility <code>text</code> (font size) + value <code>xl</code>
<code>font-bold</code>	utility <code>font</code> (weight) + value <code>bold</code>

Class	Breakdown
<code>bg-blue-500</code>	utility <code>bg</code> (background color) + color <code>blue</code> + shade <code>500</code>
<code>p-4</code>	utility <code>p</code> (padding) + scale value <code>4</code> (= 1rem)
<code>mt-6</code>	<code>m</code> (margin) + <code>t</code> (top) + <code>6</code> (= 1.5rem)
<code>rounded-lg</code>	utility <code>rounded</code> (border radius) + size <code>lg</code>
<code>shadow-md</code>	utility <code>shadow</code> + size <code>md</code>
<code>md:hover:bg-red-500</code>	at <code>md</code> screens and up, on hover, background red 500
<code>-translate-y-1</code>	negative translate on Y axis
<code>w-[350px]</code>	width with an arbitrary value

1.5 How To Understand A Class Without Memorizing It

Use this three-step method every time:

1. **Name the CSS property** you want (for example `margin-top`).
2. **Take its Tailwind prefix** (`margin` becomes `m`; `top` becomes `t`; so `mt`).
3. **Add the value** from the scale (`mt-4`) or an arbitrary one (`mt-[37px]`).

1.6 Common Naming Patterns

Pattern	Meaning	Examples
Shorthand prefix	First letters of the CSS property	<code>m</code> margin, <code>p</code> padding, <code>w</code> width, <code>h</code> height, <code>bg</code> background
Side suffix	<code>t</code> top, <code>r</code> right, <code>b</code> bottom, <code>l</code> left	<code>mt-4</code> , <code>pr-2</code> , <code>border-b</code>
Axis suffix	<code>x</code> horizontal, <code>y</code> vertical	<code>px-4</code> , <code>my-2</code> , <code>translate-x-2</code>
Logical sides	<code>s</code> start, <code>e</code> end (RTL friendly)	<code>ps-4</code> , <code>me-2</code> , <code>border-s</code>
Scale numbers	Number x 0.25rem (4px)	<code>p-1</code> = 4px, <code>p-4</code> = 16px, <code>p-8</code> = 32px
T-shirt sizes	<code>xs</code> <code>sm</code> <code>md</code> <code>lg</code> <code>xl</code> <code>2xl</code> <code>3xl</code> ...	<code>text-lg</code> , <code>shadow-xl</code> , <code>rounded-2xl</code>
Fractions	Percent widths	<code>w-1/2</code> , <code>w-2/3</code> , <code>basis-1/4</code>
Keywords	CSS keywords	<code>w-full</code> , <code>w-auto</code> , <code>h-screen</code> , <code>w-fit</code>
Color + shade	<code>{color}-{50..950}</code>	<code>text-gray-700</code> , <code>bg-red-500</code>
Negative	Leading dash	<code>-mt-4</code> , <code>-rotate-45</code> , <code>-inset-2</code>
Opacity slash	<code>/{0-100}</code>	<code>bg-blue-500/50</code>
Axis-specific split	Property family + direction	<code>overflow-x-auto</code> , <code>gap-y-4</code>

1.7 Responsive Prefixes (Overview)

Prefix any utility with a breakpoint name to apply it **at that width and above** (mobile-first):

```
<div class="text-sm md:text-lg lg:text-2xl">
  Small on mobile, large from 768px, even larger from 1024px
</div>
```

Prefix	Min width	CSS
(none)	0	applies to all sizes

Prefix	Min width	CSS
sm:	640px (40rem)	@media (min-width: 40rem)
md:	768px (48rem)	@media (min-width: 48rem)
lg:	1024px (64rem)	@media (min-width: 64rem)
xl:	1280px (80rem)	@media (min-width: 80rem)
2xl:	1536px (96rem)	@media (min-width: 96rem)

Full details are in [Section 23](#).

1.8 State Variants (Overview)

Prefix a utility with a state to apply it only in that state:

```
<button class="bg-blue-500 hover:bg-blue-600 focus:ring-2 active:scale-95 disabled:opacity-50">
  Click me
</button>
```

Variants can be stacked; they apply **left to right**: `md:hover:bg-red-500` means "at md and up, when hovered". Full details are in [Section 24](#).

1.9 Arbitrary Values (Overview)

When the theme scale does not contain the value you need, put any CSS value in square brackets:

```
<div class="w-[350px] mt-[37px] bg-[#1e293b] text-[22px] grid-cols-[200px_1fr]"></div>
```

- Use instead of spaces: `grid-cols-[200px_1fr]` becomes `200px 1fr`.
- Works with variants: `md:w-[500px]`, `hover:bg-[#123456]`.

Full details are in [Section 27](#).

1.10 Arbitrary Properties (Overview)

For a CSS property Tailwind has no utility for, write the property itself in brackets:

```
<div class="[mask-type:luminance] [--my-var:10px] hover:[mask-type:alpha]"></div>
```

Format: `[property:value]`. Works with variants too.

1.11 Theme Variables (the v4 way to customize)

In v4 you extend the design system in CSS with `@theme`. Every theme variable automatically creates utilities.

```
@import "tailwindcss";

@theme {
  --color-brand: oklch(0.65 0.2 260); /* creates bg-brand, text-brand, border-brand... */
  --font-display: "Poppins", sans-serif; /* creates font-display */
  --breakpoint-3xl: 120rem; /* creates 3xl: variant */
  --spacing: 0.25rem; /* base spacing unit (default) */
}
```

```
<h1 class="font-display text-brand 3xl:text-6xl">Custom theme</h1>
```

Tailwind v3 note: v3 customized the theme in `tailwind.config.js` under `theme.extend`. In v4 the same data lives in CSS `@theme` blocks. You can still load a JS config with `@config './tailwind.config.js'` for migration.

1.12 Tailwind v3 vs v4: What Changed (Cheat List)

Topic	Tailwind v3	Tailwind v4 (use this)
Import	<code>@tailwind base; @tailwind components; @tailwind utilities;</code>	<code>@import "tailwindcss";</code>
Config	<code>tailwind.config.js</code>	CSS <code>@theme { ... }</code>
Content paths	<code>content: [...]</code> required	Automatic detection (<code>@source</code> if needed)
Gradients	<code>bg-gradient-to-r</code>	<code>bg-linear-to-r</code> (old name still accepted as a legacy alias)
Opacity utilities	<code>bg-opacity-50</code> , <code>text-opacity-50</code>	Removed: use <code>bg-black/50</code> , <code>text-white/50</code>
Shadow scale	<code>shadow-sm</code> , <code>shadow</code> , <code>shadow-md</code>	<code>shadow-2xs</code> , <code>shadow-xs</code> , <code>shadow-sm</code> , <code>shadow-md</code> ... (v3 <code>shadow</code> = v4 <code>shadow-sm</code> ; v3 <code>shadow-sm</code> = v4 <code>shadow-xs</code>)
Inner shadow	<code>shadow-inner</code>	<code>inset-shadow-*</code>
Blur scale	<code>blur-sm</code> , <code>blur</code>	<code>blur-xs</code> , <code>blur-sm</code> ... (v3 <code>blur</code> = v4 <code>blur-sm</code>)
Radius scale	<code>rounded-sm</code> , <code>rounded</code>	<code>rounded-xs</code> , <code>rounded-sm</code> ... (v3 <code>rounded</code> = v4 <code>rounded-sm</code>)
Ring	<code>ring = 3px</code> , default blue	<code>ring = 1px</code> , default <code>currentColor</code> : use <code>ring-3</code> for old width
Outline reset	<code>outline-none</code>	<code>outline-hidden</code> (and <code>outline-none</code> now truly sets <code>outline-style: none</code>)
Default border color	<code>gray-200</code>	<code>currentColor</code> : always set a color, e.g. <code>border-gray-200</code>
Flex shrink/grow	<code>flex-shrink-0</code> , <code>flex-grow</code>	<code>shrink-0</code> , <code>grow</code>
Text overflow	<code>overflow-ellipsis</code>	<code>text-ellipsis</code>
Box decoration	<code>decoration-clone</code> , <code>decoration-slice</code>	<code>box-decoration-clone</code> , <code>box-decoration-slice</code>
Placeholder color	<code>placeholder-gray-400</code>	<code>placeholder:text-gray-400</code> (variant)
Important	<code>!font-bold</code> (prefix)	<code>font-bold!</code> (suffix; prefix still works but is legacy)
CSS variable value	<code>bg-[--brand]</code>	<code>bg(--brand)</code>
Stacked variants order	right to left	left to right
Dark mode toggle	<code>darkMode: 'class'</code> in config	<code>@custom-variant dark (...)</code> in CSS
Button cursor	<code>cursor: pointer</code> by default	<code>cursor: default</code> : add <code>cursor-pointer</code> yourself
<code>hover:</code>	Applied on touch devices too	Only when the device supports hover
<code>container</code> options	<code>center</code> , <code>padding</code> in config	Customize with <code>@utility container {...}</code>

2. Spacing System

2.1 How the Spacing Scale Works

Tailwind spacing is based on **one unit: 0.25rem (= 4px at default browser size)**. The number in the class is a **multiplier**:

```
value x 0.25rem = size
1 x 0.25rem = 0.25rem = 4px
4 x 0.25rem = 1rem = 16px
8 x 0.25rem = 2rem = 32px
```

Fast mental math: `class number x 4 = pixels`. So `p-6` = 24px, `mt-10` = 40px.

In v4, **any number works** (`p-13`, `mt-17`), because spacing is computed from the base unit. In v3 only the predefined steps existed. The same scale is used by `w-`, `h-`, `gap-`, `inset-`, `translate-` and more.

Class value	rem	px	Class value	rem	px
0	0	0	12	3	48
px	1px	1	14	3.5	56
0.5	0.125	2	16	4	64
1	0.25	4	20	5	80
1.5	0.375	6	24	6	96
2	0.5	8	28	7	112
2.5	0.625	10	32	8	128
3	0.75	12	40	10	160
3.5	0.875	14	48	12	192
4	1	16	56	14	224
5	1.25	20	64	16	256
6	1.5	24	72	18	288
8	2	32	80	20	320
10	2.5	40	96	24	384

Everyday values you will use 90% of the time: 1, 2, 3, 4, 5, 6, 8, 10, 12, 16, 20, 24.

2.2 Margin

CSS Purpose	Tailwind	Example
margin (all sides)	<code>m-{n}</code>	<code>m-4</code>
margin horizontal (left + right)	<code>mx-{n}</code>	<code>mx-4</code>
margin vertical (top + bottom)	<code>my-{n}</code>	<code>my-4</code>
margin-top	<code>mt-{n}</code>	<code>mt-4</code>
margin-right	<code>mr-{n}</code>	<code>mr-4</code>
margin-bottom	<code>mb-{n}</code>	<code>mb-4</code>

CSS Purpose	Tailwind	Example
margin-left	<code>ml-{n}</code>	<code>ml-4</code>
margin-inline-start (RTL aware)	<code>ms-{n}</code>	<code>ms-4</code>
margin-inline-end (RTL aware)	<code>me-{n}</code>	<code>me-4</code>
auto margin	<code>m-auto</code> , <code>mx-auto</code> , <code>my-auto</code> , <code>ml-auto</code> , <code>mr-auto</code>	<code>mx-auto</code>
negative margin	<code>-m-{n}</code> , <code>-mt-{n}</code> ...	<code>-mt-4</code>
arbitrary margin	<code>m-[value]</code>	<code>mt-[37px]</code>

v4.2+: logical block-axis margin utilities `mbs-{n}` (margin-block-start) and `mbe-{n}` (margin-block-end) also exist (and `pbs-` / `pbe-` for padding). Most people never need them.

2.3 Padding

CSS Purpose	Tailwind	Example
padding (all sides)	<code>p-{n}</code>	<code>p-4</code>
horizontal padding	<code>px-{n}</code>	<code>px-4</code>
vertical padding	<code>py-{n}</code>	<code>py-4</code>
padding-top	<code>pt-{n}</code>	<code>pt-4</code>
padding-right	<code>pr-{n}</code>	<code>pr-4</code>
padding-bottom	<code>pb-{n}</code>	<code>pb-4</code>
padding-left	<code>pl-{n}</code>	<code>pl-4</code>
padding-inline-start	<code>ps-{n}</code>	<code>ps-4</code>
padding-inline-end	<code>pe-{n}</code>	<code>pe-4</code>
arbitrary padding	<code>p-[value]</code>	<code>px-[18px]</code>

Padding **cannot** be `auto` and **cannot** be negative.

2.4 Auto Margins (the centering and pushing tool)

```

<!-- Center a fixed-width block horizontally -->
<div class="mx-auto w-96">Centered</div>

<!-- Push an item to the far right inside a flex row -->
<nav class="flex">
  <a>Logo</a>
  <a class="ml-auto">Login</a> <!-- takes all free space on its left -->
</nav>

<!-- Center vertically + horizontally inside a flex parent -->
<div class="flex h-screen"><div class="m-auto">Centered</div></div>

```

2.5 Negative Spacing

Put a `-` in front of the utility: `-mt-4` , `-ml-2` , `-m-1` . Also works for `inset` , `translate` , `rotate` , `skew` , `scale` and others.

```
<img class="-mt-12 rounded-full" />      <!-- pull avatar upward over a banner -->
<div class="-mx-4 px-4">Full-bleed row</div> <!-- cancel parent's horizontal padding -->
```

2.6 Gap (the modern way to space children)

gap works on **flex** and **grid** parents and puts space **between** children (not outside).

CSS Purpose	Tailwind	Example
gap (row + column)	<code>gap-{n}</code>	<code>gap-4</code>
column-gap	<code>gap-x-{n}</code>	<code>gap-x-6</code>
row-gap	<code>gap-y-{n}</code>	<code>gap-y-2</code>

```
<div class="flex gap-4">...</div>
<div class="grid grid-cols-3 gap-x-6 gap-y-2">...</div>
```

2.7 space-x and space-y

Adds margin between **direct children** of any container (no flex or grid needed).

Class	Effect
<code>space-x-{n}</code>	horizontal space between children
<code>space-y-{n}</code>	vertical space between children
<code>space-x-reverse</code> , <code>space-y-reverse</code>	use with <code>flex-row-reverse</code> / <code>flex-col-reverse</code>
<code>-space-x-{n}</code>	negative space (overlapping avatars)

```
<ul class="space-y-2">      <!-- 8px between list items -->
  <li>One</li><li>Two</li><li>Three</li>
</ul>

<div class="flex -space-x-2"> <!-- overlapping avatar stack -->
  
  
</div>
```

Prefer gap-* with flex and grid: it behaves better with wrapping and with hidden children. Use **space-*** for simple stacks or when you cannot use flex/grid.

2.8 Practical Spacing Examples

```
<!-- Card with comfortable padding and spaced contents -->
<div class="p-6 space-y-3 rounded-xl border border-gray-200">
  <h3 class="text-lg font-semibold">Title</h3>
  <p class="text-gray-600">Description</p>
  <button class="mt-2 px-4 py-2 bg-blue-500 text-white rounded-lg">Action</button>
</div>

<!-- Page section: vertical rhythm -->
<section class="py-16 md:py-24 px-4">...</section>

<!-- Button spacing: px bigger than py looks balanced -->
<button class="px-6 py-3">...</button>
```

3. Width & Height

3.1 Width

CSS	Tailwind	Notes
<code>width: 1rem</code> etc.	<code>w-4</code> , <code>w-10</code> , <code>w-64</code>	spacing scale (n x 0.25rem)
<code>width: 1px</code>	<code>w-px</code>	
<code>width: 50%</code>	<code>w-1/2</code>	fractions: <code>1/2</code> , <code>1/3</code> , <code>2/3</code> , <code>1/4</code> , <code>3/4</code> , <code>1/5</code> ... <code>11/12</code>
<code>width: 100%</code>	<code>w-full</code>	fills parent
<code>width: 100vw</code>	<code>w-screen</code>	viewport width (can cause horizontal scroll)
<code>width: 100dvw</code> / <code>svw</code> / <code>lvw</code>	<code>w-dvw</code> , <code>w-svw</code> , <code>w-lvw</code>	dynamic / small / large viewport
<code>width: auto</code>	<code>w-auto</code>	
<code>width: fit-content</code>	<code>w-fit</code>	shrink to content
<code>width: min-content</code>	<code>w-min</code>	
<code>width: max-content</code>	<code>w-max</code>	
container sizes	<code>w-3xs</code> , <code>w-xs</code> , <code>w-sm</code> , <code>w-md</code> , <code>w-lg</code> , <code>w-xl</code> , <code>w-2xl</code> ... <code>w-7xl</code>	v4: named sizes usable on <code>w-</code> too
arbitrary	<code>w-[350px]</code> , <code>w-[50vw]</code> , <code>w-[calc(100%-2rem)]</code>	
CSS variable	<code>w(--sidebar-width)</code>	v4 syntax

3.2 Height

CSS	Tailwind	Notes
<code>height: n</code>	<code>h-4</code> , <code>h-10</code> , <code>h-64</code>	spacing scale
<code>height: 100%</code>	<code>h-full</code>	parent needs a defined height
<code>height: 100vh</code>	<code>h-screen</code>	viewport height
<code>100dvh</code> / <code>svh</code> / <code>lvh</code>	<code>h-dvh</code> , <code>h-svh</code> , <code>h-lvh</code>	mobile-friendly viewport units
<code>height: auto</code>	<code>h-auto</code>	
<code>fit-content</code> / <code>min-content</code> / <code>max-content</code>	<code>h-fit</code> , <code>h-min</code> , <code>h-max</code>	
fractions	<code>h-1/2</code> , <code>h-1/3</code>	needs parent with height
arbitrary	<code>h-[500px]</code> , <code>h-[calc(100vh-4rem)]</code>	

Mobile tip: `h-screen` (100vh) can be taller than the visible area on mobile browsers because of the address bar. Prefer `h-dvh` / `min-h-dvh` for full-screen mobile layouts.

3.3 Min / Max Width and Height

CSS	Tailwind
<code>min-width: 0</code>	<code>min-w-0</code> (important to let flex children shrink and truncate)

CSS	Tailwind
<code>min-width: 100%</code>	<code>min-w-full</code>
<code>min-width: min-content / max-content / fit-content</code>	<code>min-w-min</code> , <code>min-w-max</code> , <code>min-w-fit</code>
<code>min-width: n</code>	<code>min-w-32</code> , <code>min-w-[200px]</code>
<code>max-width: none</code>	<code>max-w-none</code>
<code>max-width: 100%</code>	<code>max-w-full</code>
<code>max-width: 65ch</code>	<code>max-w-prose</code> (readable text width)
<code>max-width: 100vw</code>	<code>max-w-screen</code>
<code>min-height: 100vh</code>	<code>min-h-screen</code>
<code>min-height: 100dvh</code>	<code>min-h-dvh</code>
<code>min-height: 100%</code>	<code>min-h-full</code>
<code>min-height: 0</code>	<code>min-h-0</code>
<code>max-height: n</code>	<code>max-h-96</code> , <code>max-h-[400px]</code>
<code>max-height: 100%</code>	<code>max-h-full</code>
<code>max-height: 100vh</code>	<code>max-h-screen</code>

Named max-width scale (`max-w-*`)

Class	Width	Class	Width
<code>max-w-3xs</code>	16rem (256px)	<code>max-w-3xl</code>	48rem (768px)
<code>max-w-2xs</code>	18rem (288px)	<code>max-w-4xl</code>	56rem (896px)
<code>max-w-xs</code>	20rem (320px)	<code>max-w-5xl</code>	64rem (1024px)
<code>max-w-sm</code>	24rem (384px)	<code>max-w-6xl</code>	72rem (1152px)
<code>max-w-md</code>	28rem (448px)	<code>max-w-7xl</code>	80rem (1280px)
<code>max-w-lg</code>	32rem (512px)	<code>max-w-prose</code>	65ch
<code>max-w-xl</code>	36rem (576px)	<code>max-w-full</code>	100%
<code>max-w-2xl</code>	42rem (672px)	<code>max-w-none</code>	none

3.4 size-* (width and height together)

`size-10` sets `width` and `height` to 2.5rem. Perfect for avatars, icons and squares.

```

<img class="size-10 rounded-full" />    <!-- 40x40 circle -->
<svg class="size-5" />                 <!-- 20x20 icon -->
<div class="size-full"></div>           <!-- 100% x 100% -->
<div class="size-[72px]"></div>         <!-- arbitrary -->

```

Also available: `size-px`, `size-1/2`, `size-full`, `size-auto`, `size-fit`, `size-min`, `size-max`. For the full viewport use `w-screen` `h-screen` (or `h-dvh`).

3.5 Practical Width and Height Examples

```

<!-- Centered readable page column -->
<main class="mx-auto w-full max-w-3xl px-4">...</main>

```

```

<!-- Full-screen hero (mobile friendly) -->
<section class="min-h-dvh flex items-center justify-center">...</section>

<!-- Sidebar fixed width + flexible content -->
<div class="flex">
  <aside class="w-64 shrink-0">Sidebar</aside>
  <main class="flex-1 min-w-0">Content</main>
</div>

<!-- Scrollable list limited in height -->
<ul class="max-h-64 overflow-y-auto">...</ul>

<!-- Page height minus a 64px header -->
<div class="h-[calc(100dvh-4rem)]">...</div>

<!-- Three equal columns using fractions -->
<div class="flex"><div class="w-1/3">A</div><div class="w-1/3">B</div><div class="w-1/3">C</div></div>

```

4. Typography

4.1 Font Family

Class	CSS	Use
<code>font-sans</code>	system sans-serif stack (UI default)	body text, UI
<code>font-serif</code>	system serif stack	editorial, headlines
<code>font-mono</code>	system monospace stack	code, numbers
<code>font-[Poppins]</code>	arbitrary family name	quick one-off
<code>font-(family-name:--my-font)</code>	from a CSS variable	v4 syntax
<code>font-display</code> (custom)	defined via <code>--font-display</code> in <code>@theme</code>	your own fonts

```
@theme { --font-display: "Poppins", sans-serif; }
```

4.2 Font Size (and its built-in line height)

Each `text-*` size also sets a matching `line-height`.

Class	Font size	Line height
<code>text-xs</code>	0.75rem (12px)	1rem (16px)
<code>text-sm</code>	0.875rem (14px)	1.25rem (20px)
<code>text-base</code>	1rem (16px)	1.5rem (24px)
<code>text-lg</code>	1.125rem (18px)	1.75rem (28px)
<code>text-xl</code>	1.25rem (20px)	1.75rem (28px)
<code>text-2xl</code>	1.5rem (24px)	2rem (32px)
<code>text-3xl</code>	1.875rem (30px)	2.25rem (36px)
<code>text-4xl</code>	2.25rem (36px)	2.5rem (40px)
<code>text-5xl</code>	3rem (48px)	1
<code>text-6xl</code>	3.75rem (60px)	1
<code>text-7xl</code>	4.5rem (72px)	1
<code>text-8xl</code>	6rem (96px)	1
<code>text-9xl</code>	8rem (128px)	1

- **Custom size:** `text-[22px]`, `text-[1.35rem]`, `text-(length:--my-size)`.
- **Size + line height together:** `text-sm/6` (14px font, line-height 1.5rem), `text-xl/8`, `text-base/[1.7]`.

Heads up: `text-` is used for both **font size** and **text color** (`text-xl` vs `text-blue-500`). Tailwind tells them apart by the value.

4.3 Font Weight

Class	Weight	Class	Weight
<code>font-thin</code>	100	<code>font-semibold</code>	600
<code>font-extralight</code>	200	<code>font-bold</code>	700
<code>font-light</code>	300	<code>font-extrabold</code>	800
<code>font-normal</code>	400	<code>font-black</code>	900
<code>font-medium</code>	500	<code>font-[550]</code>	arbitrary (variable fonts)

4.4 Font Style and Smoothing

Class	CSS
<code>italic</code>	<code>font-style: italic</code>
<code>not-italic</code>	<code>font-style: normal</code>
<code>antialiased</code>	smoother text on macOS/iOS
<code>subpixel-antialiased</code>	default rendering

4.5 Text Alignment

Class	CSS
<code>text-left</code>	<code>text-align: left</code>
<code>text-center</code>	<code>text-align: center</code>
<code>text-right</code>	<code>text-align: right</code>
<code>text-justify</code>	<code>text-align: justify</code>
<code>text-start</code>	<code>text-align: start</code> (left in LTR, right in RTL)
<code>text-end</code>	<code>text-align: end</code>

Vertical alignment (for inline/table-cell content): `align-baseline`, `align-top`, `align-middle`, `align-bottom`, `align-text-top`, `align-text-bottom`, `align-sub`, `align-super`.

4.6 Line Height (leading)

Class	line-height	Best for
<code>leading-none</code>	1	big headings, single-line badges
<code>leading-tight</code>	1.25	headings
<code>leading-snug</code>	1.375	subheadings, short text
<code>leading-normal</code>	1.5	body text
<code>leading-relaxed</code>	1.625	long-form reading
<code>leading-loose</code>	2	very airy text
<code>leading-6</code>	1.5rem	exact size from the spacing scale
<code>leading-[1.7]</code>	arbitrary	

4.7 Letter Spacing (tracking)

Class	letter-spacing
<code>tracking-tighter</code>	<code>-0.05em</code>
<code>tracking-tight</code>	<code>-0.025em</code>
<code>tracking-normal</code>	<code>0em</code>
<code>tracking-wide</code>	<code>0.025em</code>
<code>tracking-wider</code>	<code>0.05em</code>
<code>tracking-widest</code>	<code>0.1em</code>
<code>tracking-[0.2em]</code>	<code>arbitrary</code>

Practical: `uppercase tracking-wider text-xs font-semibold` is the classic "label" style; `tracking-tight` makes big headlines look polished.

4.8 Text Decoration

Class	Effect
<code>underline</code>	underline
<code>overline</code>	line above text
<code>line-through</code>	strikethrough
<code>no-underline</code>	removes decoration (links)
<code>decoration-solid</code> / <code>decoration-double</code> / <code>decoration-dotted</code> / <code>decoration-dashed</code> / <code>decoration-wavy</code>	line style
<code>decoration-blue-500</code>	line color
<code>decoration-1</code> , <code>decoration-2</code> , <code>decoration-4</code> , <code>decoration-auto</code> , <code>decoration-from-font</code>	line thickness
<code>underline-offset-1</code> , <code>-2</code> , <code>-4</code> , <code>-8</code> , <code>underline-offset-auto</code>	distance from text

```
<a class="underline decoration-wavy decoration-red-500 underline-offset-4">Error link</a>
<a class="no-underline hover:underline">Underline only on hover</a>
```

4.9 Text Transform

Class	CSS
<code>uppercase</code>	<code>text-transform: uppercase</code>
<code>lowercase</code>	<code>text-transform: lowercase</code>
<code>capitalize</code>	<code>text-transform: capitalize</code>
<code>normal-case</code>	<code>text-transform: none</code>

4.10 Text Overflow, Truncation and Line Clamp

Class	Effect
<code>truncate</code>	one line, hides overflow, adds <code>...</code> (needs a constrained width)
<code>text-ellipsis</code>	<code>text-overflow: ellipsis</code> (also needs <code>overflow-hidden</code> <code>whitespace-nowrap</code>)

Class	Effect
<code>text-clip</code>	cut off without ellipsis
<code>line-clamp-1 ... line-clamp-6</code>	show N lines, then ...
<code>line-clamp-[8]</code>	arbitrary line count
<code>line-clamp-none</code>	remove clamp (for responsive reveal)

```

<p class="truncate w-48">A very long single line that will be cut with an ellipsis</p>
<p class="line-clamp-3">Paragraph that shows only three lines and then adds an ellipsis...</p>

<!-- Truncate inside a flex child: min-w-0 is REQUIRED -->
<div class="flex"><span class="min-w-0 truncate">Very long name</span><button>Edit</button></div>

```

4.11 Whitespace

Class	CSS	Behavior
<code>whitespace-normal</code>	<code>normal</code>	collapse spaces, wrap lines
<code>whitespace-nowrap</code>	<code>nowrap</code>	collapse spaces, never wrap
<code>whitespace-pre</code>	<code>pre</code>	keep spaces and newlines, never wrap
<code>whitespace-pre-line</code>	<code>pre-line</code>	collapse spaces, keep newlines, wrap
<code>whitespace-pre-wrap</code>	<code>pre-wrap</code>	keep spaces and newlines, wrap
<code>whitespace-break-spaces</code>	<code>break-spaces</code>	like pre-wrap, trailing spaces wrap too

Use `whitespace-pre-line` to display multi-line text from a textarea. Use `whitespace-nowrap` for buttons and table cells that must stay on one line.

4.12 Word Breaking and Wrapping

Class	Effect
<code>break-normal</code>	default behavior
<code>break-words</code>	break long words only if they would overflow (<code>overflow-wrap: break-word</code>)
<code>break-all</code>	break anywhere (URLs, hashes), can look ugly
<code>break-keep</code>	no breaks in CJK text
<code>wrap-break-word</code> , <code>wrap-anywhere</code> , <code>wrap-normal</code>	<code>overflow-wrap</code> variants (v4.1+)
<code>hyphens-none</code> , <code>hyphens-manual</code> , <code>hyphens-auto</code>	hyphenation (set the <code>lang</code> attribute)
<code>text-wrap</code> , <code>text-nowrap</code>	<code>text-wrap: wrap / nowrap</code>
<code>text-balance</code>	balances lines (great for headings)
<code>text-pretty</code>	avoids a single orphan word on the last line (great for paragraphs)

4.13 More Typography Utilities

Purpose	Classes
Indent first line	<code>indent-4</code> , <code>indent-8</code> , <code>-indent-4</code> , <code>indent-[2em]</code>
Numbers in tables	<code>tabular-nums</code> , <code>proportional-nums</code> , <code>lining-nums</code> , <code>oldstyle-nums</code> , <code>slashed-zero</code> , <code>diagonal-fractions</code> , <code>ordinal</code> , <code>normal-nums</code>

Purpose	Classes
List style	see Section 16
Text color	<code>text-gray-700</code> (Section 5)
Text shadow (v4.1+)	<code>text-shadow-2xs</code> , <code>text-shadow-xs</code> , <code>text-shadow-sm</code> , <code>text-shadow-md</code> , <code>text-shadow-lg</code> , <code>text-shadow-none</code> , <code>text-shadow-red-500/50</code>
Text underline offset	<code>underline-offset-4</code>
Quotes and markers	<code>marker:text-blue-500</code> , <code>before:content-['']</code>
Prose styling for CMS/markdown	<code>prose</code> class from the @tailwindcss/typography plugin

4.14 Practical Typography Recipes

```

<!-- Page title -->
<h1 class="text-4xl md:text-6xl font-bold tracking-tight text-balance text-gray-900">
  Build faster with Tailwind
</h1>

<!-- Body paragraph -->
<p class="text-base leading-relaxed text-gray-600 max-w-prose text-pretty">...</p>

<!-- Small uppercase eyebrow label -->
<span class="text-xs font-semibold uppercase tracking-widest text-blue-600">New</span>

<!-- Muted caption -->
<small class="text-sm text-gray-500">Updated 2 days ago</small>

<!-- Price with aligned digits -->
<span class="text-2xl font-bold tabular-nums">$1,249.00</span>

<!-- Link -->
<a class="font-medium text-blue-600 underline underline-offset-4 hover:text-blue-800">Read more</a>

<!-- Monospace code chip -->
<code class="rounded bg-gray-100 px-1.5 py-0.5 font-mono text-sm text-pink-600">npm install</code>

```

5. Colors

5.1 The Color Scale

Tailwind ships with a large, hand-tuned palette. Each hue has **11 shades** from **50** (almost white) to **950** (almost black).

50 100 200 300 400 500 600 700 800 900 950
lightest <----- base (500) -----> darkest

Shade	Typical use
50 , 100	subtle backgrounds, hover tints, alert backgrounds
200 , 300	borders, dividers, disabled states
400	placeholder text, icons, dark-mode text on dark
500	the "main" brand shade, buttons, focus rings
600 , 700	hover/active button, links, body text on light
800 , 900	headings, dark UI surfaces
950	near-black surfaces, dark mode backgrounds

Color families: **red** , **orange** , **amber** , **yellow** , **lime** , **green** , **emerald** , **teal** , **cyan** , **sky** , **blue** , **indigo** , **violet** , **purple** , **fuchsia** , **pink** , **rose** (vibrant) and **slate** , **gray** , **zinc** , **neutral** , **stone** (greys with different tints). v4.2 also added **mauve** , **olive** , **mist** , **taupe** (muted neutrals).

Special values: **black** , **white** , **transparent** , **current** (= **currentColor**) , **inherit** .

In v4 the default palette is defined in **OKLCH**, which gives more vivid colors on modern displays. You still use the same class names.

5.2 Color Utilities (every place a color can go)

The same pattern **{utility}-{color}-{shade}** works everywhere:

What it colors	Prefix	Example
Text	text-	text-blue-500
Background	bg-	bg-blue-500
Border (all sides)	border-	border-blue-500
Border (one side/axis)	border-t- , border-x- ...	border-t-blue-500
Ring (focus ring)	ring-	ring-blue-500
Ring offset gap color	ring-offset-	ring-offset-white
Inset ring	inset-ring-	inset-ring-blue-500
Divide (between children)	divide-	divide-gray-200
Outline	outline-	outline-blue-500
Box shadow	shadow-	shadow-blue-500/50
Inset shadow	inset-shadow-	inset-shadow-blue-500
Drop shadow filter	drop-shadow-	drop-shadow-blue-500

What it colors	Prefix	Example
Text decoration	<code>decoration-</code>	<code>decoration-blue-500</code>
Placeholder text	<code>placeholder:text-</code>	<code>placeholder:text-gray-400</code>
Form accent (checkbox, radio, range)	<code>accent-</code>	<code>accent-blue-500</code>
Text caret (cursor)	<code>caret-</code>	<code>caret-blue-500</code>
SVG fill	<code>fill-</code>	<code>fill-blue-500</code>
SVG stroke	<code>stroke-</code>	<code>stroke-blue-500</code>
Gradient stops	<code>from-</code> , <code>via-</code> , <code>to-</code>	<code>from-blue-500</code>
Selection highlight	<code>selection:bg-</code>	<code>selection:bg-yellow-200</code>

Tailwind v3 note: v3 had `placeholder-gray-400`. In v4 use the `placeholder:` variant: `placeholder:text-gray-400`.

5.3 Color Opacity (the slash syntax)

Add `/` and a number from 0 to 100 to any color utility:

```
<div class="bg-black/50"></div>           <!-- 50% black -->
<p class="text-white/80"></p>             <!-- 80% white text -->
<div class="border-blue-500/30"></div>
<div class="bg-blue-500/[0.35]"></div>    <!-- arbitrary opacity -->
<div class="bg-blue-500/(--my-alpha)"></div> <!-- from CSS variable -->
```

Tailwind v3 note: `bg-opacity-50`, `text-opacity-50`, `border-opacity-*`, `divide-opacity-*`, `ring-opacity-*` and `placeholder-opacity-*` were **removed in v4**. Use the slash syntax instead.

Opacity of the whole element (not just the color) is a different utility: `opacity-50` (Section 8).

5.4 Special Color Keywords

Class	CSS value	Use
<code>bg-transparent</code>	<code>transparent</code>	clear background
<code>text-current</code> / <code>border-current</code>	<code>currentColor</code>	inherit the text color (icons)
<code>bg-inherit</code> , <code>text-inherit</code>	<code>inherit</code>	take parent's value
<code>text-black</code> , <code>text-white</code> , <code>bg-black</code> , <code>bg-white</code>	<code>fixed</code>	

```
<button class="text-blue-600 hover:text-blue-800">
  <svg class="size-5 fill-current">...</svg> Save <!-- icon follows text color -->
</button>
```

5.5 Arbitrary Colors

Format	Example
Hex	<code>bg-[#1e293b]</code> , <code>text-[#f43f5e]</code>
Hex + alpha	<code>bg-[#1e293b80]</code>

Format	Example
RGB	<code>text-[rgb(20,30,40)]</code> , <code>bg-[rgb(20_30_40/0.5)]</code>
HSL	<code>bg-[hsl(210,40%,96%)]</code>
OKLCH	<code>bg-[oklch(0.7_0.15_200)]</code>
CSS variable	<code>bg(--brand)</code> (v4) or <code>bg-[var(--brand)]</code>
With type hint	<code>text-(color:--my-color)</code> , <code>text-[color:var(--c)]</code>

Use `_` for spaces inside the brackets. When the value is ambiguous (for example `text-[var(--x)]` could be a size or a color) add a type hint: `text-[color:var(--x)]` or `text-[length:var(--x)]`.

5.6 Defining Your Own Colors (v4)

```
@import "tailwindcss";
@theme {
  --color-brand-50:  oklch(0.97 0.02 260);
  --color-brand-500: oklch(0.62 0.2 260);
  --color-brand-900: oklch(0.3 0.1 260);
}
```

This instantly creates `bg-brand-500`, `text-brand-900`, `border-brand-50`, `ring-brand-500/40` and so on.

5.7 Color Mixing Examples

```
<div class="bg-blue-500/10 text-blue-700 border border-blue-500/30 rounded-lg p-4">Info alert</div>
<div class="bg-white/60 backdrop-blur-md border border-white/40">Glass panel</div>
<button class="bg-blue-600 hover:bg-blue-700 active:bg-blue-800 text-white">Button</button>
<input class="border-gray-300 focus:border-blue-500 focus:ring-blue-500/30 caret-blue-500 accent-blue-500" />
```

6. Backgrounds

6.1 Background Color

`bg-{color}-{shade}`, `bg-white`, `bg-black`, `bg-transparent`, `bg-current`, `bg-inherit`, `bg-black/50`, `bg-[#hex]`.

6.2 Background Image

Class	CSS
<code>bg-none</code>	<code>background-image: none</code>
<code>bg-[url(/img/hero.jpg)]</code>	image from a URL (arbitrary value)
<code>bg-[url('/img/my_hero.jpg')]</code>	quote paths with special characters
<code>bg-(image:--hero)</code>	from a CSS variable

```
<section class="bg-[url(/hero.jpg)] bg-cover bg-center bg-no-repeat min-h-96"></section>
```

In React with a bundler, import the image and use inline `style={{ backgroundImage: url(${hero}) }}` for dynamic images (Tailwind cannot see dynamic URLs).

6.3 Background Size

Class	CSS
<code>bg-auto</code>	original size
<code>bg-cover</code>	fill the area, crop if needed (most common)
<code>bg-contain</code>	fit inside, no crop
<code>bg-size-[200px_100px]</code>	arbitrary size

6.4 Background Position

`bg-center`, `bg-top`, `bg-bottom`, `bg-left`, `bg-right`, `bg-left-top`, `bg-left-bottom`, `bg-right-top`, `bg-right-bottom`, `bg-position-[center_top_1rem]` (arbitrary).

6.5 Background Repeat

`bg-repeat`, `bg-no-repeat`, `bg-repeat-x`, `bg-repeat-y`, `bg-repeat-round`, `bg-repeat-space`.

6.6 Background Attachment

Class	Effect
<code>bg-fixed</code>	stays fixed while page scrolls (parallax look)
<code>bg-local</code>	scrolls with the element's content
<code>bg-scroll</code>	scrolls with the page (default)

6.7 Background Clip and Origin

Class	CSS
<code>bg-clip-border</code>	background extends under the border (default)
<code>bg-clip-padding</code>	background stops at the border
<code>bg-clip-content</code>	background only behind content
<code>bg-clip-text</code>	background shows only inside the text shapes (gradient text)
<code>bg-origin-border</code>	positioning area starts at the border box
<code>bg-origin-padding</code>	starts at the padding box (default)
<code>bg-origin-content</code>	starts at the content box

6.8 Gradients

Linear gradients (v4: `bg-linear-*`)

Class	Direction
<code>bg-linear-to-t</code>	to top
<code>bg-linear-to-tr</code>	to top right
<code>bg-linear-to-r</code>	to right
<code>bg-linear-to-br</code>	to bottom right
<code>bg-linear-to-b</code>	to bottom
<code>bg-linear-to-bl</code>	to bottom left
<code>bg-linear-to-l</code>	to left
<code>bg-linear-to-tl</code>	to top left
<code>bg-linear-45</code>	exact angle (any number; negative with <code>-bg-linear-45</code>)
<code>bg-linear-[25deg,red_5%,yellow_60%,lime_90%]</code>	fully custom

Other gradient types (v4)

Class	Gradient
<code>bg-radial</code>	radial (circle from center)
<code>bg-radial-[at_25%_25%]</code>	radial with a custom position
<code>bg-conic</code>	conic (color wheel)
<code>bg-conic-180</code>	conic starting at an angle

Color stops: `from-{color}`, `via-{color}`, `to-{color}`, and optional positions: `from-10%`, `via-30%`, `to-90%`.

Color space (v4): add an interpolation modifier for smoother or different blends: `bg-linear-to-r/oklch`, `bg-linear-to-r/srgb`, `bg-linear-to-r/hsl`, `bg-linear-to-r/longer`.

```
<!-- Simple two-color gradient -->
<div class="bg-linear-to-r from-indigo-500 to-purple-600 p-8 text-white rounded-xl"></div>

<!-- Three stops -->
<div class="bg-linear-to-br from-pink-500 via-red-500 to-yellow-500"></div>

<!-- Fade to transparent over an image (card overlay) -->
<div class="absolute inset-0 bg-linear-to-t from-black/70 to-transparent"></div>
```

```

<!-- Gradient text -->
<h1 class="bg-linear-to-r from-blue-500 to-fuchsia-500 bg-clip-text text-transparent text-5xl font-extrabold">
  Gradient Heading
</h1>

<!-- Radial spotlight -->
<div class="bg-radial from-white to-slate-200"></div>

```

Tailwind v3 note: v3 used `bg-gradient-to-r`. In v4 the preferred name is `bg-linear-to-r`; the old `bg-gradient-to-*` names still work as legacy aliases, but use the new names in new code.

6.9 Practical Background Recipes

```

<!-- Hero with image + dark overlay -->
<section class="relative isolate min-h-dvh bg-[url(/hero.jpg)] bg-cover bg-center">
  <div class="absolute inset-0 -z-10 bg-black/50"></div>
  <div class="mx-auto max-w-3xl px-4 py-32 text-white">...</div>
</section>

<!-- Dotted pattern with arbitrary background -->
<div class="bg-[radial-gradient(#cbd5e1_1px,transparent_1px)] bg-size-[16px_16px]"></div>

<!-- Subtle page background -->
<body class="bg-gray-50 text-gray-900 dark:bg-gray-950 dark:text-gray-100">

```

7. Borders & Radius

7.1 Border Width

Class	CSS
<code>border</code>	<code>border-width: 1px</code> (all sides)
<code>border-0</code>	0
<code>border-2</code>	2px
<code>border-4</code>	4px
<code>border-8</code>	8px
<code>border-[3px]</code>	arbitrary
<code>border-x</code> , <code>border-y</code>	left+right / top+bottom (1px); also <code>border-x-2</code> , <code>border-y-4</code>
<code>border-t</code> , <code>border-r</code> , <code>border-b</code> , <code>border-l</code>	one side (1px); also <code>border-t-2</code> , <code>border-b-4</code> ...
<code>border-s</code> , <code>border-e</code>	inline start / end (RTL aware)

Important (v4): the default border color is now `currentColor`, not light gray. **Always pair `border` with a color:** `border border-gray-200`.

7.2 Border Color and Opacity

`border-{color}-{shade}`, `border-transparent`, `border-current`, side-specific `border-t-red-500`, `border-x-gray-200`, opacity with `border-blue-500/30`.

Tailwind v3 note: `border-opacity-*` is removed in v4. Use `border-black/20`.

7.3 Border Style

`border-solid`, `border-dashed`, `border-dotted`, `border-double`, `border-hidden`, `border-none`.

```
<div class="border-2 border-dashed border-gray-300 rounded-lg p-8 text-center">Drop files here</div>
```

7.4 Divide (borders between children)

Class	Effect
<code>divide-x</code> / <code>divide-x-2</code> / <code>divide-x-4</code>	vertical lines between horizontally placed children
<code>divide-y</code> / <code>divide-y-2</code>	horizontal lines between stacked children
<code>divide-gray-200</code>	color of the dividers
<code>divide-dashed</code> , <code>divide-dotted</code> , <code>divide-solid</code> , <code>divide-double</code> , <code>divide-none</code>	style
<code>divide-x-reverse</code> , <code>divide-y-reverse</code>	for reversed flex directions

```
<ul class="divide-y divide-gray-200">
  <li class="py-3">Item one</li>
```

```
<li class="py-3">Item two</li>
<li class="py-3">Item three</li>
</ul>
```

7.5 Outline and Ring

Class	Effect
<code>outline</code> / <code>outline-1</code> / <code>outline-2</code> / <code>outline-4</code> / <code>outline-8</code>	outline width
<code>outline-blue-500</code>	color
<code>outline-offset-2</code> / <code>-4</code> / <code>-8</code> / <code>-0</code>	gap between element and outline
<code>outline-dashed</code> , <code>outline-dotted</code> , <code>outline-double</code> , <code>outline-solid</code>	style
<code>outline-hidden</code>	removes the default outline but keeps it visible in forced-colors mode (accessible reset)
<code>outline-none</code>	<code>outline-style: none</code> (v4: really removes it, avoid on focusable elements)
<code>ring</code>	1px ring (box-shadow based, follows border-radius)
<code>ring-0</code> , <code>ring-1</code> , <code>ring-2</code> , <code>ring-3</code> , <code>ring-4</code> , <code>ring-8</code>	ring width
<code>ring-blue-500</code> , <code>ring-blue-500/50</code>	ring color
<code>ring-offset-2</code> , <code>ring-offset-white</code>	gap + gap color between element and ring
<code>inset-ring</code> , <code>inset-ring-2</code> , <code>inset-ring-blue-500</code>	ring drawn inside the element

```
<button class="outline-hidden focus-visible:ring-2 focus-visible:ring-blue-500 focus-visible:ring-offset-2">Accessible</button>
```

Tailwind v3 note: v3 `ring` was 3px blue. In v4 `ring` is 1px and `currentColor`; write `ring-3 ring-blue-500` to get the old look. v3 `outline-none` became `outline-hidden`.

7.6 Border Radius

Class	Radius	Pixels
<code>rounded-none</code>	0	0
<code>rounded-xs</code>	0.125rem	2px
<code>rounded-sm</code>	0.25rem	4px
<code>rounded-md</code>	0.375rem	6px
<code>rounded-lg</code>	0.5rem	8px
<code>rounded-xl</code>	0.75rem	12px
<code>rounded-2xl</code>	1rem	16px
<code>rounded-3xl</code>	1.5rem	24px
<code>rounded-4xl</code>	2rem	32px
<code>rounded-full</code>	9999px	pill or circle
<code>rounded-[10px]</code>	arbitrary	

Tailwind v3 note: v3 had bare `rounded` (4px) and `rounded-sm` (2px). In v4 those are `rounded-sm` (4px) and `rounded-xs` (2px). Use the sized names.

7.7 Individual Corners and Sides

Target	Class prefix	Example
top (both top corners)	<code>rounded-t-*</code>	<code>rounded-t-lg</code>
right	<code>rounded-r-*</code>	<code>rounded-r-lg</code>
bottom	<code>rounded-b-*</code>	<code>rounded-b-lg</code>
left	<code>rounded-l-*</code>	<code>rounded-l-lg</code>
top-left	<code>rounded-tl-*</code>	<code>rounded-tl-2xl</code>
top-right	<code>rounded-tr-*</code>	<code>rounded-tr-2xl</code>
bottom-right	<code>rounded-br-*</code>	<code>rounded-br-2xl</code>
bottom-left	<code>rounded-bl-*</code>	<code>rounded-bl-2xl</code>
logical start / end	<code>rounded-s-*</code> , <code>rounded-e-*</code>	<code>rounded-s-lg</code>
logical corners	<code>rounded-ss-*</code> , <code>rounded-se-*</code> , <code>rounded-es-*</code> , <code>rounded-ee-*</code>	<code>rounded-ss-xl</code>

```

<img class="size-24 rounded-full object-cover" />           <!-- circle avatar -->
<button class="rounded-full px-6 py-2">Pill button</button>
<div class="rounded-t-xl border border-b-0">Tab header</div>
<div class="rounded-tl-3xl rounded-br-3xl bg-blue-500 p-6">Leaf shape</div>

```

8. Shadows & Effects

8.1 Box Shadow

Class	Look	Use
<code>shadow-2xs</code>	barely there	hairline lift
<code>shadow-xs</code>	very subtle	inputs
<code>shadow-sm</code>	small	cards, buttons
<code>shadow-md</code>	medium	raised cards
<code>shadow-lg</code>	large	dropdowns, hover state
<code>shadow-xl</code>	extra large	modals
<code>shadow-2xl</code>	huge	hero images, floating panels
<code>shadow-none</code>	removes shadow	reset (for example on hover or breakpoint)
<code>shadow-[0_4px_20px_rgba(0,0,0,0.15)]</code>	custom	exact design match
<code>shadow-blue-500/50</code>	colored shadow	glow effects

Inset shadows and rings (v4): `inset-shadow-2xs`, `inset-shadow-xs`, `inset-shadow-sm`, `inset-shadow-none`, `inset-shadow-black/20`, `inset-ring-1`.

```
<div class="rounded-xl bg-white shadow-md hover:shadow-xl transition-shadow">Card</div>
<button class="bg-blue-500 text-white shadow-lg shadow-blue-500/40">Glowing button</button>
<div class="inset-shadow-sm bg-gray-100 rounded-lg p-4">Pressed look</div>
```

Tailwind v3 note: the scale shifted down one step: v3 `shadow-sm` is v4 `shadow-xs`; v3 `shadow` is v4 `shadow-sm`. v3 `shadow-inner` is replaced by `inset-shadow-*`. v3 `drop-shadow` follows the same rename (see [Section 22](#)).

Custom shadow tokens

```
@theme { --shadow-soft: 0 8px 30px rgb(0 0 0 / 0.12); } /* creates shadow-soft */
```

8.2 Opacity

`opacity-0`, `opacity-5`, `opacity-10`, `opacity-15`, `opacity-20`, `opacity-25`, `opacity-30` ... `opacity-95`, `opacity-100`, `opacity-[0.37]`.

`opacity-*` fades the **whole element including its children**. To fade only the background, use `bg-black/50`.

8.3 Blend Modes

Utility	Classes
<code>mix-blend-*</code> (element blends with what is behind)	<code>mix-blend-normal</code> , <code>multiply</code> , <code>screen</code> , <code>overlay</code> , <code>darken</code> , <code>lighten</code> , <code>color-dodge</code> , <code>color-burn</code> , <code>hard-light</code> , <code>soft-light</code> , <code>difference</code> , <code>exclusion</code> , <code>hue</code> , <code>saturation</code> , <code>color</code> , <code>luminosity</code> , <code>plus-darker</code> , <code>plus-lighter</code>
<code>bg-blend-*</code> (background layers blend)	<code>bg-blend-normal</code> , <code>multiply</code> , <code>screen</code> , <code>overlay</code> , <code>darken</code> , <code>lighten</code> , <code>color-dodge</code> , <code>color-burn</code> , <code>hard-light</code> , <code>soft-light</code> , <code>difference</code> , <code>exclusion</code> , <code>hue</code> , <code>saturation</code> , <code>color</code> , <code>luminosity</code>

```
<div class="bg-blue-500 bg-[url(/photo.jpg)] bg-blend-multiply bg-cover">Tinted photo</div>
<img class="mix-blend-multiply" /> <!-- logo that blends into a colored background -->
```

8.4 Box Decoration Break

`box-decoration-clone` / `box-decoration-slice`: controls how backgrounds, borders and padding render when an inline element wraps across lines (highlighted text).

```
<span class="box-decoration-clone bg-yellow-200 px-2 leading-loose">A highlighted phrase that wraps over
multiple lines</span>
```

Tailwind v3 note: v3 names were `decoration-clone` / `decoration-slice`.

8.5 Isolation (stacking contexts)

Class	Effect
<code>isolate</code>	creates a new stacking context (z-index children stay inside)
<code>isolation-auto</code>	default

Use `isolate` on a card so a child with `-z-10` (overlays) does not slip behind the page background.

8.6 Masks (v4.1+)

Mask utilities fade or cut parts of an element using a gradient or image as an alpha mask. Common ones: `mask-t-from-50%` (fade the top edge), `mask-b-from-50%` (fade the bottom edge), `mask-radial-from-50%` (radial fade), `mask-[url(/shape.svg)]` (custom shape). Example: fade the bottom of a scroll area with `mask-b-from-80%`. See the Tailwind docs (Masks) for the full list.

9. Flexbox

9.1 The Mental Model: Main Axis and Cross Axis

Flexbox places children along a **main axis**. The **cross axis** is perpendicular.

- `justify-*` aligns along the **main axis**
- `items-*` aligns along the **cross axis**

```
flex-row (default)
main axis ----->

+-----+
| [A] [B] [C] | ^
+-----+
|             | |
+-----+
cross axis (vertical) <-----+

justify-* = left/right placement
items-*   = top/bottom placement

flex-col
main axis (vertical)
|
| +-----+
| | [A]    |
| | [B]    |
| | [C]    |
+-----+
cross axis ----->

justify-* = top/bottom placement
items-*   = left/right placement
```

Remember: when you switch to `flex-col`, `justify` and `items` swap their visual direction.

9.2 Flex Container

Class	CSS
<code>flex</code>	<code>display: flex</code> (block-level)
<code>inline-flex</code>	<code>display: inline-flex</code>
<code>flex-row</code>	<code>flex-direction: row</code> (default)
<code>flex-row-reverse</code>	reverse order horizontally
<code>flex-col</code>	<code>flex-direction: column</code>
<code>flex-col-reverse</code>	reverse order vertically
<code>flex-wrap</code>	allow wrapping to new lines
<code>flex-nowrap</code>	never wrap (default)
<code>flex-wrap-reverse</code>	wrap upwards
<code>gap-4</code> , <code>gap-x-4</code> , <code>gap-y-2</code>	space between items

9.3 Justify Content (main axis)

Class	Result
<code>justify-start</code>	items at the start
<code>justify-center</code>	items in the center
<code>justify-end</code>	items at the end
<code>justify-between</code>	first at start, last at end, equal space between
<code>justify-around</code>	equal space around each item (half-space at edges)
<code>justify-evenly</code>	equal space everywhere including edges

Class	Result
<code>justify-stretch</code>	stretch auto-sized items to fill
<code>justify-normal</code>	default

```

justify-start  [A][B][C].....
justify-center .....[A][B][C].....
justify-end    .....[A][B][C]
justify-between [A].....[B].....[C]
justify-around  ..[A].....[B].....[C]..
justify-evenly  ...[A].....[B].....[C]...

```

9.4 Align Items (cross axis)

Class	Result
<code>items-start</code>	align to the top (in a row)
<code>items-center</code>	center vertically (in a row)
<code>items-end</code>	align to the bottom
<code>items-baseline</code>	align text baselines (mixed font sizes)
<code>items-stretch</code>	stretch to container height (default)

```

items-start  items-center  items-end  items-stretch
+-----+
|[A][B][C]| |         | |         | |[A][B][C]| | |
|         | |         | |         | |[ ]|[ ]|[ ]|
|         | |         | |         | |[ ]|[ ]|[ ]|
+-----+   +-----+   +-----+   +-----+

```

9.5 Align Content (multi-line flex only)

`content-start`, `content-center`, `content-end`, `content-between`, `content-around`, `content-evenly`, `content-stretch`, `content-normal`. These only matter when `flex-wrap` produces **multiple rows** and the container is taller than the rows.

9.6 Align Self (one item overrides `items-*`)

`self-auto`, `self-start`, `self-center`, `self-end`, `self-stretch`, `self-baseline`.

Also: `justify-self-*` and `justify-items-*` are for **grid**, see [Section 10](#).

9.7 Flex Item Sizing: flex, grow, shrink, basis

Class	CSS	Meaning
<code>flex-1</code>	<code>flex: 1 1 0%</code>	grow and shrink, start from 0 (equal-width columns)
<code>flex-auto</code>	<code>flex: 1 1 auto</code>	grow and shrink from content size
<code>flex-initial</code>	<code>flex: 0 1 auto</code>	shrink only, never grow
<code>flex-none</code>	<code>flex: none</code>	fixed, does not grow or shrink
<code>flex-[2]</code>	<code>flex: 2</code>	grow factor 2 (arbitrary)
<code>grow / grow-0</code>	<code>flex-grow: 1 / 0</code>	take up leftover space or not

Class	CSS	Meaning
<code>grow-[3]</code>	arbitrary	
<code>shrink / shrink-0</code>	<code>flex-shrink: 1 / 0</code>	allowed to shrink or not
<code>basis-1/2</code> , <code>basis-1/3</code> , <code>basis-full</code> , <code>basis-auto</code>	<code>flex-basis</code>	initial size before grow/shrink
<code>basis-64</code> , <code>basis-[200px]</code>		
<code>order-1 ... order-12</code> , <code>order-first</code> , <code>order-last</code> , <code>order-none</code>	<code>order</code>	change visual order

Tailwind v3 note: v3 also had `flex-grow` and `flex-shrink` (and `flex-grow-0`, `flex-shrink-0`). In v4 write `grow`, `shrink`, `grow-0`, `shrink-0`.

9.8 Practical Flexbox Recipes

```

<!-- Center anything (horizontally + vertically) -->
<div class="flex items-center justify-center h-64">Centered</div>

<!-- Navbar: logo left, links right -->
<nav class="flex items-center justify-between px-6 py-4">
  <a class="font-bold">Logo</a>
  <ul class="flex items-center gap-6"><li>Home</li><li>About</li></ul>
</nav>

<!-- Vertical stack with gap -->
<div class="flex flex-col gap-4">...</div>

<!-- Sidebar + content -->
<div class="flex min-h-screen">
  <aside class="w-64 shrink-0 bg-gray-900 text-white">Sidebar</aside>
  <main class="flex-1 min-w-0 p-6">Content</main>
</div>

<!-- Sticky footer: footer sits at the bottom even with little content -->
<div class="flex min-h-screen flex-col">
  <header>...</header>
  <main class="flex-1">...</main>
  <footer>...</footer>
</div>

<!-- Wrapping tag list -->
<div class="flex flex-wrap gap-2"><span class="rounded-full bg-gray-100 px-3 py-1 text-sm">Tag</span></div>

<!-- Row on desktop, column on mobile -->
<div class="flex flex-col md:flex-row gap-6">...</div>

<!-- Push last item right -->
<div class="flex items-center gap-3"><h2>Title</h2><button class="ml-auto">Action</button></div>

<!-- Icon + text button -->
<button class="inline-flex items-center gap-2 rounded-lg bg-blue-500 px-4 py-2 text-white">
  <svg class="size-4">...</svg> Save
</button>

```

10. CSS Grid

10.1 Grid Container

Class	CSS
<code>grid</code>	<code>display: grid</code>
<code>inline-grid</code>	<code>display: inline-grid</code>
<code>grid-cols-1 ... grid-cols-12</code>	<code>grid-template-columns: repeat(n, minmax(0, 1fr))</code>
<code>grid-cols-none</code>	none
<code>grid-cols-subgrid</code>	inherit parent's column tracks
<code>grid-cols-[200px_1fr]</code>	custom track list (arbitrary)
<code>grid-cols-[repeat(auto-fit,minmax(250px,1fr))]</code>	auto-responsive columns
<code>grid-rows-1 ... grid-rows-12</code> , <code>grid-rows-none</code> , <code>grid-rows-subgrid</code> , <code>grid-rows-[auto_1fr_auto]</code>	rows
<code>gap-4</code> , <code>gap-x-4</code> , <code>gap-y-4</code>	spacing between cells

In v4 any number is accepted (`grid-cols-13`).

10.2 Grid Items: Spanning and Placement

Class	CSS
<code>col-span-1 ... col-span-12</code>	span N columns
<code>col-span-full</code>	<code>grid-column: 1 / -1</code>
<code>col-auto</code>	auto placement
<code>col-start-1 ... col-start-13</code> , <code>col-start-auto</code>	start line
<code>col-end-1 ... col-end-13</code> , <code>col-end-auto</code>	end line
<code>col-[2/span_3]</code>	arbitrary
<code>row-span-1 ... row-span-12</code> , <code>row-span-full</code>	span rows
<code>row-start-*</code> , <code>row-end-*</code> , <code>row-auto</code>	row lines

```
<div class="grid grid-cols-4 gap-4">
  <div class="col-span-2">Wide</div>
  <div>Normal</div>
  <div class="row-span-2">Tall</div>
  <div class="col-start-2 col-end-4">Placed between lines 2 and 4</div>
</div>
```

10.3 Auto Flow and Auto Tracks

Class	Effect
<code>grid-flow-row</code>	fill row by row (default)
<code>grid-flow-col</code>	fill column by column

Class	Effect
<code>grid-flow-dense</code>	backfill holes
<code>grid-flow-row-dense</code> , <code>grid-flow-col-dense</code>	combined
<code>auto-cols-auto</code> , <code>auto-cols-min</code> , <code>auto-cols-max</code> , <code>auto-cols-fr</code>	size of implicit columns
<code>auto-rows-auto</code> , <code>auto-rows-min</code> , <code>auto-rows-max</code> , <code>auto-rows-fr</code>	size of implicit rows
<code>auto-rows-[minmax(100px,auto)]</code>	arbitrary

10.4 Alignment in Grid

Class	Applies to	CSS
<code>justify-items-start</code> / <code>center</code> / <code>end</code> / <code>stretch</code>	all items (inline axis)	<code>justify-items</code>
<code>justify-self-auto</code> / <code>start</code> / <code>center</code> / <code>end</code> / <code>stretch</code>	one item	<code>justify-self</code>
<code>items-start</code> / <code>center</code> / <code>end</code> / <code>stretch</code> / <code>baseline</code>	all items (block axis)	<code>align-items</code>
<code>self-start</code> / <code>center</code> / <code>end</code> / <code>stretch</code>	one item	<code>align-self</code>
<code>justify-start</code> / <code>center</code> / <code>end</code> / <code>between</code> / <code>around</code> / <code>evenly</code>	the whole grid	<code>justify-content</code>
<code>content-start</code> / <code>center</code> / <code>end</code> / <code>between</code> / <code>around</code> / <code>evenly</code>	the whole grid	<code>align-content</code>
<code>place-items-start</code> / <code>center</code> / <code>end</code> / <code>stretch</code> / <code>baseline</code>	both axes at once	<code>place-items</code>
<code>place-content-start</code> / <code>center</code> / <code>end</code> / <code>between</code> / <code>around</code> / <code>evenly</code> / <code>stretch</code>	both axes, whole grid	<code>place-content</code>
<code>place-self-auto</code> / <code>start</code> / <code>center</code> / <code>end</code> / <code>stretch</code>	one item, both axes	<code>place-self</code>

`place-items-center` is the shortest way to center content in a grid cell: `grid place-items-center min-h-screen`.

10.5 Practical Grid Layouts

```

<!-- Responsive card grid: 1 / 2 / 3 / 4 columns -->
<div class="grid grid-cols-1 gap-6 sm:grid-cols-2 lg:grid-cols-3 xl:grid-cols-4">
  <div class="rounded-xl border border-gray-200 bg-white p-6 shadow-sm">Card</div>
  <!-- repeat -->
</div>

<!-- Auto-fit cards: as many 250px+ columns as fit, no breakpoints needed -->
<div class="grid grid-cols-[repeat(auto-fit,minmax(250px,1fr))] gap-6">...</div>

<!-- Sidebar layout: fixed 240px + flexible content -->
<div class="grid grid-cols-[240px_1fr] min-h-screen">
  <aside>Sidebar</aside><main>Content</main>
</div>

<!-- Holy grail layout -->
<div class="grid min-h-screen grid-rows-[auto_1fr_auto] grid-cols-[200px_1fr]">
  <header class="col-span-2">Header</header>
  <nav>Nav</nav>
  <main>Main</main>
  <footer class="col-span-2">Footer</footer>
</div>

<!-- Dashboard: big featured card + small cards -->
<div class="grid grid-cols-3 grid-rows-2 gap-4">
  <div class="col-span-2 row-span-2">Chart</div>
  <div>Stat 1</div>
  <div>Stat 2</div>
</div>

```

```
<!-- Perfectly centered page -->  
<div class="grid min-h-screen place-items-center">Centered</div>
```

Flex vs Grid: use **flex** for one-dimensional rows or columns (navbars, button groups). Use **grid** when you control rows **and** columns (page layouts, card galleries).

11. Positioning

11.1 Position Types

Class	CSS	Behavior
<code>static</code>	<code>position: static</code>	normal flow (default), <code>top/left/z-index</code> ignored
<code>relative</code>	<code>position: relative</code>	normal flow, but can be nudged and becomes the anchor for <code>absolute</code> children
<code>absolute</code>	<code>position: absolute</code>	removed from flow, placed relative to the nearest positioned ancestor
<code>fixed</code>	<code>position: fixed</code>	placed relative to the viewport, stays while scrolling
<code>sticky</code>	<code>position: sticky</code>	scrolls normally until it hits a threshold (<code>top-0</code>), then sticks

Golden rule: put `relative` on the parent whenever a child uses `absolute`.

11.2 Inset (top, right, bottom, left)

Class	CSS
<code>inset-0</code>	<code>top, right, bottom, left: 0</code> (cover the parent)
<code>inset-x-0</code>	<code>left: 0; right: 0</code>
<code>inset-y-0</code>	<code>top: 0; bottom: 0</code>
<code>top-0</code> , <code>right-0</code> , <code>bottom-0</code> , <code>left-0</code>	one side
<code>start-0</code> , <code>end-0</code>	logical left/right (RTL aware)
<code>top-4</code> , <code>right-2</code> , <code>-top-2</code> , <code>-left-1</code>	spacing scale and negatives
<code>top-1/2</code> , <code>left-1/2</code> , <code>inset-x-1/4</code>	fractions (percent)
<code>top-full</code> , <code>bottom-full</code>	100% (place outside the edge)
<code>top-[10px]</code> , <code>inset-[5px]</code>	arbitrary
<code>inset-auto</code> , <code>top-auto</code>	auto

11.3 Z-Index

`z-0`, `z-10`, `z-20`, `z-30`, `z-40`, `z-50`, `z-auto`, `-z-10`, `z-[100]`. In v4 any integer works: `z-60`.

Suggested layering convention: content `z-0`, sticky header `z-10`, dropdown `z-20`, overlay `z-30`, modal `z-40` / `z-50`, toast `z-50`.

11.4 Positioning Recipes

```
<!-- Badge in the corner of a card -->
<div class="relative rounded-xl border p-6">
  <span class="absolute top-0 right-0 -mt-2 -mr-2 rounded-full bg-red-500 px-2 py-0.5 text-xs text-white">New</span>
</div>

<!-- Cover the parent (overlay) -->
<div class="relative">
  <img class="w-full" />
  <div class="absolute inset-0 bg-black/40"></div>
</div>
```

```

<!-- Perfectly centered absolute element -->
<div class="absolute top-1/2 left-1/2 -translate-x-1/2 -translate-y-1/2">Centered</div>

<!-- Floating action button -->
<button class="fixed bottom-4 right-4 size-14 rounded-full bg-blue-600 text-white shadow-lg">+</button>

<!-- Sticky header -->
<header class="sticky top-0 z-10 bg-white/80 backdrop-blur border-b">...</header>

<!-- Fixed full-screen modal backdrop -->
<div class="fixed inset-0 z-50 bg-black/50"></div>

<!-- Input with icon inside -->
<div class="relative">
  <svg class="absolute left-3 top-1/2 size-5 -translate-y-1/2 text-gray-400">...</svg>
  <input class="w-full rounded-lg border py-2 pl-10 pr-3" />
</div>

<!-- Dropdown menu below a button -->
<div class="relative">
  <button>Menu</button>
  <div class="absolute right-0 top-full mt-2 w-48 rounded-lg bg-white shadow-lg">...</div>
</div>

```

Sticky not working? The parent (or any ancestor) must not have `overflow-hidden` / `overflow-auto` , and you must set a threshold such as `top-0` .

12. Display

Class	CSS	When to use
<code>block</code>	<code>display: block</code>	full-width block; stack vertically; make <code><a></code> or <code></code> behave like a box
<code>inline-block</code>	<code>display: inline-block</code>	sits in a text line but accepts width, height, margin
<code>inline</code>	<code>display: inline</code>	flows with text; ignores width/height and vertical margin
<code>flex</code>	<code>display: flex</code>	one-dimensional layouts (rows/columns)
<code>inline-flex</code>	<code>display: inline-flex</code>	flex container that sits inline (buttons with icons, badges)
<code>grid</code>	<code>display: grid</code>	two-dimensional layouts
<code>inline-grid</code>	<code>display: inline-grid</code>	inline grid container
<code>table</code> , <code>table-row</code> , <code>table-cell</code> , <code>table-caption</code> , <code>table-column</code> , <code>table-column-group</code> , <code>table-header-group</code> , <code>table-footer-group</code> , <code>table-row-group</code> , <code>inline-table</code>	table display values	build table-like layouts without <code><table></code>
<code>flow-root</code>	<code>display: flow-root</code>	contain floats / prevent margin collapse
<code>contents</code>	<code>display: contents</code>	the element's box disappears; its children behave as children of the parent
<code>list-item</code>	<code>display: list-item</code>	make an element behave like an <code></code>
<code>hidden</code>	<code>display: none</code>	removes from layout and from the accessibility tree
<code>invisible</code> / <code>visible</code>	<code>visibility</code>	hidden but still takes up space
<code>sr-only</code>	visually hidden	still read by screen readers (Section 30)

```
<!-- Show on desktop only / mobile only -->
<nav class="hidden md:flex">Desktop menu</nav>
<button class="md:hidden">Hamburger</button>

<!-- Link that fills its container (large click area) -->
<a class="block px-4 py-2 hover:bg-gray-100">Menu item</a>

<!-- Toggle visibility without layout shift -->
<div class="invisible group-hover:visible">Tooltip</div>
```

Other layout helpers: `float-left`, `float-right`, `float-none`, `clear-both`, `box-border` (default in Tailwind), `box-content`, `columns-2`, `columns-3`, `columns-xs`, `break-inside-avoid` (masonry-style columns).

13. Overflow

Class	CSS	Behavior
<code>overflow-auto</code>	<code>overflow: auto</code>	scrollbars only when content overflows
<code>overflow-hidden</code>	<code>overflow: hidden</code>	clip overflowing content, no scrollbar (also rounds image corners, clears floats)
<code>overflow-clip</code>	<code>overflow: clip</code>	clip, no scroll container created
<code>overflow-visible</code>	<code>overflow: visible</code>	content spills out (default)
<code>overflow-scroll</code>	<code>overflow: scroll</code>	always show scrollbars
<code>overflow-x-auto</code> , <code>overflow-x-hidden</code> , <code>overflow-x-scroll</code> , <code>overflow-x-clip</code> , <code>overflow-x-visible</code>	horizontal only	
<code>overflow-y-auto</code> , <code>overflow-y-hidden</code> , <code>overflow-y-scroll</code> , <code>overflow-y-clip</code> , <code>overflow-y-visible</code>	vertical only	
<code>overscroll-auto</code> , <code>overscroll-contain</code> , <code>overscroll-none</code>	scroll chaining	stop a modal from scrolling the page behind it
<code>overscroll-x-contain</code> , <code>overscroll-y-none</code>	per axis	

```

<!-- Scrollable table wrapper on small screens -->
<div class="overflow-x-auto"><table class="min-w-full">...</table></div>

<!-- Rounded card that clips its image -->
<div class="overflow-hidden rounded-xl"><img class="w-full" /></div>

<!-- Scroll area -->
<div class="h-64 overflow-y-auto">...</div>

<!-- Horizontal scrolling row with snap -->
<div class="flex snap-x snap-mandatory gap-4 overflow-x-auto">
  <div class="w-64 shrink-0 snap-start">1</div>
  <div class="w-64 shrink-0 snap-start">2</div>
</div>

```

14. Object & Image Utilities

`object-*` controls how **replaced elements** (`<img>`, `<video>`) fit inside their box. It needs a set width and height.

14.1 Object Fit

Class	Behavior
<code>object-contain</code>	whole image visible, keeps ratio, may leave empty space
<code>object-cover</code>	fills the box, keeps ratio, crops the excess (most used)
<code>object-fill</code>	stretches (distorts) to fill
<code>object-none</code>	original size, not resized
<code>object-scale-down</code>	<code>none</code> or <code>contain</code> , whichever is smaller

14.2 Object Position

`object-center`, `object-top`, `object-bottom`, `object-left`, `object-right`, `object-left-top`, `object-left-bottom`, `object-right-top`, `object-right-bottom`, `object-[25%_75%]`.

14.3 Image Examples

```
<!-- Circular avatar -->


<!-- Card image with fixed height, cropped nicely -->


<!-- Responsive image -->


<!-- Logo that must never be cropped -->


<!-- Video background -->
<video class="absolute inset-0 -z-10 h-full w-full object-cover" autoplay muted loop></video>

<!-- Grayscale photo that becomes colored on hover -->

```

15. Aspect Ratio

Class	CSS
<code>aspect-auto</code>	natural ratio
<code>aspect-square</code>	<code>1 / 1</code>
<code>aspect-video</code>	<code>16 / 9</code>
<code>aspect-4/3</code> , <code>aspect-3/2</code> , <code>aspect-2/1</code> ...	any <code>a/b</code> ratio (v4 shorthand)
<code>aspect-[4/3]</code> , <code>aspect-[21/9]</code>	arbitrary ratio

```
<div class="aspect-video w-full overflow-hidden rounded-xl bg-gray-200">
  <iframe class="size-full" src="https://www.youtube.com/embed/..."></iframe>
</div>


<!-- perfect square thumbnail -->
<div class="aspect-[3/4] w-48 bg-gray-100">Portrait placeholder</div>
```

Use aspect ratio with **one dimension set** (usually width) so the other is computed. Pair with `object-cover` for images.

16. Lists

Class	Effect
<code>list-none</code>	no bullets or numbers (also remove default padding yourself)
<code>list-disc</code>	bullets
<code>list-decimal</code>	1. 2. 3.
<code>list-inside</code>	marker inside the content box (text wraps under the marker)
<code>list-outside</code>	marker outside (default)
<code>list-image-[url(/dot.svg)]</code>	custom marker image
<code>list-[upper-roman]</code> , <code>list-[lower-alpha]</code>	any CSS <code>list-style-type</code> (arbitrary)
<code>marker:text-blue-500</code>	style the marker with the <code>marker:</code> variant

Tailwind's reset removes list styling. You **must** add `list-disc` or `list-decimal` to see bullets or numbers, and usually `pl-5` (or `list-inside`) for indentation.

```
<ul class="list-disc space-y-1 pl-5 marker:text-blue-500">
  <li>First</li><li>Second</li>
</ul>

<ol class="list-decimal list-inside space-y-2">...</ol>

<ul class="space-y-3">
  <li class="flex gap-3"><svg class="mt-1 size-5 shrink-0 text-green-500">...</svg>Feature with a check icon<
</li>
</ul>
```

17. Tables

Class	CSS
<code>border-collapse</code>	merge adjacent borders (common for clean tables)
<code>border-separate</code>	separate borders (needed for rounded corners and spacing)
<code>border-spacing-2</code> , <code>border-spacing-x-4</code> , <code>border-spacing-y-2</code>	gap between cells (with <code>border-separate</code>)
<code>table-auto</code>	column widths follow content (default)
<code>table-fixed</code>	equal/specified widths, faster, enables <code>truncate</code> in cells
<code>caption-top</code>	caption above the table (default)
<code>caption-bottom</code>	caption below

```
<div class="overflow-x-auto rounded-lg border border-gray-200">
  <table class="min-w-full border-collapse text-left text-sm">
    <caption class="caption-bottom p-3 text-xs text-gray-500">Monthly users</caption>
    <thead class="bg-gray-50 text-xs uppercase tracking-wider text-gray-500">
      <tr><th class="px-4 py-3">Name</th><th class="px-4 py-3">Role</th><th class="px-4 py-3 text-right">Stat
us</th></tr>
    </thead>
    <tbody class="divide-y divide-gray-200">
      <tr class="hover:bg-gray-50 odd:bg-white even:bg-gray-50/50">
        <td class="px-4 py-3 font-medium text-gray-900">Ada</td>
        <td class="px-4 py-3 text-gray-600">Admin</td>
        <td class="px-4 py-3 text-right"><span class="rounded-full bg-green-100 px-2 py-0.5 text-xs text-
green-700">Active</span></td>
      </tr>
    </tbody>
  </table>
</div>
```

18. Forms / Inputs

Tailwind removes the browser's default form styling, so you style everything yourself. (Optionally add the official plugin: `@plugin "@tailwindcss/forms";` in your CSS to get nicer defaults.)

18.1 Form State Variants

Variant	Applies when
<code>focus:</code>	element is focused (mouse or keyboard)
<code>focus-visible:</code>	focus from the keyboard (best for buttons and links)
<code>focus-within:</code>	any child has focus (style a wrapper)
<code>hover:</code> / <code>active:</code>	pointer over / pressed
<code>disabled:</code>	<code>disabled</code> attribute present
<code>required:</code> / <code>optional:</code>	has / lacks <code>required</code>
<code>invalid:</code>	fails validation (shows immediately)
<code>user-invalid:</code> / <code>user-valid:</code>	invalid / valid after user interaction (v4.1+; nicer than <code>invalid:</code>)
<code>valid:</code>	passes validation
<code>checked:</code>	checkbox/radio checked
<code>indeterminate:</code>	indeterminate checkbox
<code>placeholder:</code>	the placeholder text itself (<code>placeholder:text-gray-400</code>)
<code>placeholder-shown:</code>	input is currently showing its placeholder (empty)
<code>read-only:</code>	<code>readonly</code> attribute
<code>autofill:</code>	browser autofilled
<code>file:</code>	the "Choose file" button of <code><input type="file"></code>
<code>aria-invalid:</code>	<code>aria-invalid="true"</code>

18.2 Text Input

```
<label class="block">
  <span class="mb-1 block text-sm font-medium text-gray-700">Email</span>
  <input
    type="email"
    placeholder="you@example.com"
    class="block w-full rounded-lg border border-gray-300 bg-white px-3 py-2 text-gray-900
      placeholder:text-gray-400
      focus:border-blue-500 focus:outline-hidden focus:ring-3 focus:ring-blue-500/30
      disabled:cursor-not-allowed disabled:bg-gray-100 disabled:text-gray-500
      user-invalid:border-red-500 user-invalid:ring-red-500/30"
  />
</label>
```

18.3 Textarea

```
<textarea rows="4" class="w-full resize-y rounded-lg border border-gray-300 p-3
  focus:border-blue-500 focus:outline-hidden focus:ring-3 focus:ring-blue-500/30"></textarea>
```

```
<!-- Auto-growing textarea (v4) -->
<textarea class="field-sizing-content min-h-20 w-full rounded-lg border p-3"></textarea>
```

Use `resize-none` to disable resizing, `resize-y` vertical only, `resize-x`, `resize`.

18.4 Select

```
<select class="w-full appearance-none rounded-lg border border-gray-300 bg-white px-3 py-2
  focus:border-blue-500 focus:outline-hidden focus:ring-3 focus:ring-blue-500/30">
  <option>One</option><option>Two</option>
</select>
```

`appearance-none` removes the native arrow. Add your own icon positioned with `absolute` and `pointer-events-none`.

18.5 Checkbox and Radio

```
<!-- Simple: color the native control with accent- -->
<label class="flex items-center gap-2">
  <input type="checkbox" class="size-4 rounded accent-blue-600" />
  <span class="text-sm">Remember me</span>
</label>

<label class="flex items-center gap-2">
  <input type="radio" name="plan" class="size-4 accent-blue-600" />
  <span class="text-sm">Monthly</span>
</label>

<!-- Fully custom checkbox using appearance-none + checked: -->
<input type="checkbox"
  class="size-5 appearance-none rounded border border-gray-300 bg-white
    checked:border-blue-600 checked:bg-blue-600
    focus-visible:ring-2 focus-visible:ring-blue-500 focus-visible:ring-offset-2" />
```

18.6 Buttons

```
<button type="button"
  class="inline-flex items-center justify-center gap-2 rounded-lg bg-blue-600 px-4 py-2
    font-medium text-white shadow-sm cursor-pointer transition
    hover:bg-blue-700 active:scale-95
    focus-visible:outline-2 focus-visible:outline-offset-2 focus-visible:outline-blue-600
    disabled:cursor-not-allowed disabled:opacity-50">
  Save
</button>
```

18.7 File Input

```
<input type="file" class="block w-full text-sm text-gray-500
  file:mr-4 file:rounded-full file:border-0 file:bg-blue-50 file:px-4 file:py-2
  file:text-sm file:font-semibold file:text-blue-700 hover:file:bg-blue-100" />
```

18.8 Range, Switch and Misc

```
<input type="range" class="w-full accent-blue-600" />

<!-- Toggle switch with a peer (see Section 25) -->
<label class="relative inline-flex cursor-pointer items-center">
```

```

<input type="checkbox" class="peer sr-only" />
<div class="h-6 w-11 rounded-full bg-gray-300 transition peer-checked:bg-blue-600
  after:absolute after:left-0.5 after:top-0.5 after:size-5 after:rounded-full after:bg-white
  after:transition peer-checked:after:translate-x-5
  peer-focus-visible:ring-2 peer-focus-visible:ring-blue-500"></div>
</label>

```

18.9 Complete Form Example

```

<form class="mx-auto max-w-md space-y-5 rounded-2xl bg-white p-8 shadow-lg">
  <h2 class="text-2xl font-bold text-gray-900">Contact us</h2>

  <div>
    <label for="name" class="mb-1 block text-sm font-medium text-gray-700">Name</label>
    <input id="name" required class="w-full rounded-lg border border-gray-300 px-3 py-2
      focus:border-blue-500 focus:outline-hidden focus:ring-3 focus:ring-blue-500/30" />
  </div>

  <div>
    <label for="msg" class="mb-1 block text-sm font-medium text-gray-700">Message</label>
    <textarea id="msg" rows="4" class="w-full rounded-lg border border-gray-300 px-3 py-2
      focus:border-blue-500 focus:outline-hidden focus:ring-3 focus:ring-blue-500/30"></textarea>
  </div>

  <label class="flex items-center gap-2 text-sm text-gray-600">
    <input type="checkbox" class="size-4 rounded accent-blue-600" /> I agree to the terms
  </label>

  <button class="w-full rounded-lg bg-blue-600 py-2.5 font-semibold text-white
    hover:bg-blue-700 focus-visible:ring-2 focus-visible:ring-blue-500 focus-visible:ring-offset-2
    disabled:opacity-50">Send</button>
</form>

```

19. Interactivity

19.1 Cursor

`cursor-auto`, `cursor-default`, `cursor-pointer`, `cursor-wait`, `cursor-text`, `cursor-move`, `cursor-help`, `cursor-not-allowed`, `cursor-none`, `cursor-context-menu`, `cursor-progress`, `cursor-cell`, `cursor-crosshair`, `cursor-vertical-text`, `cursor-alien`, `cursor-copy`, `cursor-no-drop`, `cursor-grab`, `cursor-grabbing`, `cursor-all-scroll`, `cursor-col-resize`, `cursor-row-resize`, `cursor-n-resize`, `cursor-e-resize`, `cursor-s-resize`, `cursor-w-resize`, `cursor-ne-resize`, `cursor-nw-resize`, `cursor-se-resize`, `cursor-sw-resize`, `cursor-ew-resize`, `cursor-ns-resize`, `cursor-nesw-resize`, `cursor-nwse-resize`, `cursor-zoom-in`, `cursor-zoom-out`.

Tailwind v3 note: v4 buttons use `cursor: default` (like native). Add `cursor-pointer` to clickable buttons yourself.

19.2 Pointer Events, User Select, Resize, Appearance

Class	CSS	Use
<code>pointer-events-none</code>	clicks pass through	decorative overlays, disabled look
<code>pointer-events-auto</code>	restore clicks	child inside a <code>pointer-events-none</code> parent
<code>select-none</code>	cannot select text	buttons, UI labels
<code>select-text</code>	selectable	
<code>select-all</code>	one click selects all	copyable codes
<code>select-auto</code>	default	
<code>resize</code> / <code>resize-x</code> / <code>resize-y</code> / <code>resize-none</code>	textarea resize handle	
<code>appearance-none</code>	remove native styling	custom select/checkbox
<code>appearance-auto</code>	restore	
<code>touch-none</code> , <code>touch-auto</code> , <code>touch-pan-x</code> , <code>touch-pan-y</code> , <code>touch-manipulation</code>	touch gestures	
<code>will-change-auto</code> , <code>will-change-transform</code> , <code>will-change-scroll</code> , <code>will-change-contents</code>	performance hint	animation-heavy elements

19.3 Scroll Behavior and Snap

Class	Effect
<code>scroll-smooth</code>	smooth anchor scrolling (put on <code><html></code>)
<code>scroll-auto</code>	instant scroll
<code>scroll-m-*</code> , <code>scroll-mt-20</code>	scroll margin (offset for sticky headers on anchor jumps)
<code>scroll-p-*</code> , <code>scroll-pt-4</code>	scroll padding on the container
<code>snap-x</code> , <code>snap-y</code> , <code>snap-both</code> , <code>snap-none</code>	enable snap axis on the container
<code>snap-mandatory</code> , <code>snap-proximity</code>	snap strictness

Class	Effect
<code>snap-start</code> , <code>snap-center</code> , <code>snap-end</code> , <code>snap-align-none</code>	where children align
<code>snap-always</code> , <code>snap-normal</code>	don't skip / allow skip
<code>overscroll-contain</code>	stop scroll chaining

19.4 Caret and Accent

`caret-blue-500` colors the text cursor in inputs. `accent-blue-600` colors native checkboxes, radios, range sliders and progress bars. Both support `/opacity` and arbitrary values: `caret-[#f43f5e]`, `accent-pink-500`.

```
<input class="caret-pink-500 focus:outline-hidden" />
<input type="checkbox" class="accent-emerald-500" />
<div class="select-none cursor-grab active:cursor-grabbing">Draggable</div>
```

20. Transitions & Animation

20.1 Transition Property

Class	Transitions
<code>transition</code>	a sensible default set (colors, opacity, shadow, transform, filters...)
<code>transition-all</code>	every property (convenient but can hurt performance)
<code>transition-colors</code>	color, background-color, border-color, text-decoration, fill, stroke
<code>transition-opacity</code>	opacity
<code>transition-shadow</code>	box-shadow
<code>transition-transform</code>	transform (scale, rotate, translate)
<code>transition-none</code>	no transition
<code>transition-discrete</code>	allows <code>display</code> /discrete properties to switch during transitions (v4)
<code>transition-[width]</code>	arbitrary property list

20.2 Duration, Timing, Delay

Duration	Class	Timing function	Class
75ms	<code>duration-75</code>	linear	<code>ease-linear</code>
100ms	<code>duration-100</code>	slow to fast	<code>ease-in</code>
150ms (default)	<code>duration-150</code>	fast to slow	<code>ease-out</code>
200ms	<code>duration-200</code>	slow, fast, slow	<code>ease-in-out</code>
300ms	<code>duration-300</code>	custom	<code>ease-[cubic-bezier(0.4,0,0.2,1)]</code>
500ms	<code>duration-500</code>		
700ms	<code>duration-700</code>		
1000ms	<code>duration-1000</code>		
custom	<code>duration-[2s]</code>		

Delay: `delay-75`, `delay-100`, `delay-150`, `delay-200`, `delay-300`, `delay-500`, `delay-700`, `delay-1000`, `delay-[250ms]`.

Rule of thumb: UI feedback (hover, press) `duration-150` to `duration-200`; panels and menus `duration-300`; big motion `duration-500+`.

```
<button class="bg-blue-500 transition-colors duration-200 ease-out hover:bg-blue-600">Hover me</button>
<div class="transition-transform duration-300 hover:-translate-y-1 hover:shadow-lg">Lift card</div>
<div class="transition-all duration-500 delay-100 opacity-0 translate-y-4 group-hover:opacity-100 group-hover:translate-y-0">Reveal</div>
```

20.3 Built-in Animations

Class	Effect	Use
<code>animate-spin</code>	continuous rotation	loading spinner

Class	Effect	Use
<code>animate-ping</code>	expanding fading ring	notification dot
<code>animate-pulse</code>	gentle fading in and out	skeleton loaders
<code>animate-bounce</code>	bouncing	scroll hint arrow
<code>animate-none</code>	removes animation	reset or reduced motion
<code>animate-[wiggle_1s_ease-in-out_infinite]</code>	arbitrary animation	uses your <code>@keyframes</code>

```
<svg class="size-5 animate-spin text-white" viewBox="0 0 24 24" fill="none">
  <circle class="opacity-25" cx="12" cy="12" r="10" stroke="currentColor" stroke-width="4" />
  <path class="opacity-75" fill="currentColor" d="M4 12a8 8 0 018-8v4a4 4 0 00-4 4H4z" />
</svg>

<span class="relative flex size-3">
  <span class="absolute inline-flex size-full animate-ping rounded-full bg-sky-400 opacity-75"></span>
  <span class="relative inline-flex size-3 rounded-full bg-sky-500"></span>
</span>
```

20.4 Custom Animations (concept)

Define the animation in `@theme` (keyframes go inside it), and Tailwind creates an `animate-*` utility:

```
@import "tailwindcss";

@theme {
  --animate-wiggle: wiggle 1s ease-in-out infinite;

  @keyframes wiggle {
    0%, 100% { transform: rotate(-3deg); }
    50%    { transform: rotate(3deg); }
  }
}
```

```
<div class="animate-wiggle">Wiggles forever</div>
<div class="hover:animate-wiggle">Wiggles on hover</div>
```

Tailwind v3 note: v3 defined `keyframes` and `animation` inside `tailwind.config.js` (`theme.extend`).

20.5 Animation Accessibility and Entry Animations

Variant	Purpose
<code>motion-safe:</code>	apply only if the user has not requested reduced motion
<code>motion-reduce:</code>	apply when the user prefers reduced motion
<code>starting:</code>	style an element's first frame for enter transitions (v4), for example <code>starting:opacity-0</code>

```
<div class="motion-safe:animate-bounce motion-reduce:animate-none">Respectful animation</div>
<div class="transition-opacity duration-500 starting:opacity-0">Fades in when added to the DOM</div>
```

21. Transform

In v4 Tailwind uses the individual CSS properties `scale`, `rotate` and `translate`, so these utilities **combine freely** without a `transform` class.

21.1 Scale, Rotate, Translate, Skew

Purpose	Classes	Examples
Scale (both axes)	<code>scale-0</code> , <code>scale-50</code> , <code>scale-75</code> , <code>scale-90</code> , <code>scale-95</code> , <code>scale-100</code> , <code>scale-105</code> , <code>scale-110</code> , <code>scale-125</code> , <code>scale-150</code>	<code>hover:scale-105</code>
Scale one axis	<code>scale-x-*</code> , <code>scale-y-*</code>	<code>scale-x-0</code> , <code>scale-y-110</code>
Negative / flip	<code>-scale-x-100</code> (mirror horizontally)	
Rotate	<code>rotate-0</code> , <code>rotate-1</code> , <code>rotate-2</code> , <code>rotate-3</code> , <code>rotate-6</code> , <code>rotate-12</code> , <code>rotate-45</code> , <code>rotate-90</code> , <code>rotate-180</code>	<code>rotate-45</code>
Rotate negative	<code>-rotate-45</code> , <code>-rotate-90</code> ...	
Translate X	<code>translate-x-{n}</code> , <code>-translate-x-{n}</code> , <code>translate-x-1/2</code> , <code>-translate-x-full</code>	<code>-translate-x-1/2</code>
Translate Y	<code>translate-y-{n}</code> , <code>-translate-y-{n}</code> , <code>translate-y-1/2</code> , <code>translate-y-full</code>	<code>hover:-translate-y-1</code>
Skew	<code>skew-x-{0,1,2,3,6,12}</code> , <code>skew-y-*</code> , negatives	<code>-skew-x-12</code>
Arbitrary	<code>rotate-[17deg]</code> , <code>translate-x-[30px]</code> , <code>scale-[1.15]</code>	
Reset	<code>transform-none</code>	

21.2 Transform Origin

`origin-center` (default), `origin-top`, `origin-top-right`, `origin-right`, `origin-bottom-right`, `origin-bottom`, `origin-bottom-left`, `origin-left`, `origin-top-left`, `origin-[30%_10%]`.

Example: a dropdown that grows from its top-left corner: `origin-top-left scale-95 opacity-0 group-hover:scale-100 group-hover:opacity-100 transition`.

21.3 3D Transforms (v4)

`transform-3d`, `perspective-normal`, `perspective-distant`, `rotate-x-12`, `rotate-y-12`, `translate-z-4`, `backface-hidden`, `transform-flat`. Useful for card-flip effects.

21.4 Transform Recipes

```
<!-- Hover grow -->
<div class="transition-transform duration-200 hover:scale-105">Card</div>

<!-- Lift on hover -->
<div class="transition duration-200 hover:-translate-y-1 hover:shadow-xl">Card</div>

<!-- Press effect -->
<button class="transition active:scale-95">Press me</button>

<!-- Rotate icon when open (with group or aria) -->
<svg class="size-5 transition-transform group-open:rotate-180">...</svg>
```

```
<!-- Tilted decorative card -->
<div class="rotate-3 rounded-xl bg-yellow-200 p-6 shadow-md">Sticky note</div>

<!-- Center with translate -->
<div class="absolute left-1/2 top-1/2 -translate-x-1/2 -translate-y-1/2">Center</div>

<!-- Slide-in drawer -->
<aside class="fixed inset-y-0 left-0 w-64 -translate-x-full transition-transform data-[open=true]:translate-x-0">...</aside>
```

22. Filters & Visual Effects

22.1 Filter Utilities (apply to the element)

Effect	Classes
Blur	<code>blur-none</code> , <code>blur-xs</code> (4px), <code>blur-sm</code> (8px), <code>blur-md</code> (12px), <code>blur-lg</code> (16px), <code>blur-xl</code> (24px), <code>blur-2xl</code> (40px), <code>blur-3xl</code> (64px), <code>blur-[2px]</code>
Brightness	<code>brightness-0</code> , <code>50</code> , <code>75</code> , <code>90</code> , <code>95</code> , <code>100</code> , <code>105</code> , <code>110</code> , <code>125</code> , <code>150</code> , <code>200</code>
Contrast	<code>contrast-0</code> , <code>50</code> , <code>75</code> , <code>100</code> , <code>125</code> , <code>150</code> , <code>200</code>
Grayscale	<code>grayscale</code> (100%), <code>grayscale-0</code> , <code>grayscale-50</code>
Hue rotate	<code>hue-rotate-15</code> , <code>30</code> , <code>60</code> , <code>90</code> , <code>180</code> , <code>-hue-rotate-15</code>
Invert	<code>invert</code> (100%), <code>invert-0</code> , <code>invert-50</code>
Saturate	<code>saturate-0</code> , <code>50</code> , <code>100</code> , <code>150</code> , <code>200</code>
Sepia	<code>sepia</code> , <code>sepia-0</code> , <code>sepia-50</code>
Drop shadow	<code>drop-shadow-2xs</code> , <code>drop-shadow-xs</code> , <code>drop-shadow-sm</code> , <code>drop-shadow-md</code> , <code>drop-shadow-lg</code> , <code>drop-shadow-xl</code> , <code>drop-shadow-2xl</code> , <code>drop-shadow-none</code> , <code>drop-shadow-blue-500/50</code>
Remove all	<code>filter-none</code>

`drop-shadow-*` follows the **visible shape** (works on transparent PNGs and SVGs), unlike `shadow-*` which uses the rectangular box.

22.2 Backdrop Filters (affect what is behind the element)

Effect	Classes
Backdrop blur	<code>backdrop-blur-none</code> , <code>backdrop-blur-xs</code> , <code>backdrop-blur-sm</code> , <code>backdrop-blur-md</code> , <code>backdrop-blur-lg</code> , <code>backdrop-blur-xl</code> , <code>backdrop-blur-2xl</code> , <code>backdrop-blur-3xl</code>
Brightness	<code>backdrop-brightness-50</code> , <code>75</code> , <code>90</code> , <code>100</code> , <code>110</code> , <code>125</code> , <code>150</code> , <code>200</code>
Contrast	<code>backdrop-contrast-50</code> , <code>75</code> , <code>100</code> , <code>125</code> , <code>150</code> , <code>200</code>
Grayscale	<code>backdrop-grayscale</code> , <code>backdrop-grayscale-0</code>
Hue rotate	<code>backdrop-hue-rotate-15</code> , <code>30</code> , <code>60</code> , <code>90</code> , <code>180</code>
Invert	<code>backdrop-invert</code> , <code>backdrop-invert-0</code>
Opacity	<code>backdrop-opacity-0</code> ... <code>backdrop-opacity-100</code>
Saturate	<code>backdrop-saturate-0</code> , <code>50</code> , <code>100</code> , <code>150</code> , <code>200</code>
Sepia	<code>backdrop-sepia</code> , <code>backdrop-sepia-0</code>
Remove all	<code>backdrop-filter-none</code>

Tailwind v3 note: v3 needed `filter` and `backdrop-filter` helper classes. In v4 they are applied automatically; you only write `blur-sm` or `backdrop-blur-md`. Also v3 `blur` / `drop-shadow` (bare) became `blur-sm` / `drop-shadow-sm`.

22.3 Glassmorphism Examples

```
<!-- Frosted glass card over a colorful background -->
<div class="relative min-h-screen bg-linear-to-br from-indigo-500 via-purple-500 to-pink-500 grid place-items-center p-6">
  <div class="w-full max-w-sm rounded-2xl border border-white/30 bg-white/20 p-8 text-white shadow-xl backdrop-blur-lg backdrop-saturate-150">
    <h2 class="text-2xl font-bold">Glass card</h2>
    <p class="mt-2 text-white/80">Blurred background shines through.</p>
  </div>
</div>

<!-- Frosted sticky navbar -->
<header class="sticky top-0 z-20 border-b border-white/20 bg-white/70 backdrop-blur-md dark:bg-gray-900/70">
  ..</header>

<!-- Dark glass -->
<div class="rounded-xl border border-white/10 bg-black/30 backdrop-blur-xl">...</div>

<!-- Dim and blur the page behind a modal -->
<div class="fixed inset-0 bg-black/40 backdrop-blur-sm"></div>

<!-- Disabled image look -->
<img class="grayscale opacity-60" />

<!-- Image zoom with brightness on hover -->
<img class="transition duration-300 hover:scale-105 hover:brightness-110" />
```

23. Responsive Design

23.1 Mobile-First: The Key Idea

Un-prefixed utilities apply to **all screen sizes** (mobile first). Prefixed utilities (`md:`) apply **from that breakpoint upward**. So you style the **smallest screen first**, then add overrides for larger screens.

```
<!-- Mobile: small text, 1 column. md+: bigger text, 2 columns. lg+: 3 columns -->
<h1 class="text-sm md:text-lg lg:text-2xl">Responsive text</h1>
<div class="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-4">...</div>
```

screen width:	0	640	768	1024	1280	1536
class:	(none)	sm:	md:	lg:	xl:	2xl:
	always on	640+	768+	1024+	1280+	1536+

Wrong thinking: `sm:` does **not** mean "small screens only". It means "640px and up". To target phones, use the un-prefixed class.

23.2 Breakpoints

Prefix	Min-width	CSS media query
<code>sm:</code>	640px (40rem)	<code>@media (width >= 40rem)</code>
<code>md:</code>	768px (48rem)	<code>@media (width >= 48rem)</code>
<code>lg:</code>	1024px (64rem)	<code>@media (width >= 64rem)</code>
<code>xl:</code>	1280px (80rem)	<code>@media (width >= 80rem)</code>
<code>2xl:</code>	1536px (96rem)	<code>@media (width >= 96rem)</code>

23.3 Max-Width Variants and Ranges

Variant	Applies when	Equivalent
<code>max-sm:</code>	width below 640px	<code>@media (width < 40rem)</code>
<code>max-md:</code>	below 768px	
<code>max-lg:</code>	below 1024px	
<code>max-xl:</code>	below 1280px	
<code>max-2xl:</code>	below 1536px	
<code>md:max-lg:</code>	between 768px and 1023px	range (stack variants)
<code>min-[900px]:</code>	at least 900px (arbitrary)	
<code>max-[600px]:</code>	below 600px (arbitrary)	

```
<div class="max-md:hidden">Hidden below 768px</div>
<div class="md:max-lg:grid-cols-2 grid grid-cols-1 lg:grid-cols-4">Two columns only on tablets</div>
```

23.4 Container Queries (v4 built-in)

Respond to the **parent's** width instead of the viewport. Mark the parent with `@container`, then use `@sm:`, `@md:`, `@lg:` ... on children.

```
<div class="@container">
  <div class="flex flex-col @md:flex-row @md:items-center gap-4">
    Layout changes based on the card's width, not the screen
  </div>
</div>
```

Sizes: `@3xs` (16rem), `@2xs`, `@xs`, `@sm` (24rem), `@md` (28rem), `@lg` (32rem), `@xl`, `@2xl`, ... `@7xl`; max versions `@max-md:`; named containers `@container/card` + `@md/card:`.

23.5 Custom Breakpoints

```
@theme { --breakpoint-3xl: 120rem; } /* creates 3xl: and max-3xl: */
```

23.6 Responsive Patterns

```
<!-- Stack on mobile, row on desktop -->
<div class="flex flex-col gap-6 md:flex-row">
  <div class="md:w-1/3">Sidebar</div>
  <div class="md:w-2/3">Content</div>
</div>

<!-- Show hamburger on mobile, menu on desktop -->
<button class="md:hidden">Menu</button>
<ul class="hidden md:flex gap-6">...</ul>

<!-- Responsive padding and type -->
<section class="px-4 py-12 sm:px-6 md:py-20 lg:px-8">
  <h2 class="text-2xl font-bold sm:text-3xl lg:text-5xl">Title</h2>
</section>

<!-- Centered container with a max width -->
<div class="mx-auto w-full max-w-7xl px-4">...</div>

<!-- Responsive image -->
<img class="h-48 w-full object-cover md:h-72 lg:h-96" />

<!-- Reorder on mobile -->
<div class="flex flex-col md:flex-row"><div class="order-2 md:order-1">A</div><div class="order-1 md:order-2">B</div></div>
```

Testing tip: use browser DevTools device mode and try widths 375, 768, 1024 and 1440.

24. State Variants

Any utility can be prefixed with a variant. Variants can be **stacked** (applied left to right): `md:dark:hover:bg-blue-700`.

24.1 Interaction States

Variant	When	Example
<code>hover:</code>	pointer over (only on devices that support hover in v4)	<code>hover:bg-blue-600</code>
<code>focus:</code>	element has focus	<code>focus:ring-2</code>
<code>focus-visible:</code>	keyboard focus	<code>focus-visible:outline-2</code>
<code>focus-within:</code>	a descendant has focus	<code>focus-within:ring-2</code>
<code>active:</code>	being pressed	<code>active:scale-95</code>
<code>visited:</code>	visited link	<code>visited:text-purple-600</code>
<code>target:</code>	element is the URL <code>#hash</code> target	<code>target:bg-yellow-100</code>
<code>open:</code>	<code><details></code> / <code><dialog></code> / popover is open	<code>open:bg-gray-50</code>

24.2 Form States

Variant	When	Example
<code>disabled:</code>	disabled	<code>disabled:opacity-50</code>
<code>enabled:</code>	not disabled	<code>enabled:hover:bg-blue-700</code>
<code>checked:</code>	checked box/radio	<code>checked:bg-blue-600</code>
<code>indeterminate:</code>	indeterminate checkbox	<code>indeterminate:bg-gray-400</code>
<code>default:</code>	default-selected option/checkbox	<code>default:ring-2</code>
<code>required:</code>	<code>required</code>	<code>required:border-red-300</code>
<code>optional:</code>	not required	
<code>valid:</code>	valid value	<code>valid:border-green-500</code>
<code>invalid:</code>	invalid value	<code>invalid:border-red-500</code>
<code>user-valid:</code> / <code>user-invalid:</code>	valid/invalid after interaction (v4.1+)	<code>user-invalid:border-red-500</code>
<code>in-range:</code>	value within min/max	<code>in-range:border-green-500</code>
<code>out-of-range:</code>	value outside min/max	<code>out-of-range:border-red-500</code>
<code>placeholder-shown:</code>	placeholder visible (input empty)	<code>placeholder-shown:border-gray-300</code>
<code>autofill:</code>	browser autofill	<code>autofill:bg-yellow-50</code>
<code>read-only:</code>	<code>readonly</code>	<code>read-only:bg-gray-100</code>
<code>read-write:</code>	editable	

24.3 Structural (position in the DOM)

Variant	Selects	Example
<code>first:</code>	first child	<code>first:pt-0</code>

Variant	Selects	Example
<code>last:</code>	last child	<code>last:border-b-0</code>
<code>only:</code>	only child	<code>only:rounded-lg</code>
<code>odd: / even:</code>	odd / even children	<code>odd:bg-gray-50</code>
<code>first-of-type: / last-of-type: / only-of-type:</code>	by element type	<code>first-of-type:mt-0</code>
<code>empty:</code>	no children / no text	<code>empty:hidden</code>
<code>nth-3: / nth-[3n+1]:</code>	nth-child	<code>nth-[3n+1]:bg-gray-50</code>
<code>has-[...]:</code>	parent that contains a match	<code>has-[:checked]:bg-blue-50</code>
<code>not-*:</code>	negation	<code>not-first:mt-4</code> , <code>not-hover:opacity-70</code>
<code>*:</code>	all direct children	<code>*:p-2</code>
<code>**:</code>	all descendants (v4.1+)	<code>**:text-sm</code>

24.4 Pseudo-elements

Variant	Target	Example
<code>before: / after:</code>	<code>::before</code> / <code>::after</code> (auto-adds <code>content: ''</code>)	<code>after:absolute</code> <code>after:inset-0</code>
<code>placeholder:</code>	placeholder text	<code>placeholder:italic</code>
<code>file:</code>	file input button	<code>file:bg-blue-50</code>
<code>marker:</code>	list marker	<code>marker:text-blue-500</code>
<code>selection:</code>	selected text	<code>selection:bg-yellow-200</code>
<code>first-letter: / first-line:</code>	text fragments	<code>first-letter:text-5xl</code>
<code>backdrop:</code>	<code><dialog></code> backdrop	<code>backdrop:bg-black/50</code>

24.5 Attribute, Media and Other Variants

Variant	Purpose	Example
<code>aria-checked:</code> , <code>aria-expanded:</code> , <code>aria-disabled:</code> , <code>aria-[sort=ascending]:</code>	ARIA attributes	<code>aria-expanded:rotate-180</code>
<code>data-[state=open]:</code>	data attributes	<code>data-[active=true]:bg-blue-500</code>
<code>data-active:</code>	presence of <code>data-active</code>	
<code>dark:</code>	dark mode	<code>dark:bg-gray-900</code>
<code>motion-safe: / motion-reduce:</code>	reduced motion	
<code>print:</code>	printing	<code>print:hidden</code>
<code>portrait: / landscape:</code>	orientation	
<code>rtl: / ltr:</code>	text direction	<code>rtl:space-x-reverse</code>
<code>supports-[display:grid]:</code>	feature queries	
<code>contrast-more:</code>	high-contrast preference	
<code>forced-colors:</code>	forced colors mode	
<code>pointer-fine: / pointer-coarse:</code>	mouse vs touch	<code>pointer-coarse:p-4</code>

Variant	Purpose	Example
<code>starting:</code>	entry animations	<code>starting:opacity-0</code>
<code>group-* / peer-*</code>	parent/sibling state	Section 25
<code>[&:nth-child(3)]:</code>	arbitrary selector	<code>[&>li]:py-2</code>

24.6 Practical State Examples

```

<!-- Link states -->
<a class="text-blue-600 underline-offset-4 hover:underline focus-visible:outline-2 visited:text-purple-600">Link</a>

<!-- Zebra rows + borders between all but last -->
<ul>
  <li class="odd:bg-gray-50 border-b last:border-b-0 p-3">Row</li>
</ul>

<!-- Hide empty state container -->
<div class="empty:hidden">{maybeContent}</div>

<!-- Highlight a label when its checkbox is checked -->
<label class="flex items-center gap-2 rounded-lg border p-3 has-[:checked]:border-blue-500 has-[:checked]:bg-blue-50">
  <input type="checkbox" /> Option
</label>

<!-- Drop cap -->
<p class="first-letter:float-left first-letter:mr-2 first-letter:text-5xl first-letter:font-bold">Once upon a time...</p>

<!-- Style based on a data attribute (great with React state) -->
<button data-active={isActive} class="data-[active=true]:bg-blue-600 data-[active=true]:text-white">Tab</button>

<!-- Space children except the first -->
<div class="[&*:not(:first-child)]:mt-4">...</div>

```

25. Group & Peer

25.1 Group: style children based on the PARENT'S state

Add `group` to the parent. On the child use `group-hover:`, `group-focus:`, `group-active:`, `group-disabled:`, `group-focus-within:`, `group-has-[...]:`, `group-open:`, `group-first:`, `group-odd:`, `group-aria-*`, `group-data-*`: ...

```
<!-- Card hover affects children -->
<a href="#" class="group block rounded-xl border p-6 transition hover:border-blue-500 hover:shadow-lg">
  <h3 class="font-semibold text-gray-900 group-hover:text-blue-600">Title</h3>
  <p class="text-gray-500 group-hover:text-gray-700">Description</p>
  <span class="mt-4 inline-block transition-transform group-hover:translate-x-1">Read more &rarr;</span>
</a>

<!-- Image zoom inside a clipped card -->
<div class="group overflow-hidden rounded-xl">
  
</div>

<!-- Show actions on row hover -->
<li class="group flex items-center justify-between p-3 hover:bg-gray-50">
  <span>Item</span>
  <button class="opacity-0 transition group-hover:opacity-100">Delete</button>
</li>

<!-- Details/accordion arrow -->
<details class="group">
  <summary class="flex cursor-pointer items-center justify-between">Question
    <svg class="size-4 transition-transform group-open:rotate-180">...</svg>
  </summary>
  <p>Answer</p>
</details>
```

Named groups (nested groups): give each group a name with `group/name`, then use `group-hover/name:`.

```
<ul>
  <li class="group/item hover:bg-gray-50">
    <a class="group/edit invisible group-hover/item:visible hover:bg-slate-200">
      <span class="group-hover/edit:text-gray-700">Call</span>
    </a>
  </li>
</ul>
```

25.2 Peer: style an element based on a SIBLING'S state

Add `peer` to the **earlier** sibling. The styled element must come **after** it in the DOM and use `peer-checked:`, `peer-focus:`, `peer-invalid:`, `peer-hover:`, `peer-disabled:`, `peer-placeholder-shown:`, `peer-required:`, `peer-has-[...]:` ...

```
<!-- Show an error message when the input is invalid -->
<input type="email" required class="peer w-full rounded border px-3 py-2 invalid:border-red-500" />
<p class="mt-1 hidden text-sm text-red-600 peer-invalid:block">Please enter a valid email.</p>

<!-- Radio card: highlight the card of the selected radio -->
<label>
  <input type="radio" name="plan" class="peer sr-only" />
  <div class="cursor-pointer rounded-xl border p-4 peer-checked:border-blue-600 peer-checked:bg-blue-50 peer-
```

```

focus-visible:ring-2">Pro plan</div>
</label>

<!-- Floating label input -->
<div class="relative">
  <input id="email" placeholder=" "
    class="peer w-full rounded-lg border border-gray-300 px-3 pb-2 pt-5 focus:border-blue-500 focus:outline-
hidden" />
  <label for="email"
    class="pointer-events-none absolute left-3 top-4 origin-left text-gray-500 transition-all duration-200
peer-focus:top-1.5 peer-focus:text-xs peer-focus:text-blue-600
peer-[:not(:placeholder-shown)]:top-1.5 peer-[:not(:placeholder-shown)]:text-xs">
    Email
  </label>
</div>

```

Floating label trick: the input has `placeholder=" "` (a single space) so `:placeholder-shown` tells whether it is empty.

Peer rules: the `peer` element must be a **previous sibling** (CSS can only look forward). With several peers, name them: `peer/email` and `peer-invalid/email:`.

26. Dark Mode

26.1 The dark: Variant

```
<div class="bg-white text-gray-900 dark:bg-gray-900 dark:text-white">
  <p class="text-gray-600 dark:text-gray-400">Adapts to dark mode</p>
  <button class="bg-blue-600 hover:bg-blue-700 dark:bg-blue-500 dark:hover:bg-blue-400">Button</button>
  <div class="border-gray-200 dark:border-gray-700 border">Card</div>
</div>
```

26.2 How Dark Mode Works in v4

Default (automatic): `dark:` follows the user's operating system preference through `@media (prefers-color-scheme: dark)`. No setup needed.

Manual toggle (class-based): override the variant in your CSS, then add or remove the class `dark` on `<html>`:

```
@import "tailwindcss";
@custom-variant dark (&:where(.dark, .dark *));
```

Using a data attribute instead:

```
@custom-variant dark (&:where([data-theme=dark], [data-theme=dark] *));
```

Tailwind v3 note: v3 used `darkMode: 'class'` (or `'media'`) in `tailwind.config.js`. In v4 you use `@custom-variant dark` in CSS.

26.3 React Dark Mode Toggle

```
import { useEffect, useState } from 'react'

export default function ThemeToggle() {
  const [dark, setDark] = useState(
    () => localStorage.theme === 'dark' ||
      (!('theme' in localStorage) && window.matchMedia('(prefers-color-scheme: dark)').matches)
  )

  useEffect(() => {
    document.documentElement.classList.toggle('dark', dark)
    localStorage.theme = dark ? 'dark' : 'light'
  }, [dark])

  return (
    <button onClick={() => setDark(!dark)}
      className="rounded-lg border px-3 py-1.5 text-sm dark:border-gray-600 dark:text-white">
      {dark ? 'Light' : 'Dark'} mode
    </button>
  )
}
```

(Requires the `@custom-variant dark` line above.)

26.4 Dark Mode Tips

- Pair every background with a text color: `bg-white text-gray-900 dark:bg-gray-900 dark:text-gray-100`.
- Dark surfaces: `gray-900 / gray-950`; borders: `gray-700 / gray-800`; muted text: `gray-400`.
- Lower the intensity of saturated colors in dark mode (use `400` instead of `600` for text accents).
- Define semantic color tokens once with CSS variables to avoid repeating `dark:` everywhere:

```
:root { --surface: white; --ink: #111827; }  
.dark { --surface: #0b1220; --ink: #f3f4f6; }  
@theme inline { --color-surface: var(--surface); --color-ink: var(--ink); }  
/* now: bg-surface text-ink works in both themes */
```

27. Arbitrary Values

27.1 Arbitrary Values `utility-[value]`

Use square brackets to pass **any CSS value** when the theme has no matching class.

Class	Generates
<code>w-[350px]</code>	<code>width: 350px</code>
<code>h-[500px]</code>	<code>height: 500px</code>
<code>mt-[37px]</code>	<code>margin-top: 37px</code>
<code>text-[22px]</code>	<code>font-size: 22px</code>
<code>text-[#123456]</code> / <code>bg-[#123456]</code>	colors (Tailwind detects the type)
<code>top-[10px]</code>	<code>top: 10px</code>
<code>grid-cols-[200px_1fr]</code>	<code>grid-template-columns: 200px 1fr</code>
<code>grid-cols-[repeat(auto-fit,minmax(250px,1fr))]</code>	responsive grid
<code>w-[calc(100%-2rem)]</code>	<code>width: calc(100% - 2rem)</code>
<code>h-[calc(100dvh-4rem)]</code>	<code>height: calc(100dvh - 4rem)</code>
<code>bg-[url(/img/hero.png)]</code>	background image
<code>shadow-[0_4px_20px_rgba(0,0,0,0.15)]</code>	custom shadow
<code>rotate-[17deg]</code>	<code>rotate: 17deg</code>
<code>leading-[1.7]</code>	line-height
<code>tracking-[0.2em]</code>	letter-spacing
<code>rounded-[10px]</code>	radius
<code>duration-[2s]</code>	transition duration
<code>z-[999]</code>	z-index
<code>font-[Inter]</code>	font family
<code>content-['hello']</code>	<code>content: 'hello'</code> (with <code>before:</code> / <code>after:</code>)
<code>w(--my-width)</code>	uses CSS variable (v4 shorthand for <code>w-[var(--my-width)]</code>)

Rules

- No spaces inside brackets: use `_` instead (`grid-cols-[200px_1fr]`). For a real underscore, escape it `\_`.
- `calc()` works; spaces around `+` `-` `*` `/` are added for you.
- Add a **type hint** if Tailwind cannot tell the type: `text-[length:var(--size)]`, `text-[color:var(--c)]`, or `text-[length:--size]`.
- Works with variants: `md:w-[500px]`, `hover:bg-[#123456]`, `dark:text-[#e5e7eb]`.
- Negative: `-mt-[10px]` or `mt-[-10px]`.

27.2 Arbitrary Properties `[property:value]`

For a CSS property that has no Tailwind utility (or you need it quickly):

```
<div class="[mask-type:luminance]"></div>
<div class="[--card-gap:12px] gap(--card-gap)"></div>    <!-- define & use a variable -->
```

```
<div class="[clip-path:polygon(50%_0,100%_100%,0_100%)]"></div>
<div class="hover:[mask-type:alpha]"></div>
<div class="[text-wrap:balance]"></div>
```

27.3 Arbitrary Variants `[selector]:`

```
<li class="[&:nth-child(3)]:underline">third item</li>
<ul class="[&>li]:py-2 [&_a]:text-blue-600">...</ul>      <!-- style children without classes on them -->
<div class="[@media(min-width:900px)]:flex">...</div>
<div class="[&.is-active]:font-bold">...</div>
```

`&` stands for the element itself. `[&>li]` = direct `li` children, `[&_a]` = any descendant `a` (underscore = space).

27.4 When to Use (and Not Use) Arbitrary Values

Good use	Bad use
A one-off value from a design file (<code>w-[342px]</code>)	Repeating the same value 20 times (add a theme token instead)
Complex grid templates	Replacing a scale value you could use (<code>mt-[16px]</code> instead of <code>mt-4</code>)
<code>calc()</code> layouts	Building a whole design from px values
Brand color used once	Brand color used everywhere: define <code>--color-brand</code> in <code>@theme</code>
Quick prototyping	Anything that should stay consistent across the app

28. Important Modifier

Adds `!important` to the generated declaration.

Version	Syntax	Example
Tailwind v4 (preferred)	trailing <code>!</code>	<code>text-red-500!</code> , <code>md:p-4!</code> , <code>hover:bg-blue-600!</code>
Tailwind v3 (still works in v4, legacy)	leading <code>!</code>	<code>!text-red-500</code>

```
<p class="text-gray-500 text-red-500!">Always red, even if something else tries to override it</p>
```

When it is appropriate

- Overriding styles from a **third-party library** you cannot edit (date pickers, widgets).
- Forcing a utility to beat styles injected inline or with high specificity.

When it is NOT appropriate

- To fix your own conflicting classes (remove or merge them instead: see `tailwind-merge`).
- Everywhere "just in case": it makes later overrides impossible.

29. Container

29.1 The container Class

`container` sets `width: 100%` and a `max-width` equal to the current breakpoint:

Screen	max-width
(below sm)	100%
<code>sm</code> (640px)	640px
<code>md</code> (768px)	768px
<code>lg</code> (1024px)	1024px
<code>xl</code> (1280px)	1280px
<code>2xl</code> (1536px)	1536px

It does **not** center itself or add padding by default:

```
<div class="container mx-auto px-4">...</div>
```

Most people prefer `max-w-*` for predictable widths: `mx-auto max-w-7xl px-4 sm:px-6 lg:px-8`.

29.2 Customize the container (v4)

```
@utility container {
  margin-inline: auto;
  padding-inline: 2rem;
}
```

Tailwind v3 note: v3 customized the container in `tailwind.config.js` (`container: { center: true, padding: '2rem' }`). Those options **no longer exist** in v4.

30. Accessibility

30.1 Screen Reader Utilities

Class	Effect
<code>sr-only</code>	visually hidden but available to screen readers
<code>not-sr-only</code>	undo <code>sr-only</code> (for example on focus)
<code>forced-color-adjust:none</code> / <code>forced-color-adjust:auto</code>	opt out of / into forced-colors mode

```
<!-- Icon-only button needs a text alternative -->
<button class="rounded-full p-2 hover:bg-gray-100">
  <svg class="size-5 aria-hidden=true">...</svg>
  <span class="sr-only">Close menu</span>
</button>

<!-- Skip link: visible only when focused -->
<a href="#main" class="sr-only focus:not-sr-only focus:absolute focus:left-4 focus:top-4 focus:rounded
focus:bg-white focus:px-4 focus:py-2">
  Skip to content
</a>
```

30.2 Focus States (never remove them without a replacement)

```
<!-- Good: custom visible focus ring for keyboard users -->
<button class="rounded-lg bg-blue-600 px-4 py-2 text-white outline-hidden
focus-visible:ring-2 focus-visible:ring-blue-500 focus-visible:ring-offset-2">Save</button>

<a class="rounded focus-visible:outline-2 focus-visible:outline-offset-2 focus-visible:outline-
blue-600">Link</a>
```

- Use `focus-visible:` (keyboard only) for buttons and links, `focus:` for text inputs.
- Use `outline-hidden` (not `outline-none`) when replacing the outline: it stays visible in high-contrast mode.

30.3 Accessible Buttons and Forms Checklist

- Real `<button>` for actions and `<a>` for navigation (not clickable `<div>`s).
- Every input has a `<label for="id">` (use `sr-only` if the design hides it).
- Color contrast: text on `bg-white` should be `gray-600` or darker for body text (WCAG AA 4.5:1).
- Do not rely on color alone for errors: add text or an icon.
- Disabled look: `disabled:opacity-50 disabled:cursor-not-allowed` plus the real `disabled` attribute.
- Touch targets: at least ~44x44px (`min-h-11 min-w-11`).
- Respect reduced motion: `motion-safe:animate-spin motion-reduce:animate-none`.
- Use ARIA state variants: `aria-expanded:rotate-180`, `aria-selected:bg-blue-50`, `aria-disabled:opacity-50`.

31. Common UI Patterns

Ready-to-copy components. They use plain HTML with `class`. In React, change `class` to `className`, `for` to `htmlFor`, and self-close tags (`<input />`). Icons are shown as `<svg>...</svg>` placeholders.

31.1 Navbar

```
<header class="sticky top-0 z-30 border-b border-gray-200 bg-white/80 backdrop-blur">
  <nav class="mx-auto flex max-w-7xl items-center justify-between px-4 py-3 sm:px-6 lg:px-8">
    <a href="/" class="text-xl font-bold text-blue-600">Brand</a>

    <ul class="hidden items-center gap-8 text-sm font-medium text-gray-600 md:flex">
      <li><a href="#" class="transition-colors hover:text-gray-900">Home</a></li>
      <li><a href="#" class="transition-colors hover:text-gray-900">Features</a></li>
      <li><a href="#" class="transition-colors hover:text-gray-900">Pricing</a></li>
    </ul>

    <div class="flex items-center gap-3">
      <a href="#" class="hidden text-sm font-medium text-gray-600 hover:text-gray-900 sm:block">Log in</a>
      <a href="#" class="rounded-lg bg-blue-600 px-4 py-2 text-sm font-semibold text-white hover:bg-blue-700">Sign up</a>
      <button class="rounded-lg p-2 hover:bg-gray-100 md:hidden" aria-label="Open menu"><svg class="size-6">
    ..</svg></button>
    </div>
  </nav>
</header>
```

Important classes	Why
<code>sticky top-0 z-30</code>	stays at the top while scrolling, above content
<code>bg-white/80 backdrop-blur</code>	translucent frosted bar
<code>mx-auto max-w-7xl px-4 ...</code>	centered content column with responsive side padding
<code>flex items-center justify-between</code>	logo left, actions right, vertically centered
<code>hidden md:flex</code>	desktop menu hidden on mobile, shown from 768px
<code>md:hidden</code>	hamburger only on mobile

31.2 Hero Section

```
<section class="relative isolate overflow-hidden bg-linear-to-b from-indigo-50 to-white">
  <div class="mx-auto max-w-4xl px-4 py-24 text-center sm:py-32">
    <span class="rounded-full bg-indigo-100 px-3 py-1 text-xs font-semibold uppercase tracking-wider text-indigo-700">New release</span>
    <h1 class="mt-6 text-4xl font-bold tracking-tight text-balance text-gray-900 sm:text-6xl">
      Build beautiful interfaces, faster
    </h1>
    <p class="mx-auto mt-6 max-w-2xl text-lg leading-8 text-pretty text-gray-600">
      A short supporting sentence that explains the product value in plain language.
    </p>
    <div class="mt-10 flex flex-col items-center justify-center gap-4 sm:flex-row">
      <a href="#" class="w-full rounded-lg bg-indigo-600 px-6 py-3 font-semibold text-white shadow-sm hover:bg-indigo-500 sm:w-auto">Get started</a>
      <a href="#" class="font-semibold text-gray-900">Learn more <span aria-hidden="true">&arr;</span></a>
    </div>
  </div>
</section>
```

Important classes	Why
<code>bg-linear-to-b from-indigo-50 to-white</code>	soft vertical gradient
<code>text-4xl sm:text-6xl</code>	mobile-first heading size
<code>tracking-tight text-balance</code>	polished headline wrapping
<code>mx-auto max-w-2xl</code>	centered paragraph with a readable width
<code>flex-col sm:flex-row</code>	buttons stack on phones, sit side by side after 640px

31.3 Profile Card

```
<div class="mx-auto w-full max-w-sm overflow-hidden rounded-2xl bg-white shadow-lg">
  <div class="h-28 bg-linear-to-r from-blue-500 to-purple-600"></div>
  <div class="px-6 pb-6">
    
    <h3 class="mt-3 text-xl font-bold text-gray-900">Ada Lovelace</h3>
    <p class="text-sm text-gray-500">Frontend Engineer</p>
    <p class="mt-3 text-gray-600">Building accessible interfaces with React and Tailwind.</p>
    <div class="mt-5 flex divide-x divide-gray-200 text-center">
      <div class="flex-1"><p class="font-bold">1.2k</p><p class="text-xs text-gray-500">Followers</p></div>
      <div class="flex-1"><p class="font-bold">340</p><p class="text-xs text-gray-500">Following</p></div>
      <div class="flex-1"><p class="font-bold">87</p><p class="text-xs text-gray-500">Posts</p></div>
    </div>
    <button class="mt-5 w-full rounded-lg bg-blue-600 py-2 font-medium text-white hover:bg-blue-700">Follow</button>
  </div>
</div>
```

Important classes	Why
<code>overflow-hidden rounded-2xl</code>	clips the banner to the rounded corners
<code>-mt-12 ... border-4 border-white</code>	avatar overlaps the banner with a white outline
<code>size-24 rounded-full object-cover</code>	perfect circle, image cropped not squashed
<code>flex divide-x</code>	equal stat columns with vertical separators

31.4 Product Card

```
<div class="group w-full max-w-xs overflow-hidden rounded-xl border border-gray-200 bg-white transition hover:shadow-xl">
  <div class="relative aspect-square overflow-hidden bg-gray-100">
    
    <span class="absolute left-3 top-3 rounded-full bg-red-500 px-2 py-0.5 text-xs font-semibold text-white">-20%</span>
  </div>
  <div class="space-y-2 p-4">
    <h3 class="font-semibold text-gray-900">Aero Runner</h3>
    <p class="line-clamp-2 text-sm text-gray-500">Lightweight running shoe with breathable mesh and responsive foam.</p>
    <div class="flex items-center justify-between pt-2">
      <p><span class="text-lg font-bold text-gray-900">$96</span> <span class="text-sm text-gray-400 line-through">$120</span></p>
      <button class="rounded-lg bg-gray-900 px-3 py-1.5 text-sm font-medium text-white hover:bg-gray-700">Add to cart</button>
    </div>
  </div>
</div>
```

Important classes	Why
<code>group + group-hover:scale-105</code>	image zooms when the whole card is hovered
<code>aspect-square overflow-hidden</code>	consistent square image box that clips the zoom
<code>absolute left-3 top-3</code>	badge pinned to the image corner (parent is <code>relative</code>)
<code>line-clamp-2</code>	description limited to two lines
<code>line-through</code>	struck-through old price

31.5 Login Form

```
<div class="grid min-h-dvh place-items-center bg-gray-50 px-4">
  <form class="w-full max-w-sm space-y-5 rounded-2xl bg-white p-8 shadow-lg">
    <div class="text-center">
      <h1 class="text-2xl font-bold text-gray-900">Welcome back</h1>
      <p class="mt-1 text-sm text-gray-500">Log in to your account</p>
    </div>

    <div>
      <label for="email" class="mb-1 block text-sm font-medium text-gray-700">Email</label>
      <input id="email" type="email" required placeholder="you@example.com"
        class="w-full rounded-lg border border-gray-300 px-3 py-2 placeholder:text-gray-400
        focus:border-blue-500 focus:outline-hidden focus:ring-3 focus:ring-blue-500/30" />
    </div>

    <div>
      <div class="mb-1 flex items-center justify-between">
        <label for="password" class="text-sm font-medium text-gray-700">Password</label>
        <a href="#" class="text-sm text-blue-600 hover:underline">Forgot?</a>
      </div>
      <input id="password" type="password" required
        class="w-full rounded-lg border border-gray-300 px-3 py-2
        focus:border-blue-500 focus:outline-hidden focus:ring-3 focus:ring-blue-500/30" />
    </div>

    <label class="flex items-center gap-2 text-sm text-gray-600"><input type="checkbox"
    class="size-4 rounded accent-blue-600" /> Remember me</label>

    <button
    class="w-full rounded-lg bg-blue-600 py-2.5 font-semibold text-white transition hover:bg-blue-700 focus-
    visible:ring-2 focus-visible:ring-blue-500 focus-visible:ring-offset-2">Log in</button>

    <p class="text-center text-sm text-gray-500">No account? <a href="#" class="font-medium text-blue-600
    hover:underline">Sign up</a></p>
  </form>
</div>
```

Important classes	Why
<code>grid min-h-dvh place-items-center</code>	centers the form in the viewport
<code>max-w-sm w-full</code>	fluid on phones, capped on desktop
<code>space-y-5</code>	consistent vertical rhythm between fields
<code>focus:ring-3 focus:ring-blue-500/30 focus:outline-hidden</code>	soft visible focus ring replaces the outline
<code>flex justify-between</code>	label left, link right on one row

31.6 Signup Form

```
<form class="mx-auto w-full max-w-md space-y-5 rounded-2xl bg-white p-8 shadow-lg">
  <h1 class="text-2xl font-bold text-gray-900">Create your account</h1>

  <div class="grid grid-cols-1 gap-4 sm:grid-cols-2">
    <div>
      <label for="first" class="mb-1 block text-sm font-medium text-gray-700">First name</label>
      <input id="first" class="w-full rounded-lg border border-gray-300 px-3 py-2 focus:border-blue-500 focus:outline-hidden focus:ring-3 focus:ring-blue-500/30" />
    </div>
    <div>
      <label for="last" class="mb-1 block text-sm font-medium text-gray-700">Last name</label>
      <input id="last" class="w-full rounded-lg border border-gray-300 px-3 py-2 focus:border-blue-500 focus:outline-hidden focus:ring-3 focus:ring-blue-500/30" />
    </div>
  </div>

  <div>
    <label for="mail" class="mb-1 block text-sm font-medium text-gray-700">Email</label>
    <input id="mail" type="email" required class="peer w-full rounded-lg border border-gray-300 px-3 py-2 focus:border-blue-500 focus:outline-hidden focus:ring-3 focus:ring-blue-500/30 user-invalid:border-red-500" />
    <p class="mt-1 hidden text-sm text-red-600 peer-user-invalid:block">Enter a valid email address.</p>
  </div>

  <div>
    <label for="pw" class="mb-1 block text-sm font-medium text-gray-700">Password</label>
    <input id="pw" type="password" minlength="8" required class="w-full rounded-lg border border-gray-300 px-3 py-2 focus:border-blue-500 focus:outline-hidden focus:ring-3 focus:ring-blue-500/30" />
    <p class="mt-1 text-xs text-gray-500">At least 8 characters.</p>
  </div>

  <button class="w-full rounded-lg bg-blue-600 py-2.5 font-semibold text-white hover:bg-blue-700">Sign up</button>
</form>
```

Important classes	Why
<code>grid grid-cols-1 sm:grid-cols-2 gap-4</code>	two name fields side by side from 640px
<code>peer + peer-user-invalid:block</code>	error message appears only after the user typed something invalid
<code>user-invalid:border-red-500</code>	red border after interaction (v4.1+; use <code>invalid:</code> on older v4)

31.7 Button Variants

```
<!-- Primary -->
<button
  class="rounded-lg bg-blue-600 px-4 py-2 font-medium text-white shadow-sm transition hover:bg-blue-700 active:scale-95 focus-visible:ring-2 focus-visible:ring-blue-500 focus-visible:ring-offset-2 disabled:opacity-50">Primary</button>

<!-- Secondary -->
<button class="rounded-lg bg-gray-100 px-4 py-2 font-medium text-gray-900 transition hover:bg-gray-200">Secondary</button>

<!-- Outline -->
<button class="rounded-lg border border-blue-600 px-4 py-2 font-medium text-blue-600 transition hover:bg-blue-50">Outline</button>

<!-- Ghost -->
<button class="rounded-lg px-4 py-2 font-medium text-gray-700 transition hover:bg-gray-100">Ghost</button>

<!-- Danger -->
```

```

<button class="rounded-lg bg-red-600 px-4 py-2 font-medium text-white transition hover:bg-red-700">Delete</button>

<!-- Gradient -->
<button class="rounded-lg bg-linear-to-r from-indigo-500 to-purple-600 px-5 py-2.5 font-semibold text-white shadow-md transition hover:brightness-110">Gradient</button>

<!-- Pill -->
<button class="rounded-full bg-gray-900 px-6 py-2 text-sm font-medium text-white hover:bg-gray-700">Pill</button>

<!-- Icon only -->
<button class="grid size-10 place-items-center rounded-full hover:bg-gray-100" aria-label="Like"><svg class="size-5">...</svg></button>

<!-- Link style -->
<button class="font-medium text-blue-600 underline-offset-4 hover:underline">Link button</button>

<!-- Sizes: small / medium / large -->
<button class="rounded-md px-3 py-1.5 text-sm">Small</button>
<button class="rounded-lg px-4 py-2 text-base">Medium</button>
<button class="rounded-xl px-6 py-3 text-lg">Large</button>

<!-- Loading -->
<button disabled
class="inline-flex items-center gap-2 rounded-lg bg-blue-600 px-4 py-2 text-white opacity-70 cursor-not-allowed">
  <svg class="size-4 animate-spin">...</svg> Saving...
</button>

```

Important classes	Why
<code>px-4 py-2</code>	standard button padding (wider than tall)
<code>transition hover:... active:scale-95</code>	feedback on hover and press
<code>focus-visible:ring-2 ... ring-offset-2</code>	keyboard focus indicator
<code>disabled:opacity-50</code>	disabled look

31.8 Modal

```

<!-- Backdrop -->
<div class="fixed inset-0 z-50 flex items-center justify-center bg-black/50 p-4 backdrop-blur-sm" role="dialog" aria-modal="true" aria-labelledby="modal-title">
  <!-- Panel -->
  <div class="w-full max-w-md rounded-2xl bg-white p-6 shadow-2xl">
    <div class="flex items-start justify-between">
      <h2 id="modal-title" class="text-lg font-semibold text-gray-900">Delete project?</h2>
      <button class="rounded-lg p-1 text-gray-400 hover:bg-gray-100 hover:text-gray-600" aria-label="Close"><svg class="size-5">...</svg></button>
    </div>
    <p class="mt-3 text-sm text-gray-600">This action cannot be undone. All data will be permanently removed.</p>
    <div class="mt-6 flex flex-col-reverse gap-3 sm:flex-row sm:justify-end">
      <button class="rounded-lg border border-gray-300 px-4 py-2 font-medium text-gray-700 hover:bg-gray-50">Cancel</button>
      <button class="rounded-lg bg-red-600 px-4 py-2 font-medium text-white hover:bg-red-700">Delete</button>
    </div>
  </div>
</div>

```

Important classes	Why
<code>fixed inset-0 z-50</code>	covers the whole screen above everything
<code>flex items-center justify-center</code>	centers the panel
<code>bg-black/50 backdrop-blur-sm</code>	dim and blur the page behind
<code>max-w-md w-full p-4</code> (outer)	panel never touches screen edges on phones
<code>flex-col-reverse sm:flex-row sm:justify-end</code>	primary action on top on mobile, right-aligned row on desktop

In React, render it conditionally: `{open && <Modal onClose={...} />}`. Add `overflow-hidden` to `<body>` while open to stop background scrolling.

31.9 Dropdown

```
<div class="relative inline-block text-left">
  <button class="inline-flex items-center gap-2 rounded-lg border border-gray-300 bg-white px-4 py-2 text-sm font-medium hover:bg-gray-50" aria-expanded="true" aria-haspopup="menu">
    Options <svg class="size-4">...</svg>
  </button>

  <div
    class="absolute right-0 z-20 mt-2 w-48 origin-top-right rounded-xl border border-gray-200 bg-white p-1 shadow-lg" role="menu">
    <a href="#" class="block rounded-lg px-3 py-2 text-sm text-gray-700 hover:bg-gray-100" role="menuitem">Profile</a>
    <a href="#" class="block rounded-lg px-3 py-2 text-sm text-gray-700 hover:bg-gray-100" role="menuitem">Settings</a>
    <div class="my-1 border-t border-gray-100"></div>
    <a href="#" class="block rounded-lg px-3 py-2 text-sm text-red-600 hover:bg-red-50" role="menuitem">Sign out</a>
  </div>
</div>
```

Important classes	Why
<code>relative</code> (wrapper) + <code>absolute right-0 mt-2</code> (menu)	menu is anchored under the button
<code>origin-top-right</code>	scale animations grow from the corner
<code>z-20</code>	menu sits above surrounding content
<code>block rounded-lg px-3 py-2 hover:bg-gray-100</code>	large, comfortable click target

CSS-only hover version: add `group` to the wrapper and use `invisible opacity-0 group-hover:visible group-hover:opacity-100 transition` on the menu. In React, toggle `hidden` with state.

31.10 Sidebar

```
<aside class="flex h-dvh w-64 shrink-0 flex-col border-r border-gray-200 bg-white">
  <div class="flex h-16 items-center px-6 text-xl font-bold text-blue-600">Dashboard</div>

  <nav class="flex-1 space-y-1 overflow-y-auto px-3 py-4">
    <a href="#" class="flex items-center gap-3 rounded-lg bg-blue-50 px-3 py-2 text-sm font-medium text-blue-700"><svg class="size-5">...</svg> Overview</a>
    <a href="#" class="flex items-center gap-3 rounded-lg px-3 py-2 text-sm font-medium text-gray-600 hover:bg-gray-100 hover:text-gray-900"><svg class="size-5">...</svg> Projects</a>
    <a href="#" class="flex items-center gap-3 rounded-lg px-3 py-2 text-sm font-medium text-gray-600 hover:bg-gray-100 hover:text-gray-900"><svg class="size-5">...</svg> Team</a>
  </nav>
```

```

<div class="border-t border-gray-200 p-4">
  <div class="flex items-center gap-3">
    
    <div class="min-w-0"><p class="truncate text-sm font-medium">Ada Lovelace</p><p
class="truncate text-xs text-gray-500">ada@example.com</p></div>
  </div>
</div>
</aside>

```

Important classes	Why
<code>flex h-dvh w-64 shrink-0 flex-col</code>	fixed-width full-height column that never gets squeezed
<code>flex-1 overflow-y-auto</code> (nav)	nav fills leftover space and scrolls alone
<code>bg-blue-50 text-blue-700</code> vs <code>hover:bg-gray-100</code>	active versus inactive link styles
<code>min-w-0 truncate</code>	long emails get an ellipsis inside flex

Mobile off-canvas version: `fixed inset-y-0 left-0 z-40 -translate-x-full transition-transform lg:static lg:translate-x-0`, and toggle `translate-x-0` in React state.

31.11 Dashboard Layout

```

<div class="flex min-h-dvh bg-gray-100">
  <aside class="hidden w-64 shrink-0 bg-gray-900 text-gray-200 lg:block">Sidebar</aside>

  <div class="flex min-w-0 flex-1 flex-col">
    <header class="sticky top-0 z-10 flex h-16 items-center justify-between border-b bg-white px-6">
      <h1 class="text-lg font-semibold">Overview</h1>
      
    </header>

    <main class="flex-1 space-y-6 p-6">
      <!-- stat cards -->
      <div class="grid grid-cols-1 gap-4 sm:grid-cols-2 xl:grid-cols-4">
        <div class="rounded-xl bg-white p-5 shadow-sm">
          <p class="text-sm text-gray-500">Revenue</p>
          <p class="mt-1 text-3xl font-bold tabular-nums">$48,200</p>
          <p class="mt-1 text-sm font-medium text-green-600">+12.5%</p>
        </div>
        <!-- 3 more cards -->
      </div>

      <!-- chart + side panel -->
      <div class="grid gap-6 lg:grid-cols-3">
        <section class="rounded-xl bg-white p-6 shadow-sm lg:col-span-2">Chart</section>
        <section class="rounded-xl bg-white p-6 shadow-sm">Recent activity</section>
      </div>
    </main>
  </div>
</div>

```

Important classes	Why
<code>flex min-h-dvh</code>	sidebar + content fill the viewport
<code>min-w-0 flex-1</code>	content area can shrink; tables cannot blow out the layout
<code>hidden lg:block</code>	sidebar hidden below 1024px
<code>grid-cols-1 sm:grid-cols-2 xl:grid-cols-4</code>	stat cards scale 1, 2, 4 columns
<code>lg:col-span-2</code>	chart takes two thirds on large screens

31.12 Pricing Card

```
<div class="grid max-w-5xl gap-8 md:grid-cols-3">
  <!-- Regular plan -->
  <div class="flex flex-col rounded-2xl border border-gray-200 bg-white p-8">
    <h3 class="text-lg font-semibold text-gray-900">Starter</h3>
    <p class="mt-4"><span class="text-4xl font-bold tracking-tight">$9</span><span class="text-gray-500">/
month</span></p>
    <ul class="mt-6 flex-1 space-y-3 text-sm text-gray-600">
      <li class="flex gap-3"><svg class="size-5 shrink-0 text-green-500">...</svg>1 project</li>
      <li class="flex gap-3"><svg class="size-5 shrink-0 text-green-500">...</svg>Basic analytics</li>
    </ul>
    <button class="mt-8 rounded-lg border border-blue-600 py-2.5 font-semibold text-blue-600 hover:bg-
blue-50">Choose Starter</button>
  </div>

  <!-- Highlighted plan -->
  <div class="relative flex flex-col rounded-2xl bg-blue-600 p-8 text-white shadow-xl ring-1 ring-blue-600
md:-translate-y-4">
    <span class="absolute -top-3 left-1/2 -translate-x-1/2 rounded-full bg-amber-400 px-3 py-0.5 text-xs
font-bold uppercase text-gray-900">Most popular</span>
    <h3 class="text-lg font-semibold">Pro</h3>
    <p class="mt-4"><span class="text-4xl font-bold tracking-tight">$29</span><span class="text-blue-100">/
month</span></p>
    <ul class="mt-6 flex-1 space-y-3 text-sm text-blue-50">
      <li class="flex gap-3"><svg class="size-5 shrink-0">...</svg>Unlimited projects</li>
      <li class="flex gap-3"><svg class="size-5 shrink-0">...</svg>Advanced analytics</li>
    </ul>
    <button class="mt-8 rounded-lg bg-white py-2.5 font-semibold text-blue-600 hover:bg-blue-50">Choose Pro</
button>
  </div>

  <!-- Third plan repeats the first style -->
</div>
```

Important classes	Why
<code>flex flex-col</code> + <code>flex-1</code> on the list	buttons align at the bottom of every card
<code>md:-translate-y-4</code>	raised highlighted plan on desktop
<code>absolute -top-3 left-1/2 -translate-x-1/2</code>	ribbon centered on the top edge
<code>shrink-0</code> on icons	check icons never get squished

31.13 Alert

```
<!-- Info -->
<div class="flex gap-3 rounded-lg border border-blue-200 bg-blue-50 p-4 text-sm text-blue-800" role="alert">
  <svg class="mt-0.5 size-5 shrink-0">...</svg>
  <div><p class="font-semibold">Heads up</p><p class="mt-1">Your trial ends in 3 days.</p></div>
</div>

<!-- Success -->
<div class="flex gap-3 rounded-lg border border-green-200 bg-green-50 p-4 text-sm text-green-800" role="alert">
  <p>Saved successfully.</p>
</div>

<!-- Warning -->
<div class="flex gap-3 rounded-lg border border-yellow-200 bg-yellow-50 p-4 text-sm text-yellow-800" role="al
ert">Storage almost full.</div>

<!-- Error -->
<div class="flex gap-3 rounded-lg border border-red-200 bg-red-50 p-4 text-sm text-red-800" role="alert">Paym
ent failed.</div>
```

```
<!-- Left accent -->
<div class="rounded-r-lg border-l-4 border-blue-500 bg-blue-50 p-4 text-blue-800">Accent alert</div>
```

Pattern: `bg-{color}-50` + `border-{color}-200` + `text-{color}-800` . Swap the color word for each status.

31.14 Badge

```
<span class="inline-flex items-center rounded-full bg-green-100 px-2.5 py-0.5 text-xs font-medium text-green-800">Active</span>
<span class="inline-flex items-center rounded-full bg-red-100 px-2.5 py-0.5 text-xs font-medium text-red-800">Failed</span>
<span class="inline-flex items-center rounded-md bg-gray-100 px-2 py-1 text-xs font-medium text-gray-600 ring-1 ring-inset ring-gray-500/10">Draft</span>

<!-- With dot -->
<span class="inline-flex items-center gap-1.5 rounded-full bg-emerald-50 px-2.5 py-1 text-xs font-medium text-emerald-700">
  <span class="size-1.5 rounded-full bg-emerald-500"></span> Online
</span>

<!-- Notification count on an icon -->
<div class="relative inline-block">
  <svg class="size-6">...</svg>
  <span class="absolute -right-1 -top-1 grid size-4 place-items-center rounded-full bg-red-500 text-[10px] font-bold text-white">3</span>
</div>
```

31.15 Loading Spinner

```
<!-- Border spinner -->
<div class="size-8 animate-spin rounded-full border-4 border-gray-200 border-t-blue-600" role="status" aria-label="Loading"></div>

<!-- SVG spinner (inherits text color) -->
<svg class="size-6 animate-spin text-blue-600" viewBox="0 0 24 24" fill="none" role="status" aria-label="Loading">
  <circle class="opacity-25" cx="12" cy="12" r="10" stroke="currentColor" stroke-width="4" />
  <path class="opacity-75" fill="currentColor" d="M4 12a8 8 0 018-8v4a4 4 0 00-4 4H4" />
</svg>

<!-- Full-page loader -->
<div class="fixed inset-0 grid place-items-center bg-white/70 backdrop-blur-sm">
  <div class="size-10 animate-spin rounded-full border-4 border-blue-200 border-t-blue-600"></div>
</div>

<!-- Bouncing dots -->
<div class="flex gap-1">
  <span class="size-2 animate-bounce rounded-full bg-blue-600 [animation-delay:-0.3s]"></span>
  <span class="size-2 animate-bounce rounded-full bg-blue-600 [animation-delay:-0.15s]"></span>
  <span class="size-2 animate-bounce rounded-full bg-blue-600"></span>
</div>
```

The border trick: a gray border all around, and a **colored top border** (`border-t-blue-600`) that spins.

31.16 Skeleton Loader

```
<div class="w-full max-w-sm animate-pulse rounded-xl border border-gray-200 p-4">
  <div class="h-40 rounded-lg bg-gray-200"></div>
  <div class="mt-4 space-y-3">
    <div class="h-4 w-3/4 rounded bg-gray-200"></div>
    <div class="h-4 rounded bg-gray-200"></div>
  </div>
```

```

    <div class="h-4 w-5/6 rounded bg-gray-200"></div>
  </div>
  <div class="mt-4 flex items-center gap-3">
    <div class="size-10 rounded-full bg-gray-200"></div>
    <div class="h-3 w-24 rounded bg-gray-200"></div>
  </div>
</div>

```

`animate-pulse` on the parent fades every gray block together. Match block sizes to the real content to prevent layout shift. In React: `{loading ? <Skeleton /> : <Card />}`.

31.17 Image Gallery

```

<!-- Uniform grid -->
<div class="grid grid-cols-2 gap-3 md:grid-cols-3 lg:grid-cols-4">
  <div class="group overflow-hidden rounded-xl bg-gray-100">
    
  </div>
  <!-- repeat -->
</div>

<!-- Masonry-style using CSS columns -->
<div class="columns-2 gap-3 md:columns-3 lg:columns-4">
  
  
</div>

<!-- Featured image spanning cells -->
<div class="grid grid-cols-3 gap-3">
  
  
  
</div>

<!-- Overlay caption on hover -->
<div class="group relative overflow-hidden rounded-xl">
  
  <div class="absolute inset-0 flex items-end bg-linear-to-t from-black/70 to-transparent p-4 opacity-0 transition-opacity group-hover:opacity-100">
    <p class="font-medium text-white">Caption</p>
  </div>
</div>

```

31.18 Responsive Grid

```

<!-- Fixed breakpoints -->
<div class="grid grid-cols-1 gap-6 sm:grid-cols-2 lg:grid-cols-3 xl:grid-cols-4">
  <article class="rounded-xl border border-gray-200 bg-white p-6 shadow-sm">...</article>
</div>

<!-- Auto-fit (no breakpoints) -->
<div class="grid grid-cols-[repeat(auto-fit,minmax(16rem,1fr))] gap-6">...</div>

<!-- Container-query version: reacts to the parent's width -->
<div class="@container">
  <div class="grid grid-cols-1 gap-4 @md:grid-cols-2 @3xl:grid-cols-4">...</div>
</div>

```

31.19 Footer

```
<footer class="bg-gray-900 text-gray-400">
  <div class="mx-auto max-w-7xl px-4 py-12 sm:px-6 lg:px-8">
    <div class="grid grid-cols-2 gap-8 md:grid-cols-4">
      <div class="col-span-2 md:col-span-1">
        <p class="text-xl font-bold text-white">Brand</p>
        <p class="mt-3 text-sm">Short company description goes here.</p>
      </div>
      <div>
        <h4 class="text-sm font-semibold uppercase tracking-wider text-white">Product</h4>
        <ul class="mt-4 space-y-2 text-sm">
          <li><a href="#" class="transition-colors hover:text-white">Features</a></li>
          <li><a href="#" class="transition-colors hover:text-white">Pricing</a></li>
        </ul>
      </div>
      <div>
        <h4 class="text-sm font-semibold uppercase tracking-wider text-white">Company</h4>
        <ul class="mt-4 space-y-2 text-sm">
          <li><a href="#" class="transition-colors hover:text-white">About</a></li>
          <li><a href="#" class="transition-colors hover:text-white">Contact</a></li>
        </ul>
      </div>
      <div>
        <h4 class="text-sm font-semibold uppercase tracking-wider text-white">Legal</h4>
        <ul class="mt-4 space-y-2 text-sm">
          <li><a href="#" class="transition-colors hover:text-white">Privacy</a></li>
          <li><a href="#" class="transition-colors hover:text-white">Terms</a></li>
        </ul>
      </div>
    </div>
    <div class="mt-10 flex flex-col items-center justify-between gap-4 border-t border-gray-800 pt-8 text-sm sm:flex-row">
      <p>&copy; 2026 Brand. All rights reserved.</p>
      <div class="flex gap-4"><a href="#" class="hover:text-white">Twitter</a><a href="#" class="hover:text-white">GitHub</a></div>
    </div>
  </div>
</footer>
```

Important classes	Why
<code>grid-cols-2 md:grid-cols-4 + col-span-2 md:col-span-1</code>	brand block spans the full width on mobile
<code>text-gray-400 hover:text-white</code>	muted links that brighten on hover
<code>flex-col sm:flex-row justify-between</code>	copyright and socials stack on mobile
<code>border-t border-gray-800</code>	subtle divider on dark background

32. CSS → Tailwind Conversion Table

Find the CSS you know, read the Tailwind class on the right. **n** means "a number from the spacing scale" (n x 4px).

32.1 Display and Layout

CSS	Tailwind	Notes
display: block	block	
display: inline	inline	
display: inline-block	inline-block	
display: flex	flex	
display: inline-flex	inline-flex	
display: grid	grid	
display: inline-grid	inline-grid	
display: none	hidden	
display: contents	contents	
display: table	table	also table-row, table-cell
display: flow-root	flow-root	
visibility: hidden	invisible	keeps space
visibility: visible	visible	
float: left / right / none	float-left / float-right / float-none	
clear: both	clear-both	
box-sizing: border-box	box-border	default in Tailwind
box-sizing: content-box	box-content	
columns: 3	columns-3	
isolation: isolate	isolate	

32.2 Spacing and Sizing

CSS	Tailwind	Notes
margin: 1rem	m-4	
margin: 0 auto	mx-auto	center a block
margin: 0 1rem	mx-4	
margin: 1rem 0	my-4	
margin-top: 1.5rem	mt-6	
margin-right: 0.5rem	mr-2	
margin-bottom: 2rem	mb-8	
margin-left: auto	ml-auto	push right
margin: -1rem	-m-4	negative
padding: 1rem	p-4	
padding: 0.5rem 1rem	py-2 px-4	
padding-top: 1rem	pt-4	
padding-left: 2rem	pl-8	
gap: 1rem	gap-4	flex/grid
column-gap: 1.5rem	gap-x-6	
row-gap: 0.5rem	gap-y-2	
width: 100%	w-full	
width: 50%	w-1/2	
width: 100vw	w-screen	
width: auto	w-auto	
width: fit-content	w-fit	
width: 16rem	w-64	
width: 350px	w-[350px]	arbitrary
height: 100%	h-full	

CSS	Tailwind	Notes
height: 100vh	h-screen	
height: 100dvh	h-dvh	
height: 3rem	h-12	
width: 2.5rem; height: 2.5rem	size-10	
min-width: 0	min-w-0	
max-width: 28rem	max-w-md	
max-width: 80rem	max-w-7xl	
max-width: 100%	max-w-full	
min-height: 100vh	min-h-screen	
max-height: 24rem	max-h-96	
aspect-ratio: 16 / 9	aspect-video	
aspect-ratio: 1 / 1	aspect-square	

32.3 Flexbox

CSS	Tailwind	Notes
flex-direction: row	flex-row	default
flex-direction: column	flex-col	
flex-direction: row-reverse	flex-row-reverse	
flex-wrap: wrap	flex-wrap	
flex-wrap: nowrap	flex-nowrap	
justify-content: flex-start	justify-start	
justify-content: center	justify-center	
justify-content: flex-end	justify-end	
justify-content: space-between	justify-between	
justify-content: space-around	justify-around	
justify-content: space-evenly	justify-evenly	
align-items: flex-start	items-start	
align-items: center	items-center	
align-items: flex-end	items-end	
align-items: baseline	items-baseline	

CSS	Tailwind	Notes
align-items: stretch	items-stretch	
align-content: center	content-center	
align-self: center	self-center	
align-self: flex-end	self-end	
flex: 1 1 0%	flex-1	
flex: 1 1 auto	flex-auto	
flex: 0 1 auto	flex-initial	
flex: none	flex-none	
flex-grow: 1	grow	
flex-grow: 0	grow-0	
flex-shrink: 0	shrink-0	
flex-basis: 50%	basis-1/2	
order: 1	order-1	
order: -9999	order-first	

32.4 Grid

CSS	Tailwind	Notes
grid-template-columns: repeat(3, minmax(0, 1fr))	grid-cols-3	
grid-template-columns: 200px 1fr	grid-cols-[200px_1fr]	arbitrary
grid-template-rows: repeat(2, minmax(0, 1fr))	grid-rows-2	
grid-column: span 2 / span 2	col-span-2	
grid-column: 1 / -1	col-span-full	
grid-column-start: 2	col-start-2	
grid-row: span 2 / span 2	row-span-2	
grid-auto-flow: column	grid-flow-col	
grid-auto-rows: min-content	auto-rows-min	
justify-items: center	justify-items-center	
place-items: center	place-items-center	
place-content: center	place-content-center	

CSS	Tailwind	Notes
<code>place-self: center</code>	<code>place-self-center</code>	

32.5 Position

CSS	Tailwind	Notes
<code>position: static</code>	<code>static</code>	
<code>position: relative</code>	<code>relative</code>	
<code>position: absolute</code>	<code>absolute</code>	
<code>position: fixed</code>	<code>fixed</code>	
<code>position: sticky</code>	<code>sticky</code>	
<code>top: 0</code>	<code>top-0</code>	
<code>right: 0</code>	<code>right-0</code>	
<code>bottom: 1rem</code>	<code>bottom-4</code>	
<code>left: 50%</code>	<code>left-1/2</code>	
<code>top: 0; right: 0; bottom: 0; left: 0</code>	<code>inset-0</code>	
<code>left: 0; right: 0</code>	<code>inset-x-0</code>	
<code>top: 0; bottom: 0</code>	<code>inset-y-0</code>	
<code>top: -0.5rem</code>	<code>-top-2</code>	
<code>z-index: 10</code>	<code>z-10</code>	
<code>z-index: 9999</code>	<code>z-[9999]</code>	

32.6 Typography

CSS	Tailwind	Notes
<code>font-family: sans-serif</code>	<code>font-sans</code>	
<code>font-family: serif</code>	<code>font-serif</code>	
<code>font-family: monospace</code>	<code>font-mono</code>	
<code>font-size: 0.875rem</code>	<code>text-sm</code>	
<code>font-size: 1rem</code>	<code>text-base</code>	
<code>font-size: 1.25rem</code>	<code>text-xl</code>	
<code>font-size: 1.5rem</code>	<code>text-2xl</code>	
<code>font-size: 24px</code>	<code>text-2xl</code>	(1.5rem)
<code>font-size: 22px</code>	<code>text-[22px]</code>	arbitrary

CSS	Tailwind	Notes
<code>font-weight: 300</code>	<code>font-light</code>	
<code>font-weight: 400</code>	<code>font-normal</code>	
<code>font-weight: 500</code>	<code>font-medium</code>	
<code>font-weight: 600</code>	<code>font-semibold</code>	
<code>font-weight: 700</code>	<code>font-bold</code>	
<code>font-weight: 900</code>	<code>font-black</code>	
<code>font-style: italic</code>	<code>italic</code>	
<code>font-style: normal</code>	<code>not-italic</code>	
<code>line-height: 1</code>	<code>leading-none</code>	
<code>line-height: 1.25</code>	<code>leading-tight</code>	
<code>line-height: 1.5</code>	<code>leading-normal</code>	
<code>line-height: 1.625</code>	<code>leading-relaxed</code>	
<code>line-height: 2rem</code>	<code>leading-8</code>	
<code>letter-spacing: -0.025em</code>	<code>tracking-tight</code>	
<code>letter-spacing: 0.05em</code>	<code>tracking-wider</code>	
<code>text-align: left / center / right / justify</code>	<code>text-left / text-center / text-right / text-justify</code>	
<code>text-transform: uppercase</code>	<code>uppercase</code>	
<code>text-transform: capitalize</code>	<code>capitalize</code>	
<code>text-transform: none</code>	<code>normal-case</code>	
<code>text-decoration: underline</code>	<code>underline</code>	
<code>text-decoration: line-through</code>	<code>line-through</code>	
<code>text-decoration: none</code>	<code>no-underline</code>	
<code>text-underline-offset: 4px</code>	<code>underline-offset-4</code>	
<code>text-overflow: ellipsis</code>	<code>text-ellipsis</code>	
<code>overflow: hidden; text-overflow: ellipsis; white-space: nowrap</code>	<code>truncate</code>	
<code>white-space: nowrap</code>	<code>whitespace-nowrap</code>	
<code>white-space: pre-wrap</code>	<code>whitespace-pre-wrap</code>	
<code>word-break: break-all</code>	<code>break-all</code>	
<code>overflow-wrap: break-word</code>	<code>break-words</code>	

CSS	Tailwind	Notes
<code>text-wrap: balance</code>	<code>text-balance</code>	
<code>vertical-align: middle</code>	<code>align-middle</code>	
<code>text-indent: 1rem</code>	<code>indent-4</code>	
<code>font-variant-numeric: tabular-nums</code>	<code>tabular-nums</code>	
<code>-webkit-line-clamp: 3</code>	<code>line-clamp-3</code>	
<code>list-style-type: disc</code>	<code>list-disc</code>	
<code>list-style-type: none</code>	<code>list-none</code>	

32.7 Colors and Backgrounds

CSS	Tailwind	Notes
<code>color: #3b82f6</code>	<code>text-blue-500</code>	
<code>color: white</code>	<code>text-white</code>	
<code>color: rgb(0 0 0 / 0.5)</code>	<code>text-black/50</code>	
<code>color: #1e293b</code>	<code>text-[#1e293b]</code>	arbitrary
<code>color: currentColor</code>	<code>text-current</code>	
<code>background-color: #3b82f6</code>	<code>bg-blue-500</code>	
<code>background-color: transparent</code>	<code>bg-transparent</code>	
<code>background-color: rgb(0 0 0 / 0.5)</code>	<code>bg-black/50</code>	
<code>background-image: url(...)</code>	<code>bg-[url(/img.jpg)]</code>	
<code>background-image: none</code>	<code>bg-none</code>	
<code>background-image: linear-gradient(to right, a, b)</code>	<code>bg-linear-to-r from-a to-b</code>	
<code>background-size: cover</code>	<code>bg-cover</code>	
<code>background-size: contain</code>	<code>bg-contain</code>	
<code>background-position: center</code>	<code>bg-center</code>	
<code>background-repeat: no-repeat</code>	<code>bg-no-repeat</code>	
<code>background-attachment: fixed</code>	<code>bg-fixed</code>	
<code>background-clip: text</code>	<code>bg-clip-text</code>	
<code>opacity: 0.5</code>	<code>opacity-50</code>	
<code>accent-color: #3b82f6</code>	<code>accent-blue-500</code>	
<code>caret-color: #3b82f6</code>	<code>caret-blue-500</code>	

CSS	Tailwind	Notes
<code>fill: currentColor</code>	<code>fill-current</code>	
<code>stroke: currentColor</code>	<code>stroke-current</code>	
<code>mix-blend-mode: multiply</code>	<code>mix-blend-multiply</code>	

32.8 Borders, Radius, Shadows

CSS	Tailwind	Notes
<code>border: 1px solid</code>	<code>border</code>	add a color
<code>border: 2px solid #e5e7eb</code>	<code>border-2 border-gray-200</code>	
<code>border-width: 0</code>	<code>border-0</code>	
<code>border-top: 1px solid</code>	<code>border-t</code>	
<code>border-bottom: 2px solid</code>	<code>border-b-2</code>	
<code>border-left: 4px solid</code>	<code>border-l-4</code>	
<code>border-left-width + border-right-width</code>	<code>border-x</code>	
<code>border-color: #e5e7eb</code>	<code>border-gray-200</code>	
<code>border-style: dashed</code>	<code>border-dashed</code>	
<code>border-style: dotted</code>	<code>border-dotted</code>	
<code>border-radius: 4px</code>	<code>rounded-sm</code>	
<code>border-radius: 6px</code>	<code>rounded-md</code>	
<code>border-radius: 8px</code>	<code>rounded-lg</code>	
<code>border-radius: 12px</code>	<code>rounded-xl</code>	
<code>border-radius: 16px</code>	<code>rounded-2xl</code>	
<code>border-radius: 9999px</code>	<code>rounded-full</code>	
<code>border-radius: 50%</code>	<code>rounded-full</code>	circle
<code>border-radius: 0</code>	<code>rounded-none</code>	
<code>border-top-left-radius: 8px</code>	<code>rounded-tl-lg</code>	
<code>border-radius: 10px 10px 0 0</code>	<code>rounded-t-[10px]</code>	
<code>border-collapse: collapse</code>	<code>border-collapse</code>	
<code>outline: none</code>	<code>outline-none</code>	or <code>outline-hidden</code>
<code>outline: 2px solid blue</code>	<code>outline-2 outline-blue-500</code>	
<code>outline-offset: 2px</code>	<code>outline-offset-2</code>	

CSS	Tailwind	Notes
<code>box-shadow: 0 1px 2px rgb(0 0 0 / 0.05)</code>	<code>shadow-xs</code>	
<code>box-shadow: 0 1px 3px rgb(0 0 0 / 0.1)...</code>	<code>shadow-sm</code>	
<code>box-shadow: 0 4px 6px -1px rgb(0 0 0 / 0.1)...</code>	<code>shadow-md</code>	
<code>box-shadow: 0 10px 15px -3px rgb(0 0 0 / 0.1)...</code>	<code>shadow-lg</code>	
<code>box-shadow: 0 20px 25px -5px rgb(0 0 0 / 0.1)...</code>	<code>shadow-xl</code>	
<code>box-shadow: 0 25px 50px -12px rgb(0 0 0 / 0.25)</code>	<code>shadow-2xl</code>	
<code>box-shadow: none</code>	<code>shadow-none</code>	
<code>box-shadow: 0 4px 20px rgba(0,0,0,.15)</code>	<code>shadow-[0_4px_20px_rgba(0,0,0,0.15)]</code>	
<code>box-shadow: inset 0 2px 4px ...</code>	<code>inset-shadow-sm</code>	
<code>box-shadow: 0 0 0 3px blue (focus ring)</code>	<code>ring-3 ring-blue-500</code>	

32.9 Overflow, Objects, Interactivity

CSS	Tailwind	Notes
<code>overflow: hidden</code>	<code>overflow-hidden</code>	
<code>overflow: auto</code>	<code>overflow-auto</code>	
<code>overflow: scroll</code>	<code>overflow-scroll</code>	
<code>overflow-x: auto</code>	<code>overflow-x-auto</code>	
<code>overflow-y: scroll</code>	<code>overflow-y-scroll</code>	
<code>overscroll-behavior: contain</code>	<code>overscroll-contain</code>	
<code>object-fit: cover</code>	<code>object-cover</code>	
<code>object-fit: contain</code>	<code>object-contain</code>	
<code>object-position: top</code>	<code>object-top</code>	
<code>cursor: pointer</code>	<code>cursor-pointer</code>	
<code>cursor: not-allowed</code>	<code>cursor-not-allowed</code>	
<code>cursor: grab</code>	<code>cursor-grab</code>	
<code>pointer-events: none</code>	<code>pointer-events-none</code>	
<code>user-select: none</code>	<code>select-none</code>	

CSS	Tailwind	Notes
<code>resize: none</code>	<code>resize-none</code>	
<code>appearance: none</code>	<code>appearance-none</code>	
<code>scroll-behavior: smooth</code>	<code>scroll-smooth</code>	
<code>scroll-snap-type: x mandatory</code>	<code>snap-x snap-mandatory</code>	
<code>scroll-snap-align: start</code>	<code>snap-start</code>	
<code>touch-action: none</code>	<code>touch-none</code>	
<code>will-change: transform</code>	<code>will-change-transform</code>	
<code>field-sizing: content</code>	<code>field-sizing-content</code>	

32.10 Transitions, Transforms, Filters

CSS	Tailwind	Notes
<code>transition: all 150ms</code>	<code>transition-all</code>	
<code>transition-property: color, background-color...</code>	<code>transition-colors</code>	
<code>transition-property: opacity</code>	<code>transition-opacity</code>	
<code>transition-property: transform</code>	<code>transition-transform</code>	
<code>transition-duration: 300ms</code>	<code>duration-300</code>	
<code>transition-timing-function: ease-in-out</code>	<code>ease-in-out</code>	
<code>transition-delay: 100ms</code>	<code>delay-100</code>	
<code>animation: spin 1s linear infinite</code>	<code>animate-spin</code>	
<code>animation: pulse 2s ...</code>	<code>animate-pulse</code>	
<code>animation: none</code>	<code>animate-none</code>	
<code>transform: scale(1.05)</code>	<code>scale-105</code>	
<code>transform: scaleX(-1)</code>	<code>-scale-x-100</code>	
<code>transform: rotate(45deg)</code>	<code>rotate-45</code>	
<code>transform: rotate(-12deg)</code>	<code>-rotate-12</code>	
<code>transform: translateX(1rem)</code>	<code>translate-x-4</code>	
<code>transform: translateY(-4px)</code>	<code>-translate-y-1</code>	
<code>transform: translate(-50%, -50%)</code>	<code>-translate-x-1/2 -translate-y-1/2</code>	
<code>transform: skewX(12deg)</code>	<code>skew-x-12</code>	
<code>transform-origin: top left</code>	<code>origin-top-left</code>	

CSS	Tailwind	Notes
<code>filter: blur(8px)</code>	<code>blur-sm</code>	
<code>filter: brightness(1.1)</code>	<code>brightness-110</code>	
<code>filter: grayscale(100%)</code>	<code>grayscale</code>	
<code>filter: invert(100%)</code>	<code>invert</code>	
<code>filter: drop-shadow(...)</code>	<code>drop-shadow-md</code>	
<code>backdrop-filter: blur(12px)</code>	<code>backdrop-blur-md</code>	
<code>backdrop-filter: brightness(0.5)</code>	<code>backdrop-brightness-50</code>	

32.11 Media Queries and Pseudo-classes

CSS	Tailwind	Notes
<code>@media (min-width: 640px)</code>	<code>sm:</code>	
<code>@media (min-width: 768px)</code>	<code>md:</code>	
<code>@media (min-width: 1024px)</code>	<code>lg:</code>	
<code>@media (min-width: 1280px)</code>	<code>xl:</code>	
<code>@media (max-width: 767px)</code>	<code>max-md:</code>	
<code>@media (min-width: 900px)</code>	<code>min-[900px]:</code>	
<code>@media (prefers-color-scheme: dark)</code>	<code>dark:</code>	
<code>@media (prefers-reduced-motion: reduce)</code>	<code>motion-reduce:</code>	
<code>@media print</code>	<code>print:</code>	
<code>@container (min-width: 28rem)</code>	<code>@md:</code>	
<code>:hover</code>	<code>hover:</code>	
<code>:focus</code>	<code>focus:</code>	
<code>:focus-visible</code>	<code>focus-visible:</code>	
<code>:active</code>	<code>active:</code>	
<code>:disabled</code>	<code>disabled:</code>	
<code>:checked</code>	<code>checked:</code>	
<code>:first-child</code>	<code>first:</code>	
<code>:last-child</code>	<code>last:</code>	
<code>:nth-child(odd)</code>	<code>odd:</code>	
<code>:nth-child(even)</code>	<code>even:</code>	

CSS	Tailwind	Notes
<code>::before</code>	<code>before:</code>	
<code>::after</code>	<code>after:</code>	
<code>::placeholder</code>	<code>placeholder:</code>	
<code>::selection</code>	<code>selection:</code>	
<code>content: ''</code>	<code>content-['']</code>	
<code>:has(...)</code>	<code>has-[...]:</code>	
<code>:not(...)</code>	<code>not-*</code>	

33. "I Want To Do X" Quick Finder

Search this section with Ctrl+F using the words you think in ("center", "hide", "round", "shadow"...).

33.1 Layout

I want to...	Use
I want to center something (horizontally and vertically)	<code>flex items-center justify-center</code>
I want to center with grid	<code>grid place-items-center</code>
I want to center a block horizontally	<code>mx-auto</code> (with a width, e.g. <code>max-w-md</code>)
I want to center text	<code>text-center</code>
I want to center absolutely	<code>absolute top-1/2 left-1/2 -translate-x-1/2 -translate-y-1/2</code>
I want a full-screen section	<code>min-h-screen</code> (mobile friendly: <code>min-h-dvh</code>)
I want a full-width element	<code>w-full</code>
I want a centered page container	<code>mx-auto max-w-7xl px-4 sm:px-6 lg:px-8</code>
I want two items on opposite sides	<code>flex justify-between items-center</code>
I want items in a row with space	<code>flex gap-4</code>
I want items stacked with space	<code>flex flex-col gap-4</code> or <code>space-y-4</code>
I want to push one item to the far right	<code>ml-auto</code>
I want items to wrap to the next line	<code>flex flex-wrap gap-2</code>
I want equal-width columns	<code>flex + flex-1</code> on children, or <code>grid grid-cols-3</code>
I want a sidebar and main content	<code>flex + w-64 shrink-0</code> and <code>flex-1 min-w-0</code>
I want a sticky footer	<code>flex min-h-screen flex-col + flex-1</code> on main
I want a 3-column responsive layout	<code>grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6</code>
I want a 12-column grid	<code>grid grid-cols-12 gap-4 + col-span-*</code>
I want cards that auto-fill the width	<code>grid grid-cols-[repeat(auto-fit,minmax(250px,1fr))] gap-6</code>
I want a stack on mobile and a row on desktop	<code>flex flex-col md:flex-row</code>
I want to overlap elements	<code>relative</code> parent + <code>absolute</code> child, or negative margin <code>-mt-8</code>
I want to cover the parent completely	<code>absolute inset-0</code>
I want an element fixed to the screen corner	<code>fixed bottom-4 right-4</code>
I want a sticky header	<code>sticky top-0 z-10</code>
I want to put one thing on top of another	<code>relative z-10 / z-20</code>
I want to make a square	<code>aspect-square</code> or <code>size-24</code>
I want a 16:9 box	<code>aspect-video</code>
I want to reverse the order on mobile	<code>flex flex-col-reverse</code> or <code>order-*</code>

33.2 Visibility and Responsive

I want to...	Use
I want to hide an element	<code>hidden</code>
I want to hide on mobile, show on desktop	<code>hidden md:block</code> (or <code>hidden md:flex</code>)
I want to show on mobile only	<code>md:hidden</code>
I want to hide but keep the space	<code>invisible</code>
I want to hide visually but keep for screen readers	<code>sr-only</code>
I want to change size by screen width	<code>text-sm md:text-lg lg:text-2xl</code>
I want smaller padding on mobile	<code>p-4 md:p-8</code>
I want to style only between tablet and laptop	<code>md:max-lg:...</code>
I want a different layout based on the parent's width	<code>@container</code> on the parent + <code>@md:...</code> on children
I want to hide when printing	<code>print:hidden</code>

33.3 Shape, Borders, Shadows

I want to...	Use
I want a circular image	<code>rounded-full object-cover size-24</code>
I want rounded corners	<code>rounded-lg</code> (or <code>rounded-xl</code> , <code>rounded-2xl</code>)
I want a pill-shaped button	<code>rounded-full px-6 py-2</code>
I want only top corners rounded	<code>rounded-t-lg</code>
I want a shadow card	<code>rounded-xl shadow-lg</code>
I want a subtle shadow	<code>shadow-sm</code>
I want a bigger shadow on hover	<code>shadow-md hover:shadow-xl transition-shadow</code>
I want a colored glow	<code>shadow-lg shadow-blue-500/50</code>
I want a border	<code>border border-gray-200</code>
I want a dashed border	<code>border-2 border-dashed border-gray-300</code>
I want a line between list items	<code>divide-y divide-gray-200</code>
I want a bottom border only	<code>border-b border-gray-200</code>
I want a left accent bar	<code>border-l-4 border-blue-500</code>
I want a focus ring	<code>focus:ring-2 focus:ring-blue-500</code>
I want to remove the outline accessibly	<code>outline-hidden</code> (add a ring on <code>focus-visible:</code>)
I want an inner shadow	<code>inset-shadow-sm</code>

33.4 Color and Background

I want to...	Use
I want a blue button	<code>bg-blue-600 text-white hover:bg-blue-700</code>
I want muted gray text	<code>text-gray-500</code>
I want a semi-transparent black overlay	<code>bg-black/50</code>
I want a gradient background	<code>bg-linear-to-r from-indigo-500 to-purple-600</code>

I want to...	Use
I want gradient text	<code>bg-linear-to-r from-blue-500 to-pink-500 bg-clip-text text-transparent</code>
I want a background image that covers	<code>bg-[url(/img.jpg)] bg-cover bg-center</code>
I want a fade-to-dark image overlay	<code>bg-linear-to-t from-black/70 to-transparent</code>
I want a frosted glass effect	<code>bg-white/20 backdrop-blur-lg border border-white/30</code>
I want a custom hex color	<code>bg-[#1e293b]</code>
I want dark mode colors	<code>bg-white dark:bg-gray-900 text-gray-900 dark:text-white</code>
I want an icon to match text color	<code>fill-current</code> or <code>stroke-current</code>
I want to color a checkbox	<code>accent-blue-600</code>
I want to color the typing cursor	<code>caret-blue-500</code>
I want a faded (disabled) element	<code>opacity-50 cursor-not-allowed</code>
I want a transparent background	<code>bg-transparent</code>

33.5 Text

I want to...	Use
I want a big bold heading	<code>text-4xl font-bold tracking-tight</code>
I want small gray caption text	<code>text-sm text-gray-500</code>
I want uppercase small labels	<code>text-xs font-semibold uppercase tracking-wider</code>
I want comfortable paragraph text	<code>text-base leading-relaxed text-gray-600</code>
I want a readable line length	<code>max-w-prose</code> (or <code>max-w-2xl</code>)
I want one line with an ellipsis	<code>truncate</code>
I want to show only 3 lines	<code>line-clamp-3</code>
I want text not to wrap	<code>whitespace-nowrap</code>
I want to break long words/URLs	<code>break-words</code> or <code>break-all</code>
I want balanced headline wrapping	<code>text-balance</code>
I want to underline only on hover	<code>no-underline hover:underline</code>
I want a link look	<code>text-blue-600 underline underline-offset-4 hover:text-blue-800</code>
I want strikethrough	<code>line-through</code>
I want monospace code	<code>font-mono text-sm</code>
I want numbers that align in columns	<code>tabular-nums</code>
I want a drop cap	<code>first-letter:text-5xl first-letter:font-bold first-letter:float-left</code>
I want to keep line breaks from a textarea	<code>whitespace-pre-line</code>
I want to disable text selection	<code>select-none</code>

33.6 Interaction and Animation

I want to...	Use
I want a hover animation	<code>transition hover:scale-105</code>
I want a hover lift	<code>transition hover:-translate-y-1 hover:shadow-lg</code>

I want to...	Use
I want a smooth color change on hover	<code>transition-colors duration-200 hover:bg-blue-600</code>
I want a press effect	<code>active:scale-95 transition</code>
I want a spinner	<code>animate-spin rounded-full border-4 border-gray-200 border-t-blue-600 size-8</code>
I want a skeleton loading effect	<code>animate-pulse bg-gray-200 rounded</code>
I want a pulsing notification dot	<code>animate-ping</code> on an absolute ring + a solid dot
I want a bouncing arrow	<code>animate-bounce</code>
I want a pointer cursor	<code>cursor-pointer</code>
I want a forbidden cursor	<code>cursor-not-allowed</code>
I want clicks to pass through an overlay	<code>pointer-events-none</code>
I want smooth scrolling for anchors	<code>scroll-smooth</code> on <code><html></code>
I want horizontal scroll with snapping	<code>flex overflow-x-auto snap-x snap-mandatory</code> + <code>snap-start shrink-0</code>
I want to show a child when the parent is hovered	<code>group</code> on parent + <code>group-hover:opacity-100</code> on child
I want to react to a sibling's state	<code>peer</code> on the earlier sibling + <code>peer-checked:...</code>
I want to style when a checkbox is checked	<code>checked:bg-blue-600</code> or <code>peer-checked:</code>
I want to respect reduced motion	<code>motion-safe:animate-spin motion-reduce:animate-none</code>
I want a fade-in when mounted	<code>transition-opacity starting:opacity-0</code>
I want a custom animation	define <code>--animate-name</code> in <code>@theme</code> , then <code>animate-name</code>

33.7 Images, Forms, Misc

I want to...	Use
I want an image to fill its box without distortion	<code>size-full object-cover</code>
I want an image to fit without cropping	<code>object-contain</code>
I want a responsive image	<code>max-w-full h-auto</code>
I want to zoom an image on hover	<code>overflow-hidden wrapper + transition hover:scale-110</code> on <code>img</code>
I want a grayscale image that gets color on hover	<code>grayscale hover:grayscale-0 transition</code>
I want to blur a background	<code>backdrop-blur-md</code>
I want an input that looks good	<code>rounded-lg border border-gray-300 px-3 py-2 focus:ring-3 focus:ring-blue-500/30 focus:outline-hidden focus:border-blue-500</code>
I want styled placeholder text	<code>placeholder:text-gray-400</code>
I want an input error state	<code>invalid:border-red-500</code> or <code>user-invalid:border-red-500</code>
I want a disabled input look	<code>disabled:bg-gray-100 disabled:cursor-not-allowed</code>
I want a textarea that cannot be resized	<code>resize-none</code>
I want a textarea that grows with content	<code>field-sizing-content</code>
I want a custom file button	<code>file:rounded-full file:bg-blue-50 file:px-4 file:py-2</code>

I want to...	Use
I want to space list items evenly	<code>space-y-2</code>
I want bullets back on a list	<code>list-disc pl-5</code>
I want a scrollable box	<code>max-h-64 overflow-y-auto</code>
I want a scrollable table on mobile	<code>wrapper overflow-x-auto</code>
I want a custom value that is not in Tailwind	<code>w-[350px] , mt-[37px] , bg-[#123456]</code>
I want to override with !important	<code>text-red-500!</code>
I want to style direct children	<code>*:p-2 or [>li]:py-2</code>
I want to add content with ::before	<code>before:content-[''] before:absolute</code>
I want to style based on a data attribute	<code>data-[state=open]:bg-blue-50</code>

34. Class Name Construction Guide

34.1 The Recipe

1. Which CSS property? `font-size`
2. What is Tailwind's prefix? `text`
3. What value? `24px` -> closest scale step `2xl` (`1.5rem = 24px`)
4. Join them: `text-2xl`

If no scale value matches: `text-[22px]`.

34.2 Worked Example: Heading

```
font-size: 24px;
font-weight: 700;
color: blue;
```

```
text-2xl font-bold text-blue-500
```

- `font-size: 24px`: property `font-size` uses prefix `text`. `24px` is `1.5rem`, and the `2xl` step is `1.5rem`. So `text-2xl`.
- `font-weight: 700`: prefix `font`; `700` is named `bold`. So `font-bold`.
- `color: blue`: prefix `text` (the same prefix as size, Tailwind tells them apart by value); pick a hue and shade. `blue` + `500`. So `text-blue-500`.

34.3 Worked Example: Card

```
background: white;
padding: 24px;
border-radius: 12px;
box-shadow: 0 10px 15px rgba(0,0,0,.1);
border: 1px solid #e5e7eb;
```

```
bg-white p-6 rounded-xl shadow-lg border border-gray-200
```

- `padding: 24px`: prefix `p`; `24 / 4 = 6`. So `p-6`.
- `border-radius: 12px`: prefix `rounded`; `12px` is the `xl` step. So `rounded-xl`.
- `box-shadow`: prefix `shadow`; a large soft shadow is `lg`. So `shadow-lg`.
- `border: 1px solid #e5e7eb`: `border` gives `1px solid`; `#e5e7eb` is `gray-200`. So `border border-gray-200`.

34.4 Worked Example: Centered Flex Row

```
display: flex;
justify-content: space-between;
align-items: center;
gap: 16px;
```

```
flex justify-between items-center gap-4
```

- `display: flex` becomes `flex`.
- `justify-content: space-between`: the property abbreviates to `justify`, the value `space-between` to `between`. So `justify-between`.
- `align-items: center`: `align-items` abbreviates to `items`. So `items-center`.
- `gap: 16px`: $16 / 4 = 4$. So `gap-4`.

34.5 Worked Example: Button

```
background-color: #2563eb;  
color: white;  
padding: 8px 16px;  
border-radius: 8px;  
font-weight: 500;  
transition: background-color 200ms;  
/* hover */ background-color: #1d4ed8;
```

```
bg-blue-600 text-white px-4 py-2 rounded-lg font-medium transition-colors duration-200 hover:bg-blue-700
```

- `#2563eb` is `blue-600`; `#1d4ed8` is `blue-700`.
- `padding: 8px 16px` has two values (vertical horizontal): `py-2 px-4`.
- `:hover` is the `hover:` prefix placed in front of the normal class.
- `transition: background-color 200ms` becomes `transition-colors duration-200`.

34.6 Worked Example: Absolutely Positioned Badge

```
position: absolute;  
top: -8px;  
right: -8px;  
z-index: 10;
```

```
absolute -top-2 -right-2 z-10
```

- Negative values: put `-` in front of the utility: `-top-2`.
- Remember: the parent needs `relative`.

34.7 Worked Example: Responsive Grid

```
display: grid;  
grid-template-columns: 1fr;  
gap: 24px;  
@media (min-width: 768px) { grid-template-columns: repeat(2, 1fr); }  
@media (min-width: 1024px) { grid-template-columns: repeat(3, 1fr); }
```

```
grid grid-cols-1 gap-6 md:grid-cols-2 lg:grid-cols-3
```

- `repeat(n, 1fr)` is `grid-cols-n`.
- `@media (min-width: ...)` is a breakpoint prefix; the un-prefixed class is the mobile default.

34.8 Quick Property-to-Prefix Dictionary

CSS property	Tailwind prefix	Value style
margin	m, mx, my, mt, mr, mb, ml	number (x4px) / auto
padding	p, px, py, pt, pr, pb, pl	number
width / height	w / h (size for both)	number, fraction, full, screen, auto, fit
min-width / max-width	min-w / max-w	number, md, 7xl, full, prose
font-size	text	xs ... 9xl
font-weight	font	thin ... black
font-family	font	sans, serif, mono
line-height	leading	none ... loose, number
letter-spacing	tracking	tighter ... widest
color	text	color-shade
background-color	bg	color-shade
border-width	border	none, 0, 2, 4, 8
border-color	border	color-shade
border-radius	rounded	sm ... 3xl, full
box-shadow	shadow	xs ... 2xl
opacity	opacity	0 to 100
display	(the value itself)	flex, grid, block, hidden
justify-content	justify	start, center, between ...
align-items	items	start, center, end ...
align-self	self	
flex-direction	flex	row, col
flex-wrap	flex	wrap, nowrap
gap	gap, gap-x, gap-y	number
grid-template-columns	grid-cols	1 to 12
position	(the value itself)	relative, absolute ...
top/right/bottom/left	top, right, bottom, left, inset	number, fraction
z-index	z	number
overflow	overflow	hidden, auto ...
cursor	cursor	pointer ...
transition	transition, duration, ease, delay	
transform: scale/rotate/translate	scale, rotate, translate-x/y	
object-fit	object	cover, contain
aspect-ratio	aspect	square, video, [4/3]

35. Tailwind Cheat Tables

Compact lookup tables. Scan the left column for the category, the right for every class.

35.1 Typography

Group	Classes
Family	font-sans font-serif font-mono
Size	text-xs text-sm text-base text-lg text-xl text-2xl text-3xl text-4xl text-5xl text-6xl text-7xl text-8xl text-9xl
Weight	font-thin font-extralight font-light font-normal font-medium font-semibold font-bold font-extrabold font-black
Style	italic not-italic antialiased
Align	text-left text-center text-right text-justify text-start text-end
Leading	leading-none leading-tight leading-snug leading-normal leading-relaxed leading-loose leading-6
Tracking	tracking-tighter tracking-tight tracking-normal tracking-wide tracking-wider tracking-widest
Decoration	underline overline line-through no-underline decoration-wavy decoration-2 underline-offset-4
Transform	uppercase lowercase capitalize normal-case
Overflow	truncate text-ellipsis text-clip line-clamp-1..6
Whitespace	whitespace-normal whitespace-nowrap whitespace-pre whitespace-pre-line whitespace-pre-wrap whitespace-break-spaces
Breaking	break-normal break-words break-all break-keep text-balance text-pretty text-wrap text-nowrap
Numbers	tabular-nums proportional-nums lining-nums oldstyle-nums slashed-zero ordinal

35.2 Spacing

Group	Classes
Margin	m-n mx-n my-n mt-n mr-n mb-n ml-n ms-n me-n m-auto mx-auto -m-n -mt-n
Padding	p-n px-n py-n pt-n pr-n pb-n pl-n ps-n pe-n
Gap	gap-n gap-x-n gap-y-n

Group	Classes
Space between	<code>space-x-n</code> <code>space-y-n</code> <code>space-x-reverse</code> <code>space-y-reverse</code>
Scale (n x 4px)	<code>0</code> <code>px</code> <code>0.5</code> <code>1</code> <code>1.5</code> <code>2</code> <code>2.5</code> <code>3</code> <code>3.5</code> <code>4</code> <code>5</code> <code>6</code> <code>7</code> <code>8</code> <code>9</code> <code>10</code> <code>11</code> <code>12</code> <code>14</code> <code>16</code> <code>20</code> <code>24</code> <code>28</code> <code>32</code> <code>36</code> <code>40</code> <code>44</code> <code>48</code> <code>52</code> <code>56</code> <code>60</code> <code>64</code> <code>72</code> <code>80</code> <code>96</code>

35.3 Colors

Group	Classes
Shades	<code>50</code> <code>100</code> <code>200</code> <code>300</code> <code>400</code> <code>500</code> <code>600</code> <code>700</code> <code>800</code> <code>900</code> <code>950</code>
Vivid hues	<code>red</code> <code>orange</code> <code>amber</code> <code>yellow</code> <code>lime</code> <code>green</code> <code>emerald</code> <code>teal</code> <code>cyan</code> <code>sky</code> <code>blue</code> <code>indigo</code> <code>violet</code> <code>purple</code> <code>fuchsia</code> <code>pink</code> <code>rose</code>
Neutral hues	<code>slate</code> <code>gray</code> <code>zinc</code> <code>neutral</code> <code>stone</code> (v4.2+: <code>mauve</code> <code>olive</code> <code>mist</code> <code>taupe</code>)
Special	<code>black</code> <code>white</code> <code>transparent</code> <code>current</code> <code>inherit</code>
Where colors apply	<code>text-</code> <code>bg-</code> <code>border-</code> <code>ring-</code> <code>ring-offset-</code> <code>inset-ring-</code> <code>divide-</code> <code>outline-</code> <code>shadow-</code> <code>inset-shadow-</code> <code>drop-shadow-</code> <code>decoration-</code> <code>accent-</code> <code>caret-</code> <code>fill-</code> <code>stroke-</code> <code>from-</code> <code>via-</code> <code>to-</code> <code>placeholder:text-</code> <code>selection:bg-</code>
Opacity	<code>bg-black/50</code> <code>text-white/80</code> <code>border-blue-500/30</code> <code>bg-blue-500/[0.35]</code>
Arbitrary	<code>bg-[#1e293b]</code> <code>text-[rgb(20,30,40)]</code> <code>bg-[oklch(0.7_0.15_200)]</code> <code>bg(--brand)</code>

35.4 Width

Group	Classes
Fixed	<code>w-0</code> <code>w-px</code> <code>w-1</code> <code>w-4</code> <code>w-8</code> <code>w-10</code> <code>w-12</code> <code>w-16</code> <code>w-24</code> <code>w-32</code> <code>w-48</code> <code>w-64</code> <code>w-96</code>
Fractions	<code>w-1/2</code> <code>w-1/3</code> <code>w-2/3</code> <code>w-1/4</code> <code>w-3/4</code> <code>w-1/5</code> <code>w-1/6</code> <code>w-5/12</code>
Keywords	<code>w-full</code> <code>w-screen</code> <code>w-dvw</code> <code>w-auto</code> <code>w-fit</code> <code>w-min</code> <code>w-max</code>
Min	<code>min-w-0</code> <code>min-w-full</code> <code>min-w-min</code> <code>min-w-max</code> <code>min-w-fit</code>
Max	<code>max-w-xs</code> <code>max-w-sm</code> <code>max-w-md</code> <code>max-w-lg</code> <code>max-w-xl</code> <code>max-w-2xl</code> <code>max-w-3xl</code> <code>max-w-4xl</code> <code>max-w-5xl</code> <code>max-w-6xl</code> <code>max-w-7xl</code> <code>max-w-prose</code> <code>max-w-full</code> <code>max-w-screen</code> <code>max-w-none</code>
Both	<code>size-4</code> <code>size-10</code> <code>size-full</code> <code>size-px</code> <code>size-[72px]</code>

35.5 Height

Group	Classes
Fixed	<code>h-0</code> <code>h-px</code> <code>h-4</code> <code>h-8</code> <code>h-10</code> <code>h-12</code> <code>h-16</code> <code>h-24</code> <code>h-32</code> <code>h-48</code> <code>h-64</code> <code>h-96</code>
Keywords	<code>h-full</code> <code>h-screen</code> <code>h-dvh</code> <code>h-svh</code> <code>h-lvh</code> <code>h-auto</code> <code>h-fit</code> <code>h-min</code> <code>h-max</code>
Fractions	<code>h-1/2</code> <code>h-1/3</code> <code>h-2/3</code> <code>h-1/4</code>
Min	<code>min-h-0</code> <code>min-h-full</code> <code>min-h-screen</code> <code>min-h-dvh</code> <code>min-h-fit</code>
Max	<code>max-h-48</code> <code>max-h-96</code> <code>max-h-full</code> <code>max-h-screen</code> <code>max-h-none</code>
Arbitrary	<code>h-[500px]</code> <code>h-[calc(100dvh-4rem)]</code>

35.6 Flex

Group	Classes
Display	<code>flex</code> <code>inline-flex</code>
Direction	<code>flex-row</code> <code>flex-row-reverse</code> <code>flex-col</code> <code>flex-col-reverse</code>
Wrap	<code>flex-wrap</code> <code>flex-nowrap</code> <code>flex-wrap-reverse</code>
Justify (main axis)	<code>justify-start</code> <code>justify-center</code> <code>justify-end</code> <code>justify-between</code> <code>justify-around</code> <code>justify-evenly</code>
Items (cross axis)	<code>items-start</code> <code>items-center</code> <code>items-end</code> <code>items-baseline</code> <code>items-stretch</code>
Content (multi-line)	<code>content-start</code> <code>content-center</code> <code>content-end</code> <code>content-between</code> <code>content-around</code> <code>content-evenly</code>
Self	<code>self-auto</code> <code>self-start</code> <code>self-center</code> <code>self-end</code> <code>self-stretch</code> <code>self-baseline</code>
Item sizing	<code>flex-1</code> <code>flex-auto</code> <code>flex-initial</code> <code>flex-none</code> <code>grow</code> <code>grow-0</code> <code>shrink</code> <code>shrink-0</code> <code>basis-1/2</code> <code>basis-full</code> <code>basis-auto</code>
Order	<code>order-1</code> <code>order-2</code> <code>order-first</code> <code>order-last</code> <code>order-none</code>

35.7 Grid

Group	Classes
Display	<code>grid</code> <code>inline-grid</code>
Columns	<code>grid-cols-1</code> ... <code>grid-cols-12</code> <code>grid-cols-none</code> <code>grid-cols-subgrid</code> <code>grid-cols-[200px_1fr]</code>

Group	Classes
Rows	<code>grid-rows-1</code> ... <code>grid-rows-12</code> <code>grid-rows-none</code> <code>grid-rows-[auto_1fr_auto]</code>
Column span/position	<code>col-span-1..12</code> <code>col-span-full</code> <code>col-start-1..13</code> <code>col-end-1..13</code> <code>col-auto</code>
Row span/position	<code>row-span-1..12</code> <code>row-span-full</code> <code>row-start-1..13</code> <code>row-end-1..13</code> <code>row-auto</code>
Flow	<code>grid-flow-row</code> <code>grid-flow-col</code> <code>grid-flow-dense</code> <code>grid-flow-row-dense</code> <code>grid-flow-col-dense</code>
Auto size	<code>auto-cols-auto</code> <code>auto-cols-min</code> <code>auto-cols-max</code> <code>auto-cols-fr</code> <code>auto-rows-auto</code> <code>auto-rows-min</code> <code>auto-rows-max</code> <code>auto-rows-fr</code>
Place	<code>place-items-center</code> <code>place-content-center</code> <code>place-self-center</code> <code>justify-items-*</code> <code>justify-self-*</code>

35.8 Position

Group	Classes
Type	<code>static</code> <code>relative</code> <code>absolute</code> <code>fixed</code> <code>sticky</code>
Inset	<code>inset-0</code> <code>inset-x-0</code> <code>inset-y-0</code> <code>top-0</code> <code>right-0</code> <code>bottom-0</code> <code>left-0</code> <code>start-0</code> <code>end-0</code> <code>inset-auto</code>
Offsets	<code>top-4</code> <code>-top-2</code> <code>top-1/2</code> <code>left-1/2</code> <code>top-full</code> <code>top-[10px]</code>
Z-index	<code>z-0</code> <code>z-10</code> <code>z-20</code> <code>z-30</code> <code>z-40</code> <code>z-50</code> <code>z-auto</code> <code>-z-10</code> <code>z-[100]</code>
Visibility	<code>visible</code> <code>invisible</code> <code>collapse</code>
Overflow	<code>overflow-auto</code> <code>overflow-hidden</code> <code>overflow-clip</code> <code>overflow-visible</code> <code>overflow-scroll</code> <code>overflow-x-auto</code> <code>overflow-y-auto</code>

35.9 Border

Group	Classes
Width	<code>border</code> <code>border-0</code> <code>border-2</code> <code>border-4</code> <code>border-8</code>
Sides	<code>border-x</code> <code>border-y</code> <code>border-t</code> <code>border-r</code> <code>border-b</code> <code>border-l</code> <code>border-s</code> <code>border-e</code>
Style	<code>border-solid</code> <code>border-dashed</code> <code>border-dotted</code> <code>border-double</code> <code>border-hidden</code> <code>border-none</code>
Color	<code>border-gray-200</code> <code>border-blue-500</code> <code>border-transparent</code> <code>border-current</code> <code>border-blue-500/30</code>
Divide	<code>divide-x</code> <code>divide-y</code> <code>divide-x-2</code> <code>divide-gray-200</code> <code>divide-dashed</code>
Ring	<code>ring</code> <code>ring-0</code> <code>ring-1</code> <code>ring-2</code> <code>ring-3</code> <code>ring-4</code> <code>ring-8</code> <code>ring-blue-500</code> <code>ring-offset-2</code> <code>inset-ring</code>
Outline	<code>outline</code> <code>outline-2</code> <code>outline-offset-2</code> <code>outline-dashed</code> <code>outline-hidden</code> <code>outline-none</code>

35.10 Radius

Group	Classes
Sizes	<code>rounded-none</code> <code>rounded-xs</code> <code>rounded-sm</code> <code>rounded-md</code> <code>rounded-lg</code> <code>rounded-xl</code> <code>rounded-2xl</code> <code>rounded-3xl</code> <code>rounded-4xl</code> <code>rounded-full</code>
Sides	<code>rounded-t-*</code> <code>rounded-r-*</code> <code>rounded-b-*</code> <code>rounded-l-*</code> <code>rounded-s-*</code> <code>rounded-e-*</code>
Corners	<code>rounded-tl-*</code> <code>rounded-tr-*</code> <code>rounded-br-*</code> <code>rounded-bl-*</code> <code>rounded-ss-*</code> <code>rounded-se-*</code> <code>rounded-es-*</code> <code>rounded-ee-*</code>
Pixels	<code>2px</code> <code>xs</code> , <code>4px</code> <code>sm</code> , <code>6px</code> <code>md</code> , <code>8px</code> <code>lg</code> , <code>12px</code> <code>xl</code> , <code>16px</code> <code>2xl</code> , <code>24px</code> <code>3xl</code> , <code>32px</code> <code>4xl</code>

35.11 Shadow

Group	Classes
Box shadow	<code>shadow-2xs</code> <code>shadow-xs</code> <code>shadow-sm</code> <code>shadow-md</code> <code>shadow-lg</code> <code>shadow-xl</code> <code>shadow-2xl</code> <code>shadow-none</code>
Colored	<code>shadow-blue-500/50</code> <code>shadow-black/10</code>
Inset	<code>inset-shadow-2xs</code> <code>inset-shadow-xs</code> <code>inset-shadow-sm</code> <code>inset-shadow-none</code>
Drop shadow (filter)	<code>drop-shadow-xs</code> <code>drop-shadow-sm</code> <code>drop-shadow-md</code> <code>drop-shadow-lg</code> <code>drop-shadow-xl</code> <code>drop-shadow-2xl</code> <code>drop-shadow-none</code>
Text shadow (v4.1+)	<code>text-shadow-2xs</code> <code>text-shadow-xs</code> <code>text-shadow-sm</code> <code>text-shadow-md</code> <code>text-shadow-lg</code>
Custom	<code>shadow-[0_4px_20px_rgba(0,0,0,0.15)]</code>

35.12 Responsive

Group	Classes
Min-width (mobile first)	<code>sm:</code> <code>640px</code> , <code>md:</code> <code>768px</code> , <code>lg:</code> <code>1024px</code> , <code>xl:</code> <code>1280px</code> , <code>2xl:</code> <code>1536px</code>
Max-width	<code>max-sm:</code> <code>max-md:</code> <code>max-lg:</code> <code>max-xl:</code> <code>max-2xl:</code>
Range	<code>md:max-lg:</code>
Arbitrary	<code>min-[900px]:</code> <code>max-[600px]:</code>
Container queries	<code>@container</code> <code>@sm:</code> <code>@md:</code> <code>@lg:</code> <code>@xl:</code> <code>@max-md:</code> <code>@container/name</code>
Media features	<code>dark:</code> <code>print:</code> <code>motion-safe:</code> <code>motion-reduce:</code> <code>portrait:</code> <code>landscape:</code> <code>pointer-coarse:</code> <code>contrast-more:</code>

35.13 States

Group	Classes
Pointer and focus	<code>hover:</code> <code>focus:</code> <code>focus-visible:</code> <code>focus-within:</code> <code>active:</code> <code>visited:</code> <code>target:</code>
Form	<code>disabled:</code> <code>enabled:</code> <code>checked:</code> <code>indeterminate:</code> <code>default:</code> <code>required:</code> <code>optional:</code> <code>valid:</code> <code>invalid:</code> <code>user-valid:</code> <code>user-invalid:</code> <code>in-range:</code> <code>out-of-range:</code> <code>placeholder-shown:</code> <code>autofill:</code> <code>read-only:</code>
Structure	<code>first:</code> <code>last:</code> <code>only:</code> <code>odd:</code> <code>even:</code> <code>first-of-type:</code> <code>last-of-type:</code> <code>only-of-type:</code> <code>empty:</code> <code>nth-3:</code> <code>has-[:]:</code> <code>not-*</code> <code>*</code> <code>**:</code>
Pseudo-elements	<code>before:</code> <code>after:</code> <code>placeholder:</code> <code>file:</code> <code>marker:</code> <code>selection:</code> <code>first-letter:</code> <code>first-line:</code> <code>backdrop:</code>
Parent / sibling	<code>group-hover:</code> <code>group-focus:</code> <code>group-active:</code> <code>group-disabled:</code> <code>group-has-[:]:</code> <code>peer-checked:</code> <code>peer-focus:</code> <code>peer-invalid:</code> <code>peer-disabled:</code> <code>peer-hover:</code>
Attributes	<code>aria-expanded:</code> <code>aria-checked:</code> <code>aria-selected:</code> <code>aria-disabled:</code> <code>data-[state=open]:</code> <code>open:</code>

35.14 Animation and Transition

Group	Classes
Property	<code>transition</code> <code>transition-all</code> <code>transition-colors</code> <code>transition-opacity</code> <code>transition-shadow</code> <code>transition-transform</code> <code>transition-none</code>
Duration	<code>duration-75</code> <code>duration-100</code> <code>duration-150</code> <code>duration-200</code> <code>duration-300</code> <code>duration-500</code> <code>duration-700</code> <code>duration-1000</code>
Easing	<code>ease-linear</code> <code>ease-in</code> <code>ease-out</code> <code>ease-in-out</code>
Delay	<code>delay-75</code> <code>delay-100</code> <code>delay-150</code> <code>delay-200</code> <code>delay-300</code> <code>delay-500</code> <code>delay-700</code> <code>delay-1000</code>
Keyframe animations	<code>animate-spin</code> <code>animate-ping</code> <code>animate-pulse</code> <code>animate-bounce</code> <code>animate-none</code>
Accessibility	<code>motion-safe:</code> <code>motion-reduce:</code>
Entry	<code>starting:opacity-0</code>

35.15 Transform

Group	Classes
Scale	<code>scale-0</code> <code>scale-50</code> <code>scale-75</code> <code>scale-90</code> <code>scale-95</code> <code>scale-100</code> <code>scale-105</code> <code>scale-110</code> <code>scale-125</code> <code>scale-150</code> <code>scale-x-*</code> <code>scale-y-*</code>

Group	Classes
Rotate	<code>rotate-0</code> <code>rotate-1</code> <code>rotate-2</code> <code>rotate-3</code> <code>rotate-6</code> <code>rotate-12</code> <code>rotate-45</code> <code>rotate-90</code> <code>rotate-180</code> <code>-rotate-*</code>
Translate	<code>translate-x-*</code> <code>translate-y-*</code> <code>-translate-x-*</code> <code>translate-x-1/2</code> <code>-translate-y-full</code>
Skew	<code>skew-x-*</code> <code>skew-y-*</code> <code>-skew-x-*</code>
Origin	<code>origin-center</code> <code>origin-top</code> <code>origin-top-right</code> <code>origin-right</code> <code>origin-bottom-right</code> <code>origin-bottom</code> <code>origin-bottom-left</code> <code>origin-left</code> <code>origin-top-left</code>
3D	<code>transform-3d</code> <code>perspective-*</code> <code>rotate-x-*</code> <code>rotate-y-*</code> <code>backface-hidden</code>

35.16 Effects and Filters

Group	Classes
Opacity	<code>opacity-0</code> <code>opacity-25</code> <code>opacity-50</code> <code>opacity-75</code> <code>opacity-100</code>
Blur	<code>blur-xs</code> <code>blur-sm</code> <code>blur-md</code> <code>blur-lg</code> <code>blur-xl</code> <code>blur-2xl</code> <code>blur-3xl</code>
Brightness / contrast	<code>brightness-50</code> <code>brightness-75</code> <code>brightness-110</code> <code>brightness-125</code> <code>contrast-50</code> <code>contrast-125</code> <code>contrast-150</code>
Color filters	<code>grayscale</code> <code>sepia</code> <code>invert</code> <code>saturate-0</code> <code>saturate-150</code> <code>hue-rotate-90</code>
Backdrop	<code>backdrop-blur-sm</code> <code>backdrop-blur-md</code> <code>backdrop-blur-lg</code> <code>backdrop-brightness-50</code> <code>backdrop-saturate-150</code> <code>backdrop-contrast-125</code>
Blend	<code>mix-blend-multiply</code> <code>mix-blend-screen</code> <code>mix-blend-overlay</code> <code>bg-blend-multiply</code>
Misc	<code>isolate</code> <code>box-decoration-clone</code> <code>box-decoration-slice</code>
Objects	<code>object-cover</code> <code>object-contain</code> <code>object-fill</code> <code>object-none</code> <code>object-scale-down</code> <code>object-center</code> <code>object-top</code>
Aspect	<code>aspect-auto</code> <code>aspect-square</code> <code>aspect-video</code> <code>aspect-[4/3]</code>

36. Common Mistakes

36.1 Using Invalid Class Names

Tailwind silently ignores classes it does not know: no error, no style.

Wrong (invalid or outdated)	Right
<code>text-bold</code>	<code>font-bold</code>
<code>font-24</code>	<code>text-2xl</code> or <code>text-[24px]</code>
<code>margin-4</code> , <code>padding-4</code>	<code>m-4</code> , <code>p-4</code>
<code>center</code>	<code>text-center</code> or <code>flex justify-center</code>
<code>bg-color-blue</code>	<code>bg-blue-500</code>
<code>border-radius-lg</code>	<code>rounded-lg</code>
<code>flex-center</code>	<code>flex items-center justify-center</code>
<code>w-100</code> (in v3 not valid; in v4 means 25rem)	<code>w-full</code> (or <code>w-96</code>)
<code>rounded-md-2</code>	<code>rounded-md</code>
<code>text-color-red</code>	<code>text-red-500</code>
<code>bg-opacity-50</code> (removed in v4)	<code>bg-black/50</code>
<code>flex-shrink-0</code> (removed in v4)	<code>shrink-0</code>
<code>shadow-inner</code> (v3)	<code>inset-shadow-sm</code>

How to catch them: install **Tailwind CSS IntelliSense**; unknown classes get a squiggle, and hovering any valid class shows its CSS.

36.2 Forgetting Mobile-First Behavior

```
<!-- WRONG: expecting "sm:" to mean phones. This is hidden on phones too. -->
<div class="sm:hidden">...</div>    <!-- hidden from 640px UP, visible below -->

<!-- Think it through: un-prefixed = all sizes; prefixed = from that size upward -->
<div class="block md:hidden">Mobile only</div>
<div class="hidden md:block">Desktop only</div>
```

Design the **mobile layout first** with plain classes, then add `md:` and `lg:` for larger screens. Do not start with `lg:` classes and try to patch mobile afterwards.

36.3 Confusing justify- and items-

- `justify-*` = **main axis** (horizontal in `flex-row`).
- `items-*` = **cross axis** (vertical in `flex-row`).
- After `flex-col`, they swap visual directions.

```
<div class="flex justify-center items-center">...</div>    <!-- center both ways -->
<div class="flex flex-col items-center">...</div>         <!-- column, centered horizontally -->
```

If something does not center vertically, the parent usually has no height: add `h-screen`, `min-h-screen`, or `h-64`.

36.4 Overusing Arbitrary Values

```
<!-- Over-specified: ignores the design scale -->
<div class="mt-[16px] p-[24px] text-[#3b82f6] rounded-[8px] w-[100%]">

<!-- Same result with the scale -->
<div class="mt-4 p-6 text-blue-500 rounded-lg w-full">
```

Prefer scale values. If you keep using one custom value, add it to `@theme`.

36.5 Incorrect Responsive Classes

Mistake	Fix
<code>md:text-lg text-sm</code> works but is confusing	write base first: <code>text-sm md:text-lg</code>
Using <code>sm:</code> to target phones	un-prefixed is the phone style
<code>md:hidden lg:block</code> when you wanted only desktop	<code>hidden lg:block</code>
Unsorted breakpoints in thinking (<code>lg:</code> then <code>md:</code>)	order mentally <code>sm</code> , <code>md</code> , <code>lg</code> , <code>xl</code> , <code>2xl</code>
Expecting <code>container</code> to center	add <code>mx-auto px-4</code>

36.6 Tailwind v3 vs v4 Confusion

Symptom	Cause	Fix
<code>bg-opacity-*</code> does nothing	removed in v4	<code>bg-black/50</code>
Borders look black or dark	default color is now <code>currentColor</code>	always add <code>border-gray-200</code>
<code>ring</code> looks thin	v4 <code>ring</code> = 1px	use <code>ring-3</code>
Shadows or radii look smaller	scale renamed one step	v3 <code>shadow</code> becomes <code>shadow-sm</code> ; v3 <code>rounded</code> becomes <code>rounded-sm</code>
<code>@tailwind base;</code> has no effect	v3 directive	<code>@import "tailwindcss";</code>
<code>tailwind.config.js</code> is ignored	v4 is CSS-first	use <code>@theme</code> in CSS, or <code>@config</code>
<code>dark:</code> ignores your <code>.dark</code> class	v4 default is media-based	add <code>@custom-variant dark (&:where(.dark, .dark*))</code> ;
<code>outline-none</code> removes accessibility	semantics changed	use <code>outline-hidden</code>
Buttons no longer show a pointer	v4 preflight change	add <code>cursor-pointer</code>
<code>bg-[--brand]</code> fails	old variable syntax	<code>bg(--brand)</code>
<code>hover:</code> does nothing on a phone	v4 only applies on hover-capable devices	design for tap with <code>active:</code>

Which version am I on? Check `package.json` (`"tailwindcss": "^4..."`) and your CSS entry (`@import "tailwindcss"` means v4).

36.7 Dynamic Class Names Tailwind Cannot Detect

Tailwind finds classes by scanning your source files **as plain text**. It does not run your JavaScript. A class name assembled from pieces never appears as a complete string, so its CSS is never generated.

```
// WRONG: Tailwind never sees "bg-red-500" or "bg-blue-500"
<div className={`bg-${color}-500`} />
<div className={`text-${size}`} />

// RIGHT: complete class names, chosen by logic
const colors = {
  red: 'bg-red-500',
  blue: 'bg-blue-500',
}
<div className={colors[color]} />

// RIGHT: ternary with full strings
<div className={isError ? 'text-red-600' : 'text-green-600'} />
```

For truly dynamic numbers (user-chosen pixel values, API colors) use the **style prop** or a CSS variable:

```
<div style={{ width: `${percent}%`, backgroundColor: userColor }} />
<div className="w-(--w)" style={{ '--w': `${percent}%` }} />
```

36.8 Missing Source Detection / Configuration

- **v4:** classes are detected automatically from your project (respecting `.gitignore`). If styles are missing from a library in `node_modules`, another folder, or a file type Tailwind skips, add `@source "../node_modules/some-ui-lib";` in your CSS.
- **v3:** the `content` array in `tailwind.config.js` must list every file with classes (`"./src/**/*.{js,jsx,ts,tsx}"`). A missing path means missing styles in production.
- Make sure your CSS file (with `@import "tailwindcss"`) is actually imported in `main.jsx` and the Vite plugin is registered.
- Restart the dev server after changing config.

36.9 Using `@apply` Unnecessarily

```
/* Avoid: you are just rebuilding old-style CSS and losing Tailwind's benefits */
.btn { @apply px-4 py-2 bg-blue-600 text-white rounded-lg hover:bg-blue-700; }
```

In React, **make a component** instead:

```
function Button({ children }) {
  return <button className="rounded-lg bg-blue-600 px-4 py-2 text-white hover:bg-blue-700">{children}</button>
}
```

`@apply` is fine for styling markup you cannot control (CMS content, third-party widgets) or tiny base styles. In Vue, Svelte or CSS Modules you may need `@reference "../app.css";` at the top of the stylesheet to use `@apply`.

36.10 Specificity and Override Problems

- Tailwind utilities are **flat** (single class). Order in the `class` attribute does **not** decide who wins; the order of rules in the generated stylesheet does.
- `class="p-4 p-8"` is unpredictable. Never put conflicting utilities on the same element. In React components that accept `className`, merge with `tailwind-merge` (Section 37).
- A plain CSS rule can beat utilities or lose to them depending on layers. In v4, Tailwind uses CSS `@layer`; unlayered custom CSS **beats** utilities. Put custom component CSS inside `@layer components { ... }` or use utilities.
- Third-party CSS that wins: add your own `!` modifier as a last resort (`text-red-500!`).

36.11 Important Modifier Misuse

```
<!-- Bad: using ! to cover up your own conflicting classes -->
<p class="text-gray-500 text-red-500!">

<!-- Better: just remove the conflict -->
<p class="text-red-500">
```

Using `!` widely makes every later override require another `!`. Reserve it for third-party CSS and rare inline-style fights.

36.12 Other Frequent Mistakes

Mistake	Fix
<code>h-full</code> does nothing	parent needs a height (<code>h-screen</code> , <code>min-h-screen</code> , or flex/grid sizing)
Flex child will not truncate	add <code>min-w-0</code> to the flex child
<code>absolute</code> element flies to the page corner	add <code>relative</code> to the parent
<code>sticky</code> does nothing	add <code>top-0</code> ; remove <code>overflow-hidden</code> from ancestors
<code>z-10</code> has no effect	element needs a position (<code>relative</code> , <code>absolute</code> , <code>fixed</code> , <code>sticky</code>) or be a flex/grid item
<code>w-screen</code> causes horizontal scroll	use <code>w-full</code> (the scrollbar width is included in 100vw)
Text color not changing	check you used <code>text-</code> for color and not just size
List bullets missing	Tailwind resets lists: add <code>list-disc pl-5</code>
Image stretched	add <code>object-cover</code> (or <code>object-contain</code>) with a set width and height
Hover effect jumps	add <code>transition</code> and consider <code>transform</code> instead of changing layout size
Heading sizes all look the same	the reset removes default <code>h1 - h6</code> sizes: set <code>text-*</code> and <code>font-*</code> yourself
Forgot <code><meta name="viewport" ...></code>	responsive classes will not work on phones without it
Gradient shows no color	needs <code>bg-linear-to-*</code> and a <code>from-</code> color

37. React + Tailwind Notes

37.1 className, Not class

```
// HTML: <div class="p-4 bg-blue-500 text-white rounded-lg">
<div className="p-4 bg-blue-500 text-white rounded-lg">Hello</div>
```

Other JSX differences: `for` becomes `htmlFor`; self-close `<input />` and `<img />`; inline styles take objects (`style={{ width: 20 }}`).

37.2 Conditional Classes: Ternary

```
function Button({ primary }) {
  return (
    <button className={primary ? 'bg-blue-600 text-white' : 'bg-gray-100 text-gray-900'}>
      Click
    </button>
  )
}
```

37.3 Template Literals (for combining static + conditional)

```
<button className={`rounded-lg px-4 py-2 ${active ? 'bg-blue-600 text-white' : 'bg-gray-100'}}`>
  Tab
</button>
```

Pitfalls: an empty condition can leave stray spaces (harmless); and **never build the class name from parts** (`bg-${color}-500`): see [Section 36.7](#). Template literals get unreadable with many conditions: use `clsx`.

37.4 clsx (clean conditional classes)

```
npm install clsx
```

```
import clsx from 'clsx'

<button
  className={clsx(
    'rounded-lg px-4 py-2 font-medium transition', // always
    active && 'bg-blue-600 text-white',             // if true
    !active && 'bg-gray-100 text-gray-900',          // if false
    disabled && 'cursor-not-allowed opacity-50',
    { 'ring-2 ring-blue-500': focused }             // object form
  )}
/>
```

`clsx` ignores `false`, `null` and `undefined`, so `condition && 'class'` is safe.

37.5 tailwind-merge and the cn() Helper

Problem: a reusable component with `p-4` receives `className="p-8"`. Both classes are present and the winner is unpredictable. `tailwind-merge` resolves conflicts so the **last** one wins.

```
npm install clsx tailwind-merge
```

```
// src/lib/cn.js
import { clsx } from 'clsx'
import { twMerge } from 'tailwind-merge'

export function cn(...inputs) {
  return twMerge(clsx(inputs))
}
```

```
import { cn } from '../lib/cn'

function Card({ className, children }) {
  return <div className={cn('rounded-xl bg-white p-4 shadow', className)}>{children}</div>
}

<Card className="p-8 bg-gray-50">...</Card> // result: rounded-xl shadow p-8 bg-gray-50
```

Make sure you use a `tailwind-merge` version that supports Tailwind v4 (v3.x of the package).

37.6 Variant Maps (safe dynamic styling)

Use an object whose values are **complete class strings**:

```
const variants = {
  primary: 'bg-blue-600 text-white hover:bg-blue-700',
  secondary: 'bg-gray-100 text-gray-900 hover:bg-gray-200',
  danger: 'bg-red-600 text-white hover:bg-red-700',
}
const sizes = {
  sm: 'px-3 py-1.5 text-sm',
  md: 'px-4 py-2 text-base',
  lg: 'px-6 py-3 text-lg',
}

export function Button({ variant = 'primary', size = 'md', className, ...props }) {
  return (
    <button
      className={cn(
        'inline-flex items-center justify-center rounded-lg font-medium transition',
        'focus-visible:ring-2 focus-visible:ring-blue-500 focus-visible:ring-offset-2',
        'disabled:cursor-not-allowed disabled:opacity-50',
        variants[variant],
        sizes[size],
        className
      )}
      {...props}
    />
  )
}

// <Button variant="danger" size="lg">Delete</Button>
```

For large design systems, the `class-variance-authority` (cva) library formalizes exactly this pattern.

37.7 Reusable Component Styling

- **Extract components, not CSS classes.** Repeat markup? Make a `<Card>`, `<Badge>`, `<Button>`.
- Accept a `className` prop and merge it with `cn()` so callers can tweak.

- Spread remaining props (`{...props}`) so `onClick`, `type`, `aria-*` pass through.
- Keep one place for each variant map.

```
export function Badge({ tone = 'gray', children }) {
  const tones = {
    gray: 'bg-gray-100 text-gray-700',
    green: 'bg-green-100 text-green-800',
    red: 'bg-red-100 text-red-800',
  }
  return <span className={cn('inline-flex rounded-full px-2.5 py-0.5 text-xs font-medium', tones[tone])}>{children}</span>
}
```

37.8 Styling by State with Data Attributes

Let React set a `data-*` attribute and style it with Tailwind variants. No string juggling.

```
<button
  data-active={isActive}
  className="rounded-lg px-4 py-2 data-[active=true]:bg-blue-600 data-[active=true]:text-white"
>
  Tab
</button>

<div aria-expanded={open} className="group">
  <svg className="size-4 transition-transform group-aria-expanded:rotate-180" />
</div>
```

37.9 Truly Dynamic Values

```
// Progress bar width from state: use style (or a CSS variable)
<div className="h-2 rounded-full bg-gray-200">
  <div className="h-2 rounded-full bg-blue-600 transition-all" style={{ width: `${progress}%` }} />
</div>

// CSS variable pattern: keep Tailwind classes, feed in the number
<div className="size-(--size) rounded-full bg-blue-500" style={{ '--size': `${px}px` }} />
```

37.10 Lists, Mapping and Layout in React

```
export default function ProductGrid({ products }) {
  return (
    <div className="grid grid-cols-1 gap-6 sm:grid-cols-2 lg:grid-cols-4">
      {products.map(p => (
        <article key={p.id} className="group overflow-hidden rounded-xl border border-gray-200 bg-white">
          <img className="aspect-square w-full object-cover transition duration-300 group-hover:scale-105" src={p.image} alt={p.name} />
          <div className="p-4">
            <h3 className="font-semibold text-gray-900">{p.name}</h3>
            <p className="text-gray-500">${p.price}</p>
          </div>
        </article>
      ))}
    </div>
  )
}
```

37.11 Handy React + Tailwind Checklist

- Use `className`. Never concatenate partial class names. Keep full strings.
- Use `clsx / cn` once you have more than one condition.
- Use `tailwind-merge` in components that accept `className`.
- Use `data/aria` attributes for state styling.
- Use `style` or CSS variables for runtime numbers and colors.
- Install **Tailwind CSS IntelliSense** and **prettier-plugin-tailwindcss** (add `tailwindFunctions: ["clsx", "cn"]` to Prettier config to sort classes inside those calls).
- Keep class lists readable: break long lists across lines inside template strings or `cn()` calls, grouped by purpose (layout, spacing, color, state).

Appendix: Mental Checklist When Styling Anything

1. **Layout:** `flex` / `grid` / `block` ? Direction, alignment, gap.
2. **Size:** `w-` / `h-` / `max-w-` / `aspect-` .
3. **Spacing:** `p-` inside, `m-` outside (or `gap-`).
4. **Typography:** `text-` size, `font-` weight, `leading-` , color.
5. **Colors:** `bg-` , `text-` , `border-` with a shade (50-950).
6. **Shape:** `border` , `rounded-` , `shadow-` , `ring-` .
7. **States:** `hover:` , `focus-visible:` , `disabled:` , `dark:` .
8. **Responsive:** add `md:` / `lg:` overrides **after** the mobile layout works.
9. **Motion:** `transition` , `duration-` , `hover:scale-` .
10. **Accessibility:** focus ring, labels, `sr-only` , contrast.

End of the Tailwind CSS Master Cheat Sheet (documenting Tailwind CSS v4.3).